

Universidad de Oviedo

From the Risk-Neutral Measure to Girsanov's Theorem in Stochastic Integration: Pricing of Financial Products

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Abstract

This work develops, in a self-contained and mathematically rigorous manner, the theory of European option pricing, culminating in the proof of the Black-Scholes formula for the European call *option. The approach adopted is carried out in a unified way: we first formulate the problem in the discrete setting of the multi-period binomial model, where the central conceptual ideas (no-arbitrage principle, replication, risk-neutral measure, martingale representation) are accessible with elementary probability tools; we then translate the problem and its resolution to the continuous setting of the Black-Scholes model, developing along the way the necessary apparatus of stochastic calculus. The chosen approach, via martingales rather than the classical route of partial differential equations, makes transparent the translation of the ideas from the discrete case to the continuous case, and allows the Black-Scholes formula to be presented as the natural limit of the multi-period binomial formula when the time steps become infinitely small. Thus, the development requires introducing and constructing in detail a substantial part of the stochastic calculus needed for the continuous setting: the Itô integral, Girsanov's theorem and the martingale representation theorem. These tools are not only necessary for the purposes of this work, but also play a central role in the theory of stochastic processes. The dissertation also includes the derivation of the classical Black-Scholes PDE from the formula obtained via martingales, thus showing the consistency between the two formulations.*

Keywords: no-arbitrage, martingale, Itô integral, Girsanov's theorem, martingale representation, Black-Scholes formula.

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Notation

Throughout this work the following notation will be used.

Probability spaces and filtrations

$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space.
$(\mathcal{F}_t)_{t\geq 0}$	Filtration: encodes the flow of information.
$\mathcal{F}_t^W$	Natural filtration of the Brownian motion $\{W_t\}$.
$\mathbb{P}, \mathbb{Q}$	Market measure and risk-neutral measure.
$\mathbb{Q} \sim \mathbb{P}$	The measures $\mathbb{Q}$ and $\mathbb{P}$ are equivalent.
$d\mathbb{Q}/d\mathbb{P}$	Radon-Nikodym derivative of $\mathbb{Q}$ with respect to $\mathbb{P}$.

Expectations and indicators

$\mathbb{E}_{\mathbb{P}}[X]$	Expectation of X under $\mathbb{P}$.
$\mathbb{E}_{\mathbb{Q}}[X]$	Expectation of X under $\mathbb{Q}$.
$\mathbb{E}[X \mid \mathcal{G}]$	Conditional expectation of X given $\mathcal{G}$.
$\mathbf{1}_A$	Indicator function of the event A.

Stochastic processes

$\{W_t\}_{t\geq 0}$	Standard Brownian motion under $\mathbb{P}$.
$\{\widetilde{W}_t\}$	Brownian motion under $\mathbb{Q}$ (Girsanov).
$\{S_t\}$	Price of the underlying asset.
$\widetilde{S}_t = e^{-rt} S_t$	Discounted price of the underlying.
$B_t = e^{rt}$	Price of the risk-free bond.
$\{V_t\}$	Portfolio value.
(ϕ_t, ψ_t)	Holdings of the underlying and of the bond in the portfolio.
$\{L_t\}$	Radon-Nikodym density process.

Stochastic calculus

$\int_0^t \phi_s \, dW_s$	Itô stochastic integral.
$\mathcal{H}^2[0,T]$	Processes with $\mathbb{E}[\int_0^T \phi_s^2 \, ds] < \infty$.
$\langle X, Y \rangle_t$	Quadratic covariation of X and Y.
$dX_t = a_t dt + b_t dW_t$	Differential notation for Itô processes.

Parameters of the Black-Scholes model

r	Constant risk-free interest rate.
μ, σ	Expected return and volatility of the underlying.
$T,\ K$	Maturity and *strike* of the option.
$\theta = (\mu - r)/\sigma$	Market price of risk.
$N(\cdot)$	Standard normal distribution function.
d_1, d_2	Coefficients of the Black-Scholes formula.
L^2	Space of square-integrable random variables.
$\coloneqq$	Equality by definition; a.s. stands for "almost surely".

Chapter 1

Introduction

1.1 Motivation and context

A *financial option* is a contract that grants its holder the right, but not the obligation, to buy or sell an underlying asset (a stock, an index, a currency, a commodity) at a predetermined price on a future date. The simplest option, the *European call*, grants the right to buy at a price K (the *strike*) on a date T (the *maturity*). The holder of the option will exercise the right if and only if, when the date T arrives, the price of the underlying is above K; in that case they will obtain a profit equal to the difference. The central question, from the mathematical point of view, is the following: what is the fair price of this contract at the time it is written?

The question admits a precise mathematical formulation, anchored in the *no-arbitrage principle*: the price of the contract must be such that no market participant can obtain a sure profit without taking on any risk. Under this principle, together with a mathematical description of the dynamics of the underlying, the price is uniquely determined. Determining this price, and constructing the strategy that allows the writer of the option to *replicate* its value, is the problem around which the present work is organized.

Historical note. The first mathematical formulation of the problem is due to Louis Bachelier, who in his doctoral thesis *Théorie de la spéculation* [1] modeled the dynamics of stock prices by means of Brownian motion. Bachelier's work, ignored in its time, was rediscovered in the mid-twentieth century by Paul Samuelson and other authors, who refined the model by replacing arithmetic Brownian motion with *geometric Brownian motion*, which ensures the positivity of prices. The problem was definitively solved in 1973 by Fischer Black, Myron Scholes and Robert Merton, in two seminal papers [2, 6] recognized in 1997 with the Nobel Prize in Economics awarded to Scholes and Merton (Black had died in 1995). The Black-Scholes formula, which constitutes the central result of the present work, is today one of the most influential mathematical developments of the last half century.

Why it is mathematically interesting. Beyond its practical relevance, the problem requires a rich mathematical apparatus: advanced probability (martingales, conditional expectation), stochastic analysis (Brownian motion, Itô integral and formula), functional analysis (density in L^2) and, in its PDE version, parabolic equations. The Black-Scholes formula is a meeting point of several central areas of modern mathematics.

A more general framework than it seems. It is worth anticipating an idea that runs through the whole work and goes beyond the financial problem that motivates it. The notion of fair price will lead us, both in the discrete and in the continuous model, to replace the market probability measure by an equivalent one (the *risk-neutral measure*) under which the discounted price of the underlying becomes a *martingale*, one of the fundamental concepts of modern probability. The change of measure producing that martingale rests on two results of a general nature: the Radon-Nikodym theorem, which guarantees the existence of the density between the two measures, and Girsanov's theorem, which describes that change in the setting of Brownian motion. Neither of these tools is specific to pricing: the principle of replacing a measure by an equivalent one to transform a difficult problem into a tractable one is in common use in probability and its applications, for instance in the variance reduction of Monte Carlo simulations of rare events by means of the *importance sampling* technique (see, e.g., [4], Section 4.6). In this sense, the pricing problem makes it possible to introduce mathematical tools with much wider applications.

1.2 Objectives and approach of the work

The present work sets out to develop, in a self-contained and mathematically rigorous manner, the theory of European option pricing, culminating in the proof of the Black-Scholes formula for the European *call*. The development will be carried out in a unified way: first presenting the problem in the *discrete* setting of the multi-period binomial model, where the conceptual ideas are accessible with elementary probability tools, and then transferring the problem and its solution to the *continuous* setting of the Black-Scholes model, by developing the necessary stochastic calculus.

Justification of the discrete-to-continuous plan. The chosen order of exposition is pedagogical, not historical: chronologically, the Black-Scholes formula (1973) predates the multi-period binomial model (Cox, Ross and Rubinstein, 1979). However, from the mathematical point of view the binomial model is structurally simpler: it allows the central ideas of the theory (change of measure, replication, no-arbitrage) to be articulated with elementary probability tools, without the apparatus of stochastic calculus. This fact is what justifies the plan adopted: once the logic of the problem has been assimilated in the discrete setting, the passage to the continuous one reduces to a technical translation of the tools, keeping the conceptual ideas intact.

The martingale route. The continuous Black-Scholes theory admits two classical routes: martingales (change of measure and representation theorems) and PDEs (instantaneous hedging). We opt for the former since it admits a direct translation from the discrete case. This choice has, moreover, an advantage that goes beyond mere exposition: the tools it requires, the Itô integral, Girsanov's theorem and the martingale representation theorem, are mathematical objects of interest and usefulness in their own right, and they make up a good part of the technical content of the work. The PDE route is recovered at the end (Section 5.4) as a consequence of the martingale formulation.

What the work does not address. For reasons of length and of consistency with the chosen plan, the work is restricted to *European* options on a single underlying, in the Black-Scholes model with constant parameters. Consequently, several natural extensions lie outside

its scope—American options, models with stochastic volatility or jumps, numerical methods for *payoffs* without a closed-form formula, calibration to market data—and are briefly collected in Chapter 6 as further directions.

1.3 Sources and own contribution

The expository thread of this work essentially follows that of A. Etheridge [3], from whom we take both the overall structure (the passage from the discrete binomial model to the Black-Scholes model via the martingale route) and the order in which the main ideas are introduced: the motivation of the stochastic integral from no-arbitrage, the change of measure as the crucial step of pricing, and the martingale representation theorem. The introduction to stochastic calculus, which is restricted to Brownian motion, likewise follows this structure.

As for the technical part, it relies on several introductory texts on stochastic calculus and probability theory. For Brownian motion (construction, main properties and augmented filtration) we mainly follow S. E. Shreve [8]. The central results of stochastic calculus that we use (the construction of the Itô integral and its isometry, Itô's formula and Girsanov's theorem) are presented directly for Brownian motion, following the introductory texts on the subject [7, 8] and the aforementioned Etheridge [3]. A more general formulation, valid for continuous square-integrable martingales and semimartingales, can be found in I. Karatzas and S. E. Shreve [5]. The results we use in the Brownian case are thus obtained as a particular case, with $\langle W \rangle_t = t$. For measure theory and martingale theory (conditional expectation, the Radon-Nikodym theorem and discrete-time martingales) we follow D. Williams [9]. Finally, the historical references [1, 2, 6] are used only to put the problem in context.

Since the results presented in this work are classical and well known, the contribution lies not so much in obtaining new results as in their selection, organization and development. Some proofs that the sources present in summary form, take for granted, or formulate at a higher level of generality than is needed here have been developed in more detail and adapted to the problem at hand. This is the case of the construction of the Itô integral in Chapter 4, with the isometry proved step by step, the extension to $\mathcal{H}^2$ and the continuity of the indefinite integral; of Itô's formula, detailing the analytic tools involved in its proof; and of the constructive no-arbitrage argument motivating the introduction of a new integral (Section 4.1). In addition, we have sought to keep a uniform notation across the different sources and to relate the discrete and continuous cases clearly. Thus, Girsanov's change of measure (Section 4.4), for instance, is presented as the continuous version of the adjustment of the probabilities of the different paths of the binomial model, and not as an independent result.

Some of the aspects that required the most work were, already in the discrete model, the very idea of *no-arbitrage* (Chapter 2). Understanding why the absence of sure profits without investment makes it possible to determine a price, and how this principle becomes the foundation of the theory, requires a change of perspective that is not immediate. In the continuous model, the difference between the natural Brownian filtration and its augmentation (Chapter 3) also required special attention, since the subsequent results needed a filtration satisfying certain technical properties. Finally, the appearance of the *Itô term* made it necessary to explain why the quadratic variation of Brownian motion cannot be neglected and where the second-order

correction $(dW_t)^2 = dt$, which distinguishes stochastic calculus from ordinary calculus, comes from.

1.4 Structure of the dissertation

The dissertation is organized in six chapters: introduction, the discrete part (Chapter 2), the continuous foundations (Chapters 3 and 4), the application to Black-Scholes (Chapter 5) and conclusions.

Chapter 2: the binomial model. It poses the pricing problem in the discrete setting of the multi-period binomial model. It sets out the elementary probability tools required (filtrations, conditional expectation, martingales, discrete stochastic integral) and solves the problem by means of a three-step recipe: change of measure, construction of the martingale associated with the discounted *payoff*, and application of the binomial representation theorem to identify the replicating portfolio. The chapter concludes with a brief introduction to the continuous setting, anticipating the convergence of the binomial random walk to Brownian motion.

Chapter 3: Brownian motion. It introduces Brownian motion as a mathematical object, establishes its existence as a theorem (without proof, citing the literature) and studies the properties of its paths: continuity, nowhere differentiability, infinite total variation and finite quadratic variation equal to the length of the interval. This last property is the technical piece that will motivate the development of the stochastic calculus of the following chapter.

Chapter 4: stochastic calculus. It constitutes the technical core of the work. It constructs the Itô stochastic integral first on simple processes and then, by density, on the space $\mathcal{H}^2$. It proves Itô's formula for functions of Brownian motion and states its general version. It formulates the change of measure in continuous time by means of Girsanov's theorem, and states the Brownian martingale representation theorem. Each of these pieces is the translation to the continuous setting of an object introduced in the discrete case in Chapter 2.

Chapter 5: the Black-Scholes model. It applies the tools of Chapter 4 to the pricing problem in the Black-Scholes model, carrying out the three-step recipe in its continuous version. It obtains the general formula for the price of a European derivative and specializes it to the *payoff* of the European *call*, explicitly deriving the classical Black-Scholes formula with its coefficients d_1 and d_2. The chapter concludes by deriving the Black-Scholes PDE from the formula obtained, showing the consistency with the classical route.

Chapter 6: conclusions. It summarizes the work carried out, presents the convergence of the multi-period binomial formula to the Black-Scholes formula (as the culmination of the relationship between the two models in quantitative terms) and briefly discusses the natural further directions of the work.

Chapter 2

Option pricing in discrete time

2.1 Financial concepts and the no-arbitrage principle

2.1.1 Basic financial definitions

Throughout this work we place ourselves in an idealized market in which two basic types of financial instruments are traded: a risk-free asset and a risky asset. Derivative contracts will be built on them; their value ultimately depends on the future behavior of the underlying asset.

The risk-free bond. We denote by B_t the price at time $t \geq 0$ of the risk-free asset, normalized so that $B_0 = 1$. This asset represents an investment without uncertainty (in practice, an interest-bearing deposit or a government bond) whose value grows deterministically at a fixed interest rate. Before writing down its expression, it is worth pausing to see how this growth is modeled.

Suppose one monetary unit is invested at an interest rate r per unit of time and that interest is reinvested (*compounded*) n times per unit of time. Each subperiod, of length $1/n$, accrues the proportional rate r/n; since interest is reinvested, the capital is multiplied in each subperiod by the factor $1 + r/n$, so that after one unit of time (n subperiods) the initial monetary unit has become

$$\left(1 + \frac{r}{n}\right)^n$$

This is *compound interest* with frequency n.

Continuous compounding arises when interest is reinvested instantaneously, that is, when the frequency n is made arbitrarily large. Taking the limit yields the classical expression

$$\lim_{n\to\infty} \left(1 + \frac{r}{n}\right)^n = e^r \tag{2.1}$$

The growth factor per unit of time is therefore e^r, and the one corresponding to an arbitrary horizon $t > 0$ is obtained by composition:

$$B_t = (e^r)^t = e^{rt}.$$

We shall therefore adopt

$$B_t = e^{rt}, \qquad r > 0, \qquad B_0 = 1, \tag{2.2}$$

as our model for the risk-free asset. The name *risk-free* is justified by the fact that the return

is known with certainty at the initial time: investing one monetary unit at $t = 0$ guarantees e^{rT} units at $t = T$, with no uncertainty about the outcome.

The choice of continuous compounding entails no loss of generality with respect to the usual discrete regimes: it is merely a convenient way of expressing the same phenomenon. Indeed, if a discrete regime offers an effective rate r' per unit of time, so that one monetary unit becomes $1 + r'$ after one unit of time, the continuous rate r producing the same growth factor is the one satisfying $e^r = 1 + r'$, that is,

$$r = \log(1 + r').$$

For instance, an effective annual rate of 5% ($r' = 0.05$) is equivalent to a continuous rate $r = \log(1.05) \approx 0.0488$, that is, 4.88%. From now on, r will always denote the continuously compounded rate.

The risky stock. By contrast, the price of the risky asset (which we shall generically call the *stock*) is modeled as a stochastic process $\{S_t\}_{t\geq 0}$ defined on a suitable probability space. The specific dynamics of $\{S_t\}$ will be made precise as the work progresses: in this chapter it will take values in a finite set at each time (binomial model), whereas in Chapters 3 and 5 we shall consider continuous-time models.

Derivative contracts. A *derivative contract* is an agreement between two parties whose *payoff* (the amount they must exchange at a future date) depends on the price of the underlying asset. Three derivatives will appear repeatedly throughout this work; they are introduced in the following definitions, where $T > 0$ denotes the maturity date and $K > 0$ the price agreed in the contract.

Definition 2.1 (Forward contract)**.** A *forward* contract on the stock S with maturity T and delivery price K is an agreement whereby the buyer is obliged to purchase the asset S at time T at the price K, fixed in advance. The *payoff* to the buyer at T is the random variable

$$S_T - K.$$

The *forward* allows both parties to fix today the price of a transaction that will be executed in the future, thereby removing the uncertainty about the price at which the asset will be bought or sold at T. It is, for this reason, a common hedging tool: whoever will need the asset later is protected against a rise in its price, and whoever intends to dispose of it, against a fall.

The *forward* is a contract *of obligation*: both parties must honor the agreement, so the *payoff* may be negative if $S_T < K$. The natural question is which value of K makes the contract *fair*, in the sense that it can be entered into at no cost at $t = 0$.

Before defining them, it is worth explaining what the two options that follow are for. Unlike the *forward*, which binds both parties, an option works in practice as *insurance* against movements in the price of the underlying. A *call* is of interest to someone who expects to buy the asset in the future and wants protection against a rise: it allows them to lock in a purchase price today, benefiting from the rise if it occurs, but without being bound if the price falls. A *put*, on the other hand, is of interest to someone who owns the asset or plans to sell it and fears a fall: it guarantees a selling price, protecting their gains against declines. Besides this hedging use,

options are also employed for speculative purposes, since they allow one to bet on the direction of the price with a small initial outlay.

Definition 2.2 (European *call* option)**.** A European *call option* on S with maturity T and strike price K gives the buyer the right (but not the obligation) to purchase the asset at T at the price K. Its *payoff* at T is the random variable

$$(S_T - K)^+ \coloneqq \max\{S_T - K,\, 0\}.$$

Definition 2.3 (European *put* option)**.** A European *put option* on S with maturity T and strike price K gives the buyer the right (but not the obligation) to sell the asset at T at the price K. Its *payoff* at T is the random variable

$$(K - S_T)^+ \coloneqq \max\{K - S_T,\, 0\}.$$

Unlike the *forward*, the *call* and the *put* are contracts *of right*: the holder decides at maturity whether or not to exercise, choosing whichever is more favorable. Their *payoff* is therefore nonnegative and, unlike the *forward*, whose fair price at $t = 0$ turns out to be zero, an option always has strictly positive value when it is written.

Vocabulary. In standard terminology, the asset S is called the *underlying*, the price K the *strike*, and T the *maturity*. The qualifier *European* indicates that exercise is only possible at T, in contrast with American options (exercisable at any $t \in [0, T]$), which lie outside the scope of this work.

2.1.2 The no-arbitrage principle

Before turning to the specific model that will occupy the rest of the chapter, it is worth pausing on the principle underlying the whole pricing theory. This principle, known as *absence of arbitrage* or the *no-arbitrage principle*, is not a mathematical result we are going to prove, but an assumption about the market that we take as our starting point.

The informal idea. An *arbitrage opportunity* is, intuitively, an investment strategy which, starting from zero initial capital, guarantees the investor a nonnegative profit in every possible scenario and a strictly positive one in at least one of them. In other words: a way of obtaining something from nothing, without taking on any risk.

We shall assume that the market admits no arbitrage opportunities. The justification is economic: any real opportunity would be detected and exploited immediately, causing it to disappear; arbitrages, if they exist, are transient phenomena. The mathematical formalization is deferred to Section 2.2, once the model has been specified.

The replication principle. The no-arbitrage assumption can be reformulated in a more operational way. The following statement, although informal, captures its fundamental idea.

> *If two portfolios of assets produce exactly the same future payments in all possible scenarios, then they must have the same value at any earlier time.*

The justification is immediate: if at some time $t < T$ one of the portfolios were worth more than the other, selling the expensive one and buying the cheap one would produce a sure gain without taking on any risk, contradicting the assumption. We shall call this consequence the *replication principle*.

This principle suggests a general strategy for pricing derivative contracts: given an option or, more generally, any contract whose *payoff* at T is a known random variable C_T, it suffices to construct a portfolio (made up of assets whose prices we do know) whose value at T coincides with C_T in every scenario. We then say that the portfolio *replicates* the contract. By the replication principle, the fair price of the derivative at any earlier time is necessarily equal to the value of this replicating portfolio. This is the central idea of the whole theory we shall develop: both in the discrete and in the continuous setting, the problem of finding the price of a derivative will ultimately reduce to the problem of constructing a replicating portfolio.

On the role of probabilities. It is worth making a remark that will gain significance over the course of the chapter: the replication argument, as we have formulated it, never makes use of the probabilities attached to each future scenario. Only the *possible* values of the asset come into play, not the probability of their occurrence. As we shall see, this fact has deep consequences: the fair price of a derivative is independent of the market's subjective beliefs about the future. It is one of the most surprising and beautiful ideas in the theory.

2.1.3 Pricing the forward as a warm-up case

As a first application of the replication principle, we address the simplest possible problem: the pricing of a *forward*. Throughout this subsection we assume that the two assets in the market can be traded without restrictions at any time $t \in [0, T]$, with no transaction costs and in arbitrary fractional amounts. These assumptions are idealizations, but they will suffice for the arguments that follow to hold.

The fair price of a *forward* with *strike* K. Consider a *forward* on the stock S with maturity T and delivery price K. Its *payoff* to the buyer at T is $S_T - K$. We seek the fair price V_0 of the contract at the initial time: the amount the buyer must pay the writer at $t = 0$ in exchange for receiving $S_T - K$ at $t = T$.

Consider the following portfolio, set up at $t = 0$:

- one unit of the stock is bought, at the price S_0;
- an amount Ke^{-rT} of the risk-free asset is borrowed (that is, issued).

The net cost of setting it up at $t = 0$ is $\Pi_0 = S_0 - Ke^{-rT}$. At $t = T$, the unit of stock will be worth S_T, while the debt incurred will have grown to $Ke^{-rT} \cdot e^{rT} = K$. Consequently, the value of the portfolio at $t = T$ is

$$\Pi_T = S_T - K,$$

which coincides with the *payoff* of the *forward*. By the replication principle,

$$V_0 = S_0 - Ke^{-rT}. \tag{2.3}$$

It is worth checking that the argument is consistent, in the sense that any deviation from the price (2.3) would produce an arbitrage opportunity. If the contract were sold at $V_0' > S_0 - Ke^{-rT}$, an investor could: at $t = 0$, sell the *forward* (receiving V_0'), set up the replicating portfolio (at a cost $S_0 - Ke^{-rT}$) and keep the remainder $V_0' - (S_0 - Ke^{-rT}) > 0$. At $t = T$, the payments associated with the *forward* and with the portfolio cancel out, so the investor would have obtained a sure profit without taking on any risk. The symmetric situation $V_0' < S_0 - Ke^{-rT}$ leads to an analogous arbitrage.

The fair strike and the *forward* price. In practice, *forwards* are usually entered into with $V_0 = 0$. Imposing this condition in (2.3) allows us to solve for the delivery price that makes the contract fair:

$$K = S_0 e^{rT}. \tag{2.4}$$

This quantity is called the *forward price* of the stock for the maturity T, and it is interpreted as the value at $t = T$ that one unit of stock bought today would have if compounded at the risk-free rate.

Two general lessons.

Remark 2.4 (Independence from the dynamics of the underlying). The derivation of the *forward* price has made no use whatsoever of the specific model governing the evolution of the stock price: we have assumed neither that $\{S_t\}$ follows a particular process, nor even that it takes values in a specific set. Formula (2.3) is valid in any arbitrage-free market in which the bond and the stock can be traded without restrictions. The ultimate reason is that the *payoff* of the *forward* depends *linearly* on the underlying, and therefore replication can be carried out by a static strategy. This universality is, however, an exception: as soon as the *payoff* of the derivative is nonlinear in S_T (as is the case for the *call* and the *put*), the replication argument ceases to be static and comes to require a specific model for the dynamics of the underlying.

Remark 2.5 (Irrelevance of subjective probabilities). No probability associated with the future evolution of the stock appears in formula (2.3). The fair price of the *forward* does not depend on how likely it is, according to the market's beliefs, that the stock will go up or down, nor on its expected mean return, nor on its volatility. This independence from subjective probabilities will reappear, in a subtler form, in the nontrivial models we study next.

Taking this initial example as a reference, we can now address the general problem in its simplest, though by no means trivial, version.

2.2 The one-period binary model

The simplest model that retains the essential ingredients of the problem is the *one-period binary* model: the market is observed only at $t = 0$ and $t = T$, and the price of the underlying takes only two possible values at T.

2.2.1 Description of the model and formalization of no-arbitrage

Description of the model. The market consists of a risk-free bond, with $B_t = e^{rt}$ and $r > 0$, and a stock whose initial price $S_0 > 0$ is known and whose final price S_T is a random variable

taking only two values. Specifically, we fix two constants u, d with $0 < d < u$ and assume that

$$S_T = \begin{cases} S_0\, u & \text{with probability } p, \\ S_0\, d & \text{with probability } 1-p, \end{cases}$$

for some $p \in (0,1)$. We call u and d the *up* and *down factors*, and p the *subjective probability* of an up move.

To fix ideas, we formulate the model in the language of probability theory. Let $\Omega = \{\omega_u, \omega_d\}$ be the sample space, $\mathcal{F} = 2^\Omega$, and let $\mathbb{P}$ be the probability on $(\Omega, \mathcal{F})$ defined by $\mathbb{P}(\{\omega_u\}) = p$ and $\mathbb{P}(\{\omega_d\}) = 1-p$. On $(\Omega, \mathcal{F}, \mathbb{P})$, the final price of the stock is the random variable

$$S_T(\omega_u) = S_0\, u, \qquad S_T(\omega_d) = S_0\, d.$$

We call $\mathbb{P}$ the *market measure* or *physical measure*.

Although we include p in the description, its only role is to specify which scenarios we regard as *possible* (those with strictly positive probability). The numerical value of p will play no part in any computation.

Formalization of the notion of arbitrage.

Definition 2.6 (Strategy or portfolio)**.** A *strategy* in the one-period binary model is a pair $(\phi, \psi) \in \mathbb{R}^2$, where ϕ represents the number of units of the stock and ψ the number of units of the bond held in the portfolio during the period $[0, T]$.

The value of the strategy at each time is

$$V_0(\phi,\psi) = \phi\, S_0 + \psi,$$
$$V_T(\phi,\psi) = \phi\, S_T + \psi\, e^{rT}.$$

We allow ϕ and ψ to take any real value, including negative ones: $\phi < 0$ corresponds to *going short* in the stock, and $\psi < 0$ to borrowing an amount $|\psi|$ from the bank. This possibility is essential for the theory: without it, not every option could be replicated.

Definition 2.7 (Arbitrage opportunity)**.** A strategy (ϕ, ψ) is an *arbitrage opportunity* if:

(i) $V_0(\phi,\psi) = 0$;

(ii) $V_T(\phi,\psi) \geq 0$ almost surely under $\mathbb{P}$;

(iii) $\mathbb{P}\big(\{V_T(\phi,\psi) > 0\}\big) > 0$.

We say that the model is *arbitrage-free* if it admits no strategy of this kind.

Theorem 2.8 (Characterization of no-arbitrage in the binary model)**.** *The one-period binary model is arbitrage-free if and only if*

$$d < e^{rT} < u. \tag{2.5}$$

Proof. ($\Rightarrow$) We argue by contraposition: if condition (2.5) fails, we exhibit an arbitrage opportunity.

Suppose $e^{rT} \le d$. Consider the strategy $(\phi, \psi) = (1, -S_0)$, which consists in buying one unit of stock financed with the bond. Its initial value is $V_0 = S_0 - S_0 = 0$, and its final value is

$$V_T = S_T - S_0\, e^{rT} = S_0\big(S_T/S_0 - e^{rT}\big),$$

which equals $S_0(u - e^{rT}) > 0$ at ω_u and $S_0(d - e^{rT}) \ge 0$ at ω_d. The strategy is an arbitrage opportunity. Similarly, if $e^{rT} \ge u$, the strategy $(\phi, \psi) = (-1, S_0)$ yields an arbitrage.

($\Leftarrow$) Assume (2.5). Let (ϕ, ψ) satisfy $V_0 = 0$, that is, $\psi = -\phi\, S_0$. Its final value is

$$V_T(\omega_u) = \phi\, S_0\big(u - e^{rT}\big),$$
$$V_T(\omega_d) = \phi\, S_0\big(d - e^{rT}\big).$$

By (2.5), the factors $u - e^{rT}$ and $d - e^{rT}$ have opposite signs. The sign of V_T is therefore opposite in the two states, except in the trivial case $\phi = 0$, which implies $V_T \equiv 0$. There is no strategy $(\phi, \psi) \neq (0, 0)$ with $V_0 = 0$, $V_T \ge 0$ and $V_T > 0$ in at least one state. □

Condition (2.5) has a natural economic interpretation: the return on the stock must be able to exceed that of the risk-free bond (hence $u > e^{rT}$) but must also be able to fall below it (hence $d < e^{rT}$). Otherwise trivial arbitrages would arise. The assumption requires the stock to be genuinely *risky* relative to the bond. We assume it from now on.

2.2.2 Construction of the replicating portfolio

Consider a derivative with *payoff* C_T defined on $(\Omega, \mathcal{F}, \mathbb{P})$. By construction, C_T takes at most two values:

$$C_u := C_T(\omega_u), \qquad C_d := C_T(\omega_d).$$

We seek a strategy $(\phi, \psi) \in \mathbb{R}^2$ such that

$$V_T(\phi, \psi)(\omega) = C_T(\omega) \quad \text{for all } \omega \in \Omega. \tag{2.6}$$

A strategy satisfying (2.6) is called a *replicating portfolio* for the derivative.

Condition (2.6), written out explicitly in each state, becomes the linear system

$$\begin{cases} \phi\, S_0\, u + \psi\, e^{rT} = C_u, \\ \phi\, S_0\, d + \psi\, e^{rT} = C_d. \end{cases} \tag{2.7}$$

The coefficient matrix has determinant $S_0\, e^{rT}\,(u - d) > 0$, and therefore the system has a unique solution:

$$\phi = \frac{C_u - C_d}{S_0\,(u - d)}, \qquad \psi = e^{-rT}\,\frac{u\, C_d - d\, C_u}{u - d}. \tag{2.8}$$

Theorem 2.9 (Replication in the one-period binary model)**.** *In the arbitrage-free one-period binary model, every derivative with* payoff C_T *admits a unique replicating portfolio* $(\phi, \psi) \in \mathbb{R}^2$,

given by (2.8). *The fair price of the derivative at* $t = 0$ *is*

$$V_0 = \phi\, S_0 + \psi = \frac{C_u - C_d}{u - d} + e^{-rT}\, \frac{u\, C_d - d\, C_u}{u - d}. \tag{2.9}$$

Proof. The existence and uniqueness of (ϕ, ψ) were established in the preceding discussion. To see that $V_0 = \phi\, S_0 + \psi$ is the fair price, suppose the derivative trades at $V_0' > V_0$. Selling the derivative and setting up the replicating portfolio requires an initial investment $V_0 - V_0' < 0$, leaving a positive remainder. At $t = T$, the *payoff* of the derivative and the value of the portfolio cancel out, producing a sure profit. Similarly for $V_0' < V_0$. Both situations contradict the no-arbitrage assumption. □

Two remarks. Formula (2.9) does not contain the subjective probability p: the price depends only on $S_0, u, d, r, T, C_u, C_d$. On the other hand, the quantity ϕ obtained in (2.8) is the ratio between the change in the *payoff* and the change in the price of the underlying: it measures the sensitivity of the derivative to movements of the underlying and is known in the financial literature as the *delta*.

2.2.3 Emergence of the risk-neutral measure

Formula (2.9) admits an elementary algebraic manipulation that brings out a new (and central) object of the theory. Regrouping the terms in C_u and C_d,

$$V_0 = e^{-rT}\left[\frac{e^{rT} - d}{u - d}\, C_u + \frac{u - e^{rT}}{u - d}\, C_d\right].$$

Defining

$$q \coloneqq \frac{e^{rT} - d}{u - d}, \tag{2.10}$$

and noting that $1 - q = (u - e^{rT})/(u - d)$, the formula takes the compact form

$$V_0 = e^{-rT}\left[\, q\, C_u + (1 - q)\, C_d \,\right]. \tag{2.11}$$

Properties of the ratio q**.** The no-arbitrage condition $d < e^{rT} < u$ implies that both $e^{rT} - d > 0$ and $u - d > 0$, and that $u - e^{rT} > 0$. Consequently, $q \in (0, 1)$. The converse statement is also easily verified. This equivalence between $q \in (0, 1)$ and the no-arbitrage condition (2.5) is important: what the fact $q \in (0, 1)$ implies is that the coefficients appearing in formula (2.11) are nonnegative numbers adding up to one, that is, *they are the probabilities of a measure on* Ω.

Definition of the risk-neutral measure. The preceding remark justifies introducing, on the space $(\Omega, \mathcal{F})$, a new probability measure

$$\mathbb{Q}(\{\omega_u\}) = q, \qquad \mathbb{Q}(\{\omega_d\}) = 1 - q. \tag{2.12}$$

Definition 2.10 (Risk-neutral measure)**.** The probability $\mathbb{Q}$ defined in (2.12) is called the *risk-neutral measure* (or *equivalent martingale measure*) of the one-period binary model.

Note that $\mathbb{Q}$ and $\mathbb{P}$ assign strictly positive probability to the same events. In the language of measure theory, they are *equivalent*, written $\mathbb{Q} \sim \mathbb{P}$. This equivalence, trivial here, will be a delicate condition in the continuous models of Chapter 5.

The fundamental property of $\mathbb{Q}$. Let us compute the expectation (under $\mathbb{Q}$) of the discounted stock price at $t = T$:

$$\mathbb{E}_{\mathbb{Q}}\big[e^{-rT}\, S_T\big] = e^{-rT}\, S_0\, \big[q\, u + (1-q)\, d\big].$$

Substituting $q = (e^{rT} - d)/(u - d)$, a direct computation shows that $q\, u + (1-q)\, d = e^{rT}$, and hence

$$\mathbb{E}_{\mathbb{Q}}\big[e^{-rT}\, S_T\big] = S_0. \tag{2.13}$$

Identity (2.13) contains a central idea. If we introduce the notation $\widetilde{S}_t := e^{-rt} S_t$ for the discounted price, it can be rewritten as $\mathbb{E}_{\mathbb{Q}}[\widetilde{S}_T] = \widetilde{S}_0$, a property that in Section 2.3.4 we shall recognize as the preliminary version of the *martingale* property. This is why $\mathbb{Q}$ is also called the equivalent martingale measure.

As for the adjective *risk-neutral*, its origin is the following. Rewriting (2.13) without discounting, $\mathbb{E}_{\mathbb{Q}}[S_T] = S_0\, e^{rT}$: under $\mathbb{Q}$, the mean return on the stock coincides with that of the risk-free bond. That is, $\mathbb{Q}$ is the measure under which an investor would be indifferent between holding the risky asset and the risk-free asset, since both would offer the same expected return. In reality (under $\mathbb{P}$) investors are risk-averse and demand a premium for holding risky assets; the measure $\mathbb{Q}$ removes that premium by construction.

The pricing formula as a discounted expectation. By the definition (2.12) of $\mathbb{Q}$, the expression $q\, C_u + (1-q)\, C_d$ is exactly the expectation of C_T under $\mathbb{Q}$. The pricing formula then takes its definitive form

$$\boxed{V_0 = e^{-rT}\, \mathbb{E}_{\mathbb{Q}}[C_T].} \tag{2.14}$$

The fair price of a derivative at $t = 0$ is the discounted value of the expectation of its *payoff* under the risk-neutral measure $\mathbb{Q}$. The formula reduces the pricing problem to a computation of expectations, once the appropriate measure has been identified.

Let us insist on a remarkable feature of (2.14): we have arrived at it without ever using the subjective probability p.

2.2.4 Independence from subjective probabilities

An intuition worth correcting. For a reader with a background in probability, the first natural answer to the question "what is the fair price of a derivative with random *payoff* C_T?" would be: *the discounted expected value of its* payoff, that is,

$$V_0^{\text{naive}} := e^{-rT}\, \mathbb{E}_{\mathbb{P}}[C_T] = e^{-rT}\, \big[p\, C_u + (1-p)\, C_d\big]. \tag{2.15}$$

However, it suffices to compare (2.15) with the correct price (2.14) to notice that the two coincide if and only if $p = q$. Except in this fortuitous case, the two prices differ.

What is wrong with the intuitive reasoning? Formula (2.15) would only be reasonable if

investors were indifferent to risk, in which case they would value any asset by its discounted expected value. In practice, investors are risk-averse and demand premiums. If (2.15) were used to set the price of the derivative, arbitrage opportunities would arise[1].

Pricing by replication, not by expectation. The argument that led us to (2.14) does not rest on any computation of expectations under the market measure, but on the explicit construction of a replicating portfolio. What determines the price of the derivative is the possibility of *reproducing* its *payoff* by means of a combination of bonds and stocks; the subjective probability p does not come into play because the replicating portfolio reproduces the *payoff* in *all possible scenarios*, with no need to weight them by their probabilities.

The role of probabilities in the theory. The above might suggest that probabilities disappear from the pricing problem, but this is not so: the *relevant* probabilities are not those of the market, but those of the risk-neutral measure $\mathbb{Q}$. Under $\mathbb{Q}$, the price of the derivative is recovered as the discounted expectation of the *payoff*. The initial probabilistic intuition was not wrong in its form (the price is indeed a discounted expectation) but in the choice of the measure under which it must be taken.

This replacement of $\mathbb{P}$ by $\mathbb{Q}$ is the idea that will structure the subsequent development, both in the multi-period binomial model and in Black-Scholes.

2.2.5 The trinomial model and incompleteness

The natural generalization of the binary model consists in allowing more values for the underlying. We briefly study the trinomial case to show that replication breaks down, a phenomenon that motivates the notion of *completeness* of a market.

We keep the two assets of the model and assume that at $t = T$ the price takes values in $\{S_0\,u_1, S_0\,u_2, S_0\,u_3\}$ with $0 < u_3 < u_2 < u_1$, all with strictly positive probability. We assume the model is arbitrage-free (which requires $u_3 < e^{rT} < u_1$).

Given a derivative with *payoff* C_T taking the values $C_i := C_T(\omega_i)$ for $i = 1, 2, 3$, the replication condition $V_T(\phi, \psi) = C_T$ becomes a linear system of *three* equations in two unknowns:

$$\phi\, S_0\, u_i + \psi\, e^{rT} = C_i, \quad i = 1, 2, 3.$$

For an arbitrary *payoff*, the vector $(C_1, C_2, C_3)^\top$ does not in general belong to the two-dimensional subspace of $\mathbb{R}^3$ spanned by the columns of the system matrix, and therefore the system has no solution. The conclusion is immediate: in the trinomial model, the replication method fails, and the no-arbitrage principle, by itself, does not provide a unique price.

Definition 2.11 (Complete market, preliminary version)**.** A market is said to be *complete* if every derivative on the assets of the market admits a replicating portfolio.

In an incomplete market, the no-arbitrage principle still constrains the price of a derivative, but not uniquely: it leaves an *interval* of arbitrage-free prices, in general nondegenerate. To

[1] If $p > q$, then $V_0^{\text{naive}} > V_0$ (in the case $C_u > C_d$), and an investor could sell the derivative at the price V_0^{naive}, set up the replicating portfolio at cost V_0, and pocket the difference without taking on any risk. The situation $p < q$ produces a symmetric arbitrage.

fix a price within the interval, additional considerations must be invoked (investor preferences, equilibrium models, utility criteria) that go beyond the pure no-arbitrage principle.

Throughout this work we shall only deal with complete models. Both the multi-period binomial model (Section 2.3) and the Black-Scholes model (Chapter 5) will be complete; in the continuous setting, completeness will be guaranteed by the Brownian martingale representation theorem (Chapter 4).

2.3 The multi-period binomial model and the probabilistic machinery

The one-period binary model is insufficient as a realistic model of a market. Its natural generalization, the *multi-period binomial* model, allows the market to be observed at several intermediate times. Its treatment requires introducing filtrations, conditional expectations, martingales and predictable processes, concepts that formalize natural financial ideas of the model.

2.3.1 Description of the multi-period binomial model

Discretization of the time horizon. Fix $T > 0$ and an integer $N \geq 1$. We subdivide $[0, T]$ into N subintervals of length $\delta t := T/N$, and consider the times $t_k := k\,\delta t$ for $k = 0, 1, \ldots, N$. The market is observed only at these times. The bond evolves deterministically: $B_k := e^{r\,t_k}$.

The dynamics of the underlying. The stock has a known initial price $S_0 > 0$. We fix constants u, d with $0 < d < u$ and assume that between two consecutive times the price is multiplied by u with probability p or by d with probability $1 - p$, independently of the behavior in the previous steps:

$$S_{k+1} = \begin{cases} S_k\, u & \text{with probability } p, \\ S_k\, d & \text{with probability } 1 - p, \end{cases}$$

The structure is visualized by a recombining binomial tree (Figure 2.1): at time t_k, the price can take $k + 1$ distinct values, $S_0\, u^j\, d^{k-j}$ for $j = 0, \ldots, k$.

The probability space. As sample space we take the set of all possible paths:

$$\Omega := \{u, d\}^N,$$

where each $\omega = (\omega_1, \ldots, \omega_N)$ with $\omega_j \in \{u, d\}$ represents a complete history of the market. We take $\mathcal{F} := 2^\Omega$ and, on $(\Omega, \mathcal{F})$, we define the market measure $\mathbb{P}$ by

$$\mathbb{P}(\{\omega\}) := p^{\#\{j:\omega_j = u\}}\,(1 - p)^{\#\{j:\omega_j = d\}},$$

which encodes the independence of the steps. The price at time t_k is the random variable $S_k(\omega) := S_0 \prod_{j=1}^{k} f(\omega_j)$, where $f(u) = u$ and $f(d) = d$.

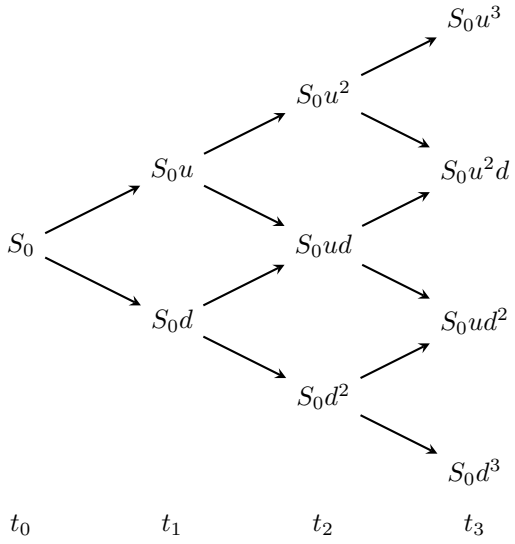


Figure 2.1: Multi-period binomial tree with $N = 3$ steps. The recombining structure (the paths ud and du meet at the same node $S_0\,ud$) reduces the number of nodes from 2^k to $k+1$.

2.3.2 Filtered probability spaces

In what follows we assume familiarity with the basic notions of probability (measurable space, σ-algebra, random variable, expectation) covered in any *STEM* degree.

σ-algebras as information. The idea is that a σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ on $(\Omega, \mathcal{F})$ can be interpreted as the set of those events whose occurrence an observer is able to determine: if $A \in \mathcal{G}$, the observer can answer the question "has A occurred?". The full σ-algebra $\mathcal{F}$ corresponds to the omniscient observer; sub-σ-algebras $\mathcal{G} \subseteq \mathcal{F}$ correspond to observers with partial information.

The σ-algebra $\mathcal{F}_k$ of the binomial model. Let us illustrate this reading on the multi-period binomial model. For each $k \in \{0, 1, \dots, N\}$, we define $\mathcal{F}_k$ as the σ-algebra generated by the partition of Ω that groups together the paths agreeing in their first k coordinates:

$$A_{(\omega_1,\dots,\omega_k)} := \{\omega' \in \Omega : \omega'_j = \omega_j,\ j = 1, \dots, k\}.$$

There are 2^k such blocks (the *atoms* of $\mathcal{F}_k$). At the extremes, $\mathcal{F}_0 = \{\emptyset, \Omega\}$ is the trivial σ-algebra (having observed nothing, no event can be distinguished from the whole world) and $\mathcal{F}_N = \mathcal{F}$ (after observing the complete path, everything is known). On the tree of Figure 2.1, $A \in \mathcal{F}_k$ if and only if we can determine whether $\omega \in A$ from the first k steps.

The sequence $(\mathcal{F}_k)_{k=0}^N$ is *increasing*: $\mathcal{F}_k \subseteq \mathcal{F}_{k+1}$ for all k. This motivates the following definition.

Definition 2.12 (Filtration)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *filtration* on this space is a sequence $(\mathcal{F}_k)_{k=0}^N$ of sub-σ-algebras of $\mathcal{F}$ with $\mathcal{F}_k \subseteq \mathcal{F}_{k+1} \subseteq \mathcal{F}$ for all k. The quadruple $(\Omega, \mathcal{F}, (\mathcal{F}_k), \mathbb{P})$ is called a *filtered probability space*.

From the financial point of view, the filtration of the binomial model is the *market filtration*.

Definition 2.13 (Adapted process)**.** A *stochastic process* is a sequence $(X_k)_{k=0}^N$ of random variables on $(\Omega, \mathcal{F})$. The process is *adapted* to the filtration $(\mathcal{F}_k)$ if, for each k, X_k is $\mathcal{F}_k$-measurable.

Measurability with respect to $\mathcal{F}_k$ formalizes the fact that the value of X_k is known after observing what has happened up to time t_k. In the binomial model, S_k is by construction a function of the first k coordinates and hence $\mathcal{F}_k$-measurable: the price process is adapted.

Definition 2.14 (Natural filtration)**.** The *natural filtration* of a process $(X_k)_{k=0}^N$ is $(\mathcal{F}_k^X)_{k=0}^N$ with $\mathcal{F}_k^X := \sigma(X_0, X_1, \ldots, X_k)$, the smallest σ-algebra making the variables $X_0, \ldots, X_k$ measurable.

It is the smallest filtration with respect to which (X_k) is adapted. In the multi-period binomial model, the market filtration coincides with the natural filtration of the price, and both will be identified and denoted simply by $(\mathcal{F}_k)$.

2.3.3 Conditional expectation

Starting point. The reader will recall the elementary notion of conditional expectation given an event $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$: the number $\mathbb{E}[X \mid B] := \mathbb{E}[X\,\mathbf{1}_B]/\mathbb{P}(B)$, which represents the mean value of X in those scenarios in which B has occurred. The notion we shall need is of a different nature: we do not want to condition on an individual event, but on a whole σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, and the result will no longer be a number but a random variable.

Motivation on the tree. Let X be an $\mathcal{F}_3$-measurable random variable on the binomial tree with $N = 3$ steps, and let us ask for its mean value "given the information $\mathcal{F}_2$". Intuitively, standing at time t_2 the observer has reached one of the 4 atoms of $\mathcal{F}_2$; in each of them there are two compatible paths, corresponding to the two outcomes of the third step. The conditional expectation given $\mathcal{F}_2$ must assign to each atom the mean value of X over the two compatible paths. It is an $\mathcal{F}_2$-measurable random variable (constant on each atom of $\mathcal{F}_2$). The axiomatic definition captures this idea without resorting to the particular structure of the model.

Kolmogorov's definition.

Definition 2.15 (Conditional expectation)**.** Let X be an integrable random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra. A random variable Y is a *conditional expectation of X given $\mathcal{G}$* if:

(i) Y is $\mathcal{G}$-measurable;

(ii) for all $A \in \mathcal{G}$, $\displaystyle\int_A Y\, d\mathbb{P} = \int_A X\, d\mathbb{P}$.

Theorem 2.16 (Existence and uniqueness)**.** *Let X be integrable and $\mathcal{G} \subseteq \mathcal{F}$. There exists a conditional expectation of X given $\mathcal{G}$, and any two conditional expectations coincide $\mathbb{P}$-almost surely.*

The proof relies on the Radon-Nikodym theorem applied to the measure $\nu(A) := \int_A X\, d\mathbb{P}$ on $(\Omega, \mathcal{G})$, which is absolutely continuous with respect to $\mathbb{P}\restriction_{\mathcal{G}}$ and admits a density Y satisfying (i) and (ii). It can be found in [9]. Since the conditional expectation is unique up to almost sure

equality, we shall speak of "*the*" conditional expectation of X given $\mathcal{G}$, denoted $\mathbb{E}[X \mid \mathcal{G}]$, with the understanding that any identity in which it appears holds $\mathbb{P}$-almost surely.

Condition (i) formalizes that the conditional expectation can be determined from the information $\mathcal{G}$. Condition (ii) is the *conservation of averages*: on any $A \in \mathcal{G}$, the integrals of $\mathbb{E}[X \mid \mathcal{G}]$ and X coincide. Taking $A = \Omega$, $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$.

Fundamental properties. In what follows, X, Y are integrable random variables and $\mathcal{G}, \mathcal{H} \subseteq \mathcal{F}$ are sub-σ-algebras.

Proposition 2.17 (Properties of conditional expectation)**.** *The following properties hold $\mathbb{P}$-almost surely.*

(i) Linearity*: $\mathbb{E}[aX + bY \mid \mathcal{G}] = a\,\mathbb{E}[X \mid \mathcal{G}] + b\,\mathbb{E}[Y \mid \mathcal{G}]$ for any $a, b \in \mathbb{R}$.*

(ii) Known measurable variables*: if X is $\mathcal{G}$-measurable, then $\mathbb{E}[X \mid \mathcal{G}] = X$.*

(iii) Taking out what is known*: if Y is $\mathcal{G}$-measurable and XY is integrable, then $\mathbb{E}[XY \mid \mathcal{G}] = Y\,\mathbb{E}[X \mid \mathcal{G}]$.*

(iv) Tower property*: if $\mathcal{H} \subseteq \mathcal{G}$, then $\mathbb{E}\big[\mathbb{E}[X \mid \mathcal{G}] \,\big|\, \mathcal{H}\big] = \mathbb{E}[X \mid \mathcal{H}]$.*

(v) *Independence: if X is independent of $\mathcal{G}$, then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$.*

By way of illustration (and because it will be the most frequently used property) we prove the tower property.

Proof of the tower property. Let $Z \coloneqq \mathbb{E}\big[\mathbb{E}[X \mid \mathcal{G}] \,\big|\, \mathcal{H}\big]$. We must check that Z satisfies the two conditions of Definition 2.15 applied to X and $\mathcal{H}$. The first is immediate: Z is $\mathcal{H}$-measurable by construction. For the second, let $A \in \mathcal{H}$. Since $\mathcal{H} \subseteq \mathcal{G}$, also $A \in \mathcal{G}$. Applying (ii) first to Z with $\mathcal{H}$ and then to $\mathbb{E}[X \mid \mathcal{G}]$ with $\mathcal{G}$,

$$\int_A Z \, d\mathbb{P} = \int_A \mathbb{E}[X \mid \mathcal{G}] \, d\mathbb{P} = \int_A X \, d\mathbb{P}.$$

By uniqueness, $Z = \mathbb{E}[X \mid \mathcal{H}]$ almost surely. □

The intuitive reading of the tower property is the following: conditioning first on richer information $\mathcal{G}$ and then on coarser information $\mathcal{H}$ is equivalent to conditioning directly on $\mathcal{H}$.

Shorthand notation. If $Y_1, \dots, Y_k$ are random variables, we write $\mathbb{E}[X \mid Y_1, \dots, Y_k] \coloneqq \mathbb{E}[X \mid \sigma(Y_1, \dots, Y_k)]$.

2.3.4 Discrete-time martingales

The property $\mathbb{E}_{\mathbb{Q}}[\widetilde{S}_T] = \widetilde{S}_0$ (eq. (2.13)) that characterized the risk-neutral measure of the one-period model admits a natural generalization: the concept of *martingale*.

Definition and first properties. We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_k), \mathbb{P})$.

Definition 2.18 (Martingale)**.** A process $(M_k)_{k=0}^N$ adapted to $(\mathcal{F}_k)$ is a *martingale* with respect to $\mathbb{P}$ if:

(i) M_k is integrable for all k;

(ii) $\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] = M_k$ almost surely, for $k = 0, \dots, N-1$.

Condition (ii) can be read as follows: the best prediction of the value of the process at time $k+1$, given the information up to k, is the current value M_k. A martingale is an "unbiased" process. Iterating (ii) and applying the tower property, for $0 \le j \le k \le N$,

$$\mathbb{E}[M_k \mid \mathcal{F}_j] = M_j \quad \text{a.s.} \tag{2.16}$$

In particular, $\mathbb{E}[M_k] = \mathbb{E}[M_0]$ for all k: the expectation of a martingale is constant in time.

The martingale property depends on the measure: replacing $\mathbb{P}$ by a suitable measure $\mathbb{Q}$ will turn the discounted price of the underlying into a martingale.

The risk-neutral measure in the multi-period model. The no-arbitrage condition becomes, taking $\delta t = T/N$,

$$d < e^{r\delta t} < u, \tag{2.17}$$

which we assume from now on. We define

$$q \coloneqq \frac{e^{r\delta t} - d}{u - d} \in (0,1), \tag{2.18}$$

in analogy with the one-period case. On $(\Omega, \mathcal{F})$, we define the measure $\mathbb{Q}$ by

$$\mathbb{Q}(\{\omega\}) \coloneqq q^{\#\{j:\omega_j = u\}} \, (1-q)^{\#\{j:\omega_j = d\}}. \tag{2.19}$$

Under $\mathbb{Q}$, the steps are independent with probability q of an up move in each. Since $q \in (0,1)$ and $p \in (0,1)$, the measures $\mathbb{Q}$ and $\mathbb{P}$ assign strictly positive probability to the same paths, that is, $\mathbb{Q} \sim \mathbb{P}$.

The discounted price as a martingale under $\mathbb{Q}$. Let $\widetilde{S}_k \coloneqq e^{-r\,t_k} S_k$ be the discounted price.

Theorem 2.19 (Martingale property of the discounted price)**.** *In the multi-period binomial model, under the measure $\mathbb{Q}$ of* (2.19)*, the process $(\widetilde{S}_k)_{k=0}^N$ is a martingale with respect to $(\mathcal{F}_k)$.*

Proof. Adaptedness is immediate since $\widetilde{S}_k$ is a function of S_k, which is $\mathcal{F}_k$-measurable. Integrability is trivial. For the martingale property, write $S_{k+1} = S_k \, f_{k+1}$, where f_{k+1} equals u with probability q and d with probability $1-q$ under $\mathbb{Q}$, independently of $\mathcal{F}_k$. Then

$$\widetilde{S}_{k+1} = e^{-r\delta t} \, \widetilde{S}_k \, f_{k+1}.$$

Applying "taking out what is known" (Proposition 2.17) and independence,

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}[\widetilde{S}_{k+1} \mid \mathcal{F}_k] &= e^{-r\delta t} \, \widetilde{S}_k \, \mathbb{E}_{\mathbb{Q}}[f_{k+1} \mid \mathcal{F}_k] = e^{-r\delta t} \, \widetilde{S}_k \, \mathbb{E}_{\mathbb{Q}}[f_{k+1}] \\ &= e^{-r\delta t} \, \widetilde{S}_k \, \big[q\,u + (1-q)\,d\big] = e^{-r\delta t} \, \widetilde{S}_k \, e^{r\delta t} = \widetilde{S}_k. \end{aligned}$$

□

The measure $\mathbb{Q}$ receives the same names as in the one-period case: *risk-neutral measure* or *equivalent martingale measure*. Its uniqueness will be proved in Subsection 2.3.6.

2.3.5 Predictable processes and the discrete stochastic integral

Definition 2.20 (Predictable process)**.** A process $(H_k)_{k=1}^N$ is *predictable* with respect to the filtration $(\mathcal{F}_k)$ if, for each $k = 1, \dots, N$, H_k is $\mathcal{F}_{k-1}$-measurable.

The difference from the notion of adapted process (Definition 2.13) is a shift of a single index, but its content is essential: an adapted process may use the information available *up to* t_k, whereas a predictable one may only use that available *before* t_k. In financial language, H_k is chosen at time t_{k-1}, before knowing how the market will move in the next step. It is this restriction that rules out strategies "with foresight".

Definition 2.21 (Discrete stochastic integral)**.** Let $(M_k)_{k=0}^N$ be adapted and $(H_k)_{k=1}^N$ predictable. The *discrete stochastic integral* of H with respect to M is the process

$$(H \cdot M)_n \coloneqq \sum_{k=1}^{n} H_k \, (M_k - M_{k-1}), \qquad n = 0, 1, \dots, N, \tag{2.20}$$

with $(H \cdot M)_0 \coloneqq 0$.

The formula has a suggestive financial reading: if M_k is the price of an asset and H_k the quantity held during $(t_{k-1}, t_k]$, then $H_k \, (M_k - M_{k-1})$ is the gain generated in that period, and $(H \cdot M)_n$ is the total gain up to t_n.

The name "discrete stochastic integral" anticipates the role of this object in the continuous setting: replacing the sum by an integral and the increments $M_k - M_{k-1}$ by infinitesimal increments dM_t, the formula becomes $\int_0^t H_s \, dM_s$, the *Itô integral* of Chapter 4.

Theorem 2.22 (Stability of martingales under stochastic integration)**.** *Let (M_k) be a $\mathbb{P}$-martingale with respect to $(\mathcal{F}_k)$ and (H_k) predictable and bounded. Then $\big((H \cdot M)_n\big)_{n=0}^N$ is a $\mathbb{P}$-martingale with respect to the same filtration.*

Proof. Adaptedness and integrability. For each n, $(H \cdot M)_n$ is a sum of products $H_k \, (M_k - M_{k-1})$ with $k \le n$, all of them $\mathcal{F}_n$-measurable. Integrability follows from the boundedness of H and the integrability of M.

Martingale property. Since $(H \cdot M)_{n+1} = (H \cdot M)_n + H_{n+1}(M_{n+1} - M_n)$, by linearity

$$\mathbb{E}\big[(H \cdot M)_{n+1} \,\big|\, \mathcal{F}_n\big] = (H \cdot M)_n + \mathbb{E}\big[H_{n+1}(M_{n+1} - M_n) \,\big|\, \mathcal{F}_n\big],$$

where we have used that $(H \cdot M)_n$ is $\mathcal{F}_n$-measurable. Since H_{n+1} is $\mathcal{F}_n$-measurable (by predictability), the property "taking out what is known" gives

$$\mathbb{E}\big[H_{n+1}(M_{n+1} - M_n) \,\big|\, \mathcal{F}_n\big] = H_{n+1} \, \mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n] = 0,$$

where the last equality uses the martingale property of (M_k). We conclude that $\mathbb{E}[(H \cdot M)_{n+1} \mid \mathcal{F}_n] = (H \cdot M)_n$. □

The fact that predictability of the integrand + martingale property of the integrator $\Rightarrow$ martingale property of the integral is the structural idea that will reappear in the continuous setting. As an immediate consequence, $\mathbb{E}[(H \cdot M)_n] = 0$ for all n: the expected gain of any predictable strategy on a martingale is zero.

2.3.6 The binomial representation theorem

We close the section with the result that ties the whole theory together: the *binomial representation theorem*. It states that every $\mathbb{Q}$-martingale can be written as a discrete stochastic integral with respect to the discounted price of the underlying. It is the discrete and constructive version of the *Brownian martingale representation theorem* of Chapter 4: the proof we give here provides an explicit formula for the integrand, whereas in the continuous setting we shall only have its existence.

We work on the multi-period binomial model endowed with the risk-neutral measure $\mathbb{Q}$ and the filtration $(\mathcal{F}_k)$. We write $\widetilde{S}_k = e^{-rk\,\delta t} S_k$.

Theorem 2.23 (Binomial representation). *Let $(M_k)_{k=0}^N$ be a $\mathbb{Q}$-martingale with respect to $(\mathcal{F}_k)$. There exists a unique predictable process $(\phi_k)_{k=1}^N$ such that*

$$M_n = M_0 + \sum_{k=1}^{n} \phi_k \,(\widetilde{S}_k - \widetilde{S}_{k-1}), \qquad n = 0, 1, \dots, N. \tag{2.21}$$

Proof. We construct ϕ node by node. Let $k \in \{1, \dots, N\}$ and let A be an atom of $\mathcal{F}_{k-1}$. By the structure of the model, A decomposes as the disjoint union of two atoms A_u, A_d of $\mathcal{F}_k$. On each of them the $\mathcal{F}_k$-measurable variables take a constant value; denote by M_{k-1}^A, $M_k^{A_u}$, $M_k^{A_d}$, $\widetilde{S}_{k-1}^A$, $\widetilde{S}_k^{A_u}$, $\widetilde{S}_k^{A_d}$ the corresponding values. We want ϕ_k constant on A (denote it ϕ_k^A) such that (2.21) holds on A_u and A_d:

$$\begin{cases} M_k^{A_u} = M_{k-1}^A + \phi_k^A \left(\widetilde{S}_k^{A_u} - \widetilde{S}_{k-1}^A\right), \\ M_k^{A_d} = M_{k-1}^A + \phi_k^A \left(\widetilde{S}_k^{A_d} - \widetilde{S}_{k-1}^A\right). \end{cases}$$

Subtracting eliminates M_{k-1}^A:

$$M_k^{A_u} - M_k^{A_d} = \phi_k^A \left(\widetilde{S}_k^{A_u} - \widetilde{S}_k^{A_d}\right).$$

Since $\widetilde{S}_k^{A_u} \neq \widetilde{S}_k^{A_d}$,

$$\phi_k^A = \frac{M_k^{A_u} - M_k^{A_d}}{\widetilde{S}_k^{A_u} - \widetilde{S}_k^{A_d}}. \tag{2.22}$$

Formula (2.22) determines ϕ_k uniquely on each atom of $\mathcal{F}_{k-1}$. The first equation is satisfied by construction; it remains to verify the second. Multiplying the first by q and the second by $1-q$ and adding,

$$q\, M_k^{A_u} + (1-q)\, M_k^{A_d} = M_{k-1}^A + \phi_k^A \left[q\, \widetilde{S}_k^{A_u} + (1-q)\, \widetilde{S}_k^{A_d} - \widetilde{S}_{k-1}^A\right].$$

The left-hand side is $\mathbb{E}_{\mathbb{Q}}[M_k \mid \mathcal{F}_{k-1}] \restriction_A = M_{k-1}^A$ (martingale property of M); the bracket on the right-hand side is $\mathbb{E}_{\mathbb{Q}}[\widetilde{S}_k \mid \mathcal{F}_{k-1}] \restriction_A - \widetilde{S}_{k-1}^A = \widetilde{S}_{k-1}^A - \widetilde{S}_{k-1}^A = 0$ by the martingale property of $(\widetilde{S}_k)$

(Theorem 2.19). The identity holds trivially, and combining it with the first equation recovers the second. Repeating this on each atom of each $\mathcal{F}_{k-1}$ yields a predictable process $(\phi_k)_{k=1}^N$ satisfying (2.21). Uniqueness is immediate from (2.22). □

Reading of the integrand. Formula (2.22) expresses ϕ_k as the ratio of the change in M to the change in the discounted price between the two children of the atom. If we identify M with the discounted value of a derivative, ϕ_k is the quantity of stock that must be held in order to absorb its possible change: it is the *delta* of the derivative at each node.

Two consequences. The first is the *uniqueness of the risk-neutral measure*, anticipated in Subsection 2.3.4. Let $\mathbb{Q}'$ be another measure under which $(\widetilde{S}_k)$ is a martingale. For each atom A of $\mathcal{F}_{k-1}$, the martingale condition

$$\mathbb{Q}'(A_u \mid A)\,\widetilde{S}_k^{A_u} + \mathbb{Q}'(A_d \mid A)\,\widetilde{S}_k^{A_d} = \widetilde{S}_{k-1}^{A},$$

is linear in $\mathbb{Q}'(A_u \mid A) \in (0,1)$ with positive coefficients, and has a unique solution, which coincides with q. Consequently $\mathbb{Q}' = \mathbb{Q}$. The second consequence is the *completeness of the multi-period binomial model*, which will be fully justified in Subsection 2.4.1: since every $\mathbb{Q}$-martingale can be represented as a stochastic integral with respect to $\widetilde{S}$, the $\mathbb{Q}$-martingale associated with the discounted value of a derivative will give rise, via representation, to the integrand that constitutes the stock holding of the replicating portfolio.

2.4 Solution of the problem in discrete time

We now bring together the results developed in Section 2.3 to obtain the fair price formula and the replicating strategy. The construction is carried out in three steps.

2.4.1 The three steps toward replication

Self-financing strategies. The idea is that a self-financing strategy is one in which rebalancing the portfolio requires neither injecting nor withdrawing capital from outside.

Definition 2.24 (Self-financing strategy)**.** A *strategy* in the multi-period binomial model is a pair of processes $\big((\phi_k),(\psi_k)\big)$ predictable with respect to $(\mathcal{F}_k)$. Its *value* at t_k is $V_k := \phi_k\,S_k + \psi_k\,B_k$ (with $V_0 := \phi_1\,S_0 + \psi_1\,B_0$). The strategy is *self-financing* if, for all $k = 1,\dots,N-1$,

$$\phi_k\,S_k + \psi_k\,B_k = \phi_{k+1}\,S_k + \psi_{k+1}\,B_k. \tag{2.23}$$

Proposition 2.25 (Characterization via the discounted process)**.** *Let $\big((\phi_k),(\psi_k)\big)$ be a strategy and $\widetilde{V}_k := e^{-r\,t_k}V_k$. The strategy is self-financing if and only if*

$$\widetilde{V}_n = V_0 + \sum_{k=1}^{n} \phi_k\,\big(\widetilde{S}_k - \widetilde{S}_{k-1}\big), \qquad n = 0,1,\dots,N. \tag{2.24}$$

Proof. ($\Rightarrow$) Let us compute $\widetilde{V}_k - \widetilde{V}_{k-1}$. Since $\widetilde{V}_k = \phi_k\,\widetilde{S}_k + \psi_k$,

$$\widetilde{V}_k - \widetilde{V}_{k-1} = \phi_k\,(\widetilde{S}_k - \widetilde{S}_{k-1}) + (\phi_k - \phi_{k-1})\,\widetilde{S}_{k-1} + (\psi_k - \psi_{k-1}).$$

The self-financing condition (2.23) at t_{k-1}, divided by B_{k-1}, implies $(\phi_k - \phi_{k-1})\widetilde{S}_{k-1} + (\psi_k - \psi_{k-1}) = 0$. Consequently $\widetilde{V}_k - \widetilde{V}_{k-1} = \phi_k(\widetilde{S}_k - \widetilde{S}_{k-1})$. Summing yields (2.24).

($\Leftarrow$) If (2.24) holds, subtracting the cases n and $n-1$ and combining with $\widetilde{V}_k = \phi_k\,\widetilde{S}_k + \psi_k$ recovers (2.23). □

The proposition links the financial notion of "self-financing strategy" with the mathematical notion of "discrete stochastic integral". In particular, combining it with Theorem 2.22, the discounted value of every self-financing strategy is a $\mathbb{Q}$-martingale.

The three-step recipe. Let C_T be an $\mathcal{F}_N$-measurable random variable (the *payoff* of the derivative).

Recipe for pricing by replication.

Step 1. *Change of measure.* Determine the unique measure $\mathbb{Q}$ under which $(\widetilde{S}_k)$ is a martingale (Section 2.3.4).

Step 2. *Construction of the discounted value.* Define

$$M_k \coloneqq \mathbb{E}_{\mathbb{Q}}\big[e^{-rT}\,C_T \,\big|\, \mathcal{F}_k\big]. \tag{2.25}$$

By construction, (M_k) is a $\mathbb{Q}$-martingale with $M_N = e^{-rT}C_T$.

Step 3. *Application of the representation.* By Theorem 2.23, there exists a unique predictable (ϕ_k) such that $M_k = M_0 + (\phi \cdot \widetilde{S})_k$. Define $\psi_k \coloneqq M_k - \phi_k\widetilde{S}_k$.

Theorem 2.26 (Pricing by replication in the multi-period binomial model)**.** *In the arbitrage-free multi-period binomial model, every derivative with* payoff C_T *admits a self-financing replicating strategy. The fair price at t_n is*

$$V_n = \mathbb{E}_{\mathbb{Q}}\big[e^{-r(T-t_n)}\,C_T \,\big|\, \mathcal{F}_n\big], \tag{2.26}$$

and at $t = 0$, $V_0 = \mathbb{E}_{\mathbb{Q}}[e^{-rT}\,C_T]$.

Proof. Step 1. The existence and uniqueness of $\mathbb{Q}$ were established in Section 2.3.4.

Step 2. The process (M_k) defined in (2.25) is a $\mathbb{Q}$-martingale by the tower property: for $j \le k$,

$$\mathbb{E}_{\mathbb{Q}}[M_k \mid \mathcal{F}_j] = \mathbb{E}_{\mathbb{Q}}\big[\mathbb{E}_{\mathbb{Q}}[e^{-rT}C_T \mid \mathcal{F}_k] \,\big|\, \mathcal{F}_j\big] = \mathbb{E}_{\mathbb{Q}}[e^{-rT}C_T \mid \mathcal{F}_j] = M_j.$$

Step 3. By Theorem 2.23 there exists a predictable (ϕ_k) such that $M_n = M_0 + \sum_{k=1}^{n} \phi_k(\widetilde{S}_k - \widetilde{S}_{k-1})$. We define $\psi_k \coloneqq M_k - \phi_k\,\widetilde{S}_k$, which is predictable (a combination of $\mathcal{F}_{k-1}$-measurable quantities), and $V_n \coloneqq e^{rt_n}M_n$. By Proposition 2.25, the representation of M_n as a stochastic integral is equivalent to the self-financing condition with $V_0 = M_0$ and $\widetilde{V}_n = M_n$. The strategy is self-financing and replicates the derivative, since $V_N = e^{rT}\,M_N = e^{rT}\,e^{-rT}\,C_T = C_T$.

Finally, by the no-arbitrage principle, the fair price of the derivative at t_n is the value of the portfolio at that time:

$$V_n = e^{rt_n}\, M_n = \mathbb{E}_{\mathbb{Q}}\big[e^{-r(T-t_n)}\, C_T \mid \mathcal{F}_n\big].$$

□

Formula (2.26) generalizes (2.14) to the multi-period setting: the fair price is the discounted value of the expectation of the *payoff* under the risk-neutral measure, conditioned on the information at the time of pricing.

2.4.2 Passage to continuous time

We close the chapter with a look ahead. The multi-period binomial model has an artificial feature: the market is observed only at a finite number of equally spaced times. This subsection argues, in a deliberately heuristic way, that the progressive refinement of the time partition (letting $N \to \infty$ with T fixed and $\delta t = T/N \to 0$) leads to a continuous-time model whose underlying follows a geometric Brownian motion. The rigorous construction of this model, as well as the associated pricing formula (the Black-Scholes formula), will be the subject of the following chapters.

Calibration of the factors u and d. If we took u, d constant (independent of δt), the price paths would become dramatically erratic as N increases. A sensible choice requires the log-returns at each step to be compatible with a stable behavior in the limit. Specifically, let $\sigma > 0$ and $\nu \in \mathbb{R}$ be two constants that will represent the *volatility* and the *trend* of the price in the limiting model. Let us calibrate

$$u = e^{\sigma\sqrt{\delta t}}, \qquad d = e^{-\sigma\sqrt{\delta t}}, \tag{2.27}$$

and the subjective probability as $p = \frac{1}{2}(1 + \frac{\nu}{\sigma}\sqrt{\delta t})$. Under this calibration, the log-return $\log(S_{k+1}/S_k)$ equals $+\sigma\sqrt{\delta t}$ with probability p and $-\sigma\sqrt{\delta t}$ with probability $1-p$. Its expectation is $\nu\,\delta t$ and its variance $\sigma^2\,\delta t + O(\delta t^2)$.

Convergence to geometric Brownian motion. We have

$$S_N = S_0\, \exp\left(\sum_{k=1}^{N} \log \frac{S_k}{S_{k-1}}\right).$$

The sum in the exponent is a sum of independent identically distributed variables, each with expectation $\nu\delta t$ and variance $\sigma^2\delta t + O(\delta t^2)$. The total sum has expectation νT and variance tending to $\sigma^2 T$ as $N \to \infty$. By the Central Limit Theorem,

$$\log\frac{S_T}{S_0} \xrightarrow{d} \mathcal{N}(\nu T,\, \sigma^2 T).$$

Equivalently,

$$S_T = S_0\, \exp\big(\nu T + \sigma\, W_T\big), \tag{2.28}$$

with $W_T \sim \mathcal{N}(0,T)$. The argument extends to any intermediate time $t \in [0,T]$, giving

$$S_t = S_0 \exp\bigl(\nu t + \sigma W_t\bigr),$$

with $W_t \sim \mathcal{N}(0,t)$. The family $\{W_t\}_{t\geq 0}$ emerging in the limit is what is known as *Brownian motion*: a continuous-time stochastic process with independent Gaussian increments and continuous paths. Its rigorous construction will be addressed in Chapter 3. The process S_t defined by (2.28) is called *geometric Brownian motion*, and it will be the canonical model for the stock price in the continuous setting.

Under the risk-neutral measure $\mathbb{Q}$ of the discrete model, an analogous computation (replacing p by q as defined in (2.18) and expanding in powers of δt) shows that the drift of the log-price under $\mathbb{Q}$ becomes $r - \frac{1}{2}\sigma^2$ instead of ν. This *change of drift* induced by the change of measure is the discrete and heuristic version of the phenomenon that in the continuous setting will give rise to *Girsanov's theorem*, one of the central pieces of Chapter 4.

Summary of the chapter. Starting from the static pricing of a *forward* by a replication argument, we have progressed to a complete and rigorous formula, in the multi-period binomial model, for the price of any European derivative, expressed as the discounted expectation of the *payoff* under a risk-neutral measure, together with a self-financing strategy that replicates it. The final three-step articulation is the scheme that will reappear in the continuous setting of Black-Scholes. Each of the mathematical concepts introduced (filtrations, conditional expectation, martingales, predictable processes, discrete stochastic integral) has a continuous-time version that we shall employ in Chapter 4. Finally, the introduction given in this subsection suggests that the underlying in the continuous setting will be a process with independent Gaussian increments and continuous paths: *Brownian motion*, to which we devote Chapter 3.

Chapter 3

Brownian motion

This chapter introduces *Brownian motion*, the stochastic foundation of the continuous-time pricing theory. The financial motivation now recedes into the background: the tools we shall need require a certain amount of self-contained development before we return to apply them.

3.1 Definition and existence

3.1.1 Formal definition of Brownian motion

Definition 3.1 (Standard Brownian motion)**.** A *standard Brownian motion* on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a family $\{W_t\}_{t\geq 0}$ of real random variables satisfying:

(i) $W_0 = 0$ almost surely;

(ii) for any $0 \leq s < t$, the increment $W_t - W_s$ has distribution $\mathcal{N}(0,\, t-s)$;

(iii) increments over disjoint intervals are independent, that is, for every $n \geq 1$ and every choice of times $0 \leq t_0 < t_1 < \cdots < t_n$, the random variables

$$W_{t_1} - W_{t_0},\ W_{t_2} - W_{t_1},\ \ldots,\ W_{t_n} - W_{t_{n-1}}$$

are mutually independent;

(iv) the paths $t \mapsto W_t(\omega)$ are continuous functions almost surely.

Conditions (ii)–(iii) inherit, via the CLT, the Gaussian and independent nature of the steps of the rescaled random walk; (i) is a mere normalization (any process satisfying (ii)–(iv) can be shifted so as to start at zero) and (iv) is the condition that qualitatively distinguishes Brownian motion: continuous paths, without jumps.

It is worth noting that conditions (i)–(iii) are *distributional*: they specify the joint law of the process, whereas (iv) is *pathwise*: it requires each realization $t \mapsto W_t(\omega)$ to be a continuous function.

3.1.2 Existence of Brownian motion

The preceding definition immediately raises the question of whether the mathematical object it defines actually exists. The answer is affirmative, but its proof is not trivial: it requires constructing a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a family of random variables $\{W_t\}_{t\geq 0}$ on it satisfying all the properties simultaneously, which does not follow from elementary facts of probability.

Theorem 3.2 (Existence of Brownian motion)**.** *There exist a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a family of random variables $\{W_t\}_{t\geq 0}$ on it satisfying conditions (i)–(iv) of Definition 3.1.*

The rigorous proof of this theorem was originally provided by Norbert Wiener in 1923 and, in a more elementary version based on the Schauder system of functions, by Paul Lévy in 1939. The essential idea of Lévy's construction is the following: the values of W on the dyadic rationals of the interval $[0, 1]$ are defined by an infinite series of successive adjustments, controlling that the partial paths converge uniformly with probability one; the process so constructed is then extended by continuity to the whole interval. The detailed proof can be found in [5] or [9]. From this point on, we take the existence of Brownian motion for granted and focus on studying its properties.

Remark 3.3 (On the continuity condition in the definition). Many authors define Brownian motion by requiring only (i)–(iii), which are distributional in nature: as already pointed out, specifying (i)–(iii) amounts to fixing the finite-dimensional distributions of the process (the joint laws of $(W_{t_1}, \ldots, W_{t_n})$, which turn out to be Gaussian with zero mean and covariance $\mathbb{E}[W_s W_t] = \min(s, t)$). Once these distributions are fixed, the *Kolmogorov extension theorem* guarantees the existence of a process with those distributions. However, this result does not guarantee that its paths are continuous: Kolmogorov's construction yields the process from its finite-dimensional distributions, but by itself it provides no information on the regularity of the paths $t \mapsto W_t(\omega)$.

To obtain continuous paths one further step is needed. By *Doob's theorem*, the process admits a separable version, that is, a process with the same finite-dimensional distributions whose behavior is determined by the values it takes on a countable dense set of times. It can then be shown that this version has continuous paths almost surely. This is the usual way of establishing the existence of a continuous Brownian motion in a course on stochastic processes, and it constitutes an alternative to Lévy's explicit construction presented above.

Requiring (iv) from the outset, as we do here, is therefore not an extra condition: continuity is obtained at no additional cost by passing to the separable version, and indeed several authors choose to build it into the definition. This choice is moreover convenient for our purposes, since all the stochastic calculus we develop from Chapter 4 onward, starting with the very construction of the Itô integral, depends on the continuity of the Brownian paths.

3.2 Properties of the paths

In this section we study three pathwise properties of Brownian motion: nowhere differentiability, unbounded variation, and the explicit computation of the quadratic variation.

3.2.1 Nowhere differentiability of the paths

The pathwise continuity of Brownian motion might suggest a picture of smooth functions; the reality is the opposite: the paths are so irregular that they admit a derivative at no point.

Heuristic argument. Suppose, arguing by contradiction, that the path of a Brownian motion were differentiable at some time t_0. This would mean that the difference quotient

$$\frac{W_{t_0+h} - W_{t_0}}{h}$$

has a finite limit as $h \to 0$. But by condition (ii) of Definition 3.1, $W_{t_0+h} - W_{t_0}$ has distribution $\mathcal{N}(0, h)$, and therefore the difference quotient has distribution $\mathcal{N}(0,\, 1/h)$. Its variance diverges as $h \to 0$: the quotient cannot converge to anything finite.

The argument is heuristic (we gloss over the difference between convergence in distribution and almost sure convergence) but it captures the underlying phenomenon. The rigorous proof requires technical refinements, in particular the Borel-Cantelli lemma applied to estimates on the increments at dyadic scales, and can be found in [5].

Theorem 3.4 (Nowhere differentiability of Brownian paths)**.** *Let $\{W_t\}_{t\geq 0}$ be a standard Brownian motion. With probability one, the path $t \mapsto W_t(\omega)$ is differentiable at no point $t \geq 0$.*

Brownian paths are therefore continuous everywhere but differentiable nowhere, a combination that will motivate the construction of the *ad hoc* calculus of Chapter 4.

3.2.2 Unbounded variation of the paths

We next prove a stronger quantitative property: Brownian paths have, with probability one, *unbounded variation* on every interval.

Total variation of a function. Let us first recall the classical notion.

Definition 3.5 (Total variation)**.** Let $f : [0, T] \to \mathbb{R}$ be a real function defined on a compact interval. The *total variation* of f on $[0, T]$ is

$$V_T(f) \coloneqq \sup_{\pi} \sum_{k=0}^{n-1} |f(t_{k+1}) - f(t_k)|,$$

where the supremum is taken over all partitions $\pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ of the interval. The function f is said to be of *bounded variation* on $[0, T]$ if $V_T(f) < \infty$.

Differentiable functions with bounded derivative are of bounded variation (the variation is the integral of $|f'|$), as are monotone functions. For a function of bounded variation, the Riemann-Stieltjes integral $\int_0^T g\, df$ is unambiguously defined for every continuous integrand g. It is precisely this circumstance that fails in the case of Brownian motion.

Heuristic argument. Let $\pi_n = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ be the uniform partition of $[0, T]$ into n subintervals of length T/n. By condition (ii) of Definition 3.1, each increment $W_{t_{k+1}} - W_{t_k}$

has distribution $\mathcal{N}(0,\, T/n)$, and therefore its absolute value satisfies

$$\mathbb{E}\big[|W_{t_{k+1}} - W_{t_k}|\big] = \sqrt{\frac{2T}{\pi\, n}}.$$

Summing over the n subintervals,

$$\mathbb{E}\left[\sum_{k=0}^{n-1} |W_{t_{k+1}} - W_{t_k}|\right] = n\sqrt{\frac{2T}{\pi\, n}} = \sqrt{\frac{2Tn}{\pi}}.$$

The expectation diverges as $n \to \infty$: refining the partition makes the sum of the absolute values of the increments grow without bound, instead of approaching a finite value as would happen for a function of bounded variation. The reasoning therefore suggests that the Brownian path cannot be of bounded variation.

The above argument is heuristic (it proves divergence in expectation, not almost surely) but it clearly displays the phenomenon. The rigorous proof can be obtained, among other routes, as a corollary of the computation of the quadratic variation that we address in the next subsection (see the remark at the end of that subsection).

Theorem 3.6 (Unbounded variation of Brownian paths)**.** *Let $\{W_t\}_{t\geq 0}$ be a standard Brownian motion and $T > 0$. With probability one, the path $t \mapsto W_t(\omega)$ has unbounded variation on $[0, T]$, that is, $V_T(W) = \infty$ almost surely.*

Consequence: the failure of Riemann-Stieltjes. The unbounded variation of the paths has an immediate consequence for the possibility of defining integrals against Brownian motion. The classical Riemann-Stieltjes construction of $\int_0^T g\, df$ requires f to be of bounded variation in order for the sums

$$\sum_{k=0}^{n-1} g(\tau_k)\,\big(f(t_{k+1}) - f(t_k)\big)$$

to converge to the same limit regardless of the choice of the points $\tau_k \in [t_k, t_{k+1}]$. When $f = W$, this result fails: the limit depends on the choice of the τ_k. Constructing an integral $\int_0^T g_s\, dW_s$ will therefore require fixing a specific convention for the evaluation points and verifying that the resulting construction is well defined.

3.2.3 Quadratic variation of Brownian motion

Despite the irregularity of the paths, the *sum of the squares* of the increments exhibits a remarkably regular behavior: it converges to a finite limit as the partition is refined. This quantity, the *quadratic variation*, is the central result of the chapter.

Theorem 3.7 (Quadratic variation of Brownian motion)**.** *Let $\{W_t\}_{t\geq 0}$ be a standard Brownian motion and $T > 0$. For each partition $\pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ of $[0, T]$, denote its mesh by $|\pi| \coloneqq \max_k(t_{k+1} - t_k)$ and consider the random variable*

$$Q_\pi \coloneqq \sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2.$$

Then, as $|\pi| \to 0$*,*

$$Q_\pi \xrightarrow{L^2} T, \tag{3.1}$$

that is, $\mathbb{E}\big[(Q_\pi - T)^2\big] \to 0$*.*

The quantity Q_π is, by definition, a random variable (its value depends on the path ω). The theorem states that, in the limit of fine partitions, this random variable ceases to be random: it converges to a deterministic constant, T, equal to the length of the interval. This is at first sight a surprising result, since one would expect a sum of squares of Brownian increments (each with nonzero variance) to retain some degree of randomness in the limit. The reason this does not happen is that, as the partition is refined, the contributions of the subintervals become ever more numerous and ever smaller; the law of large numbers, applied to the sum, produces the averaging to the constant.

Proof. The strategy consists in computing directly the expectation and the variance of Q_π, and showing that in the limit the former tends to T and the latter to zero.

Step 1: computation of the expectation. By condition (ii) of Definition 3.1, each increment $W_{t_{k+1}} - W_{t_k}$ has distribution $\mathcal{N}(0,\, t_{k+1} - t_k)$, and therefore

$$\mathbb{E}\big[(W_{t_{k+1}} - W_{t_k})^2\big] = t_{k+1} - t_k.$$

By linearity of expectation,

$$\mathbb{E}[Q_\pi] = \sum_{k=0}^{n-1} (t_{k+1} - t_k) = T. \tag{3.2}$$

The expectation of Q_π is exactly T, with no error: this result is *exact*, not asymptotic, and it holds for any partition.

Step 2: computation of the variance. By definition,

$$\mathrm{Var}(Q_\pi) = \mathbb{E}\Big[\big(Q_\pi - T\big)^2\Big] = \mathbb{E}\left[\left(\sum_{k=0}^{n-1} \big[(W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)\big]\right)^2\right].$$

Expanding the square,

$$\begin{aligned}\mathrm{Var}(Q_\pi) = \sum_{k=0}^{n-1} \mathbb{E}\Big[\big((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)\big)^2\Big] \\ + 2\sum_{j<k} \mathbb{E}\Big[\big((W_{t_{j+1}} - W_{t_j})^2 - (t_{j+1} - t_j)\big) \\ \cdot \big((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)\big)\Big].\end{aligned}$$

The cross terms ($j < k$) vanish: by the independence of increments over disjoint intervals (condition (iii) of Definition 3.1), the variables $(W_{t_{j+1}} - W_{t_j})^2 - (t_{j+1} - t_j)$ and $(W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)$ are independent and, moreover, both have zero expectation (by (3.2) applied to a single subinterval). Hence their product has zero expectation, and only the diagonal terms

survive:

$$\operatorname{Var}(Q_\pi) = \sum_{k=0}^{n-1} \mathbb{E}\Big[\big((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)\big)^2\Big].$$

For each summand, write $\Delta_k := t_{k+1} - t_k$ and expand:

$$\begin{aligned}\mathbb{E}\Big[\big((W_{t_{k+1}} - W_{t_k})^2 - \Delta_k\big)^2\Big] &= \mathbb{E}\big[(W_{t_{k+1}} - W_{t_k})^4\big] - 2\,\Delta_k\,\mathbb{E}\big[(W_{t_{k+1}} - W_{t_k})^2\big] + \Delta_k^2 \\ &= 3\,\Delta_k^2 - 2\,\Delta_k^2 + \Delta_k^2 = 2\,\Delta_k^2,\end{aligned}$$

where we have used that the fourth moment of a $\mathcal{N}(0, \Delta_k)$ is $3\,\Delta_k^2$.

Substituting,

$$\operatorname{Var}(Q_\pi) = 2\sum_{k=0}^{n-1} \Delta_k^2. \tag{3.3}$$

Step 3: passage to the limit. To bound the above sum, we observe that

$$\sum_{k=0}^{n-1} \Delta_k^2 \le \max_k \Delta_k \sum_{k=0}^{n-1} \Delta_k = |\pi|\,T.$$

Consequently, $\operatorname{Var}(Q_\pi) \le 2\,|\pi|\,T \to 0$ as $|\pi| \to 0$. Combining with $\mathbb{E}[Q_\pi] = T$ from Step 1,

$$\mathbb{E}\big[(Q_\pi - T)^2\big] = \operatorname{Var}(Q_\pi) \to 0,$$

which is the L^2 convergence stated in (3.1). □

The heuristic rule $(dW_t)^2 = dt$. Theorem 3.7 admits a particularly illuminating informal interpretation. The statement that $\sum_k (W_{t_{k+1}} - W_{t_k})^2 \to T$ as the partition is refined can be read, formally, as

$$\int_0^T (dW_t)^2 = T,$$

or, in differential form, as the rule

$$(dW_t)^2 = dt. \tag{3.4}$$

Identity (3.4) is, obviously, an informal shorthand that by itself has no precise mathematical meaning. But it captures a fundamental fact: although the "infinitesimal increment" dW_t is of order $\sqrt{dt}$ (its standard deviation), its square is of order dt, not dt^2 as it would be for a smooth function. In classical calculus the terms quadratic in dt are negligible; for Brownian motion, they are not.

Consequence: unbounded variation. We close the section by returning to the claim announced in Subsection 3.2.2: the finite and nonzero quadratic variation of Brownian motion implies, with probability one, its unbounded variation. The argument is as follows. On any partition π,

$$Q_\pi = \sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2 \le \max_k |W_{t_{k+1}} - W_{t_k}| \cdot \sum_{k=0}^{n-1} |W_{t_{k+1}} - W_{t_k}|.$$

If the path were of bounded variation, the second factor would be finite; by the uniform continuity of W on $[0,T]$ (guaranteed by condition (iv) of Definition 3.1), the first factor would tend to zero as $|\pi| \to 0$. The product would therefore tend to zero, contradicting Theorem 3.7, which ensures $Q_\pi \to T > 0$. Consequently, the path cannot be of bounded variation.

3.3 The Brownian filtration and continuous martingales

We close the chapter with the filtration associated with Brownian motion and the canonical examples of continuous martingales. The section is brief: its content is essentially the adaptation to the continuous setting of the discrete notions of Chapter 2.

3.3.1 The Brownian filtration and associated martingales

The natural filtration of Brownian motion. The filtration (Definition 2.12) generalizes to the continuous setting by replacing the sequence $(\mathcal{F}_k)_{k=0}^N$ with a family $(\mathcal{F}_t)_{t\geq 0}$ having the same monotonicity property $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$.

Definition 3.8 (Natural filtration of Brownian motion)**.** Let $\{W_t\}_{t\geq 0}$ be a standard Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$. The *natural filtration* of W is the family $(\mathcal{F}_t^W)_{t\geq 0}$ defined by

$$\mathcal{F}_t^W := \sigma\big(W_s : 0 \leq s \leq t\big),$$

that is, the σ-algebra generated by the values of the process on the interval $[0,t]$.

The natural filtration encodes precisely the information generated by Brownian motion, but it has two technical limitations that should be corrected before using it as an operational framework. The first is that it does not necessarily contain the $\mathbb{P}$-null sets of $\mathcal{F}$: a modification of an adapted process on a set of measure zero may fail to be adapted, which prevents us from handling *almost sure* questions with the necessary flexibility. The second is that *right-continuity* (that is, the identity $\mathcal{F}_t^W = \bigcap_{s>t} \mathcal{F}_s^W$) is not guaranteed, a property that is indispensable for handling stopping times and for the existence of continuous versions of processes such as the Itô integral (Chapter 4).

The standard solution consists in passing to a slightly enlarged filtration that does satisfy both properties.

Definition 3.9 (Augmented filtration of Brownian motion)**.** Let $\mathcal{N} := \{N \in \mathcal{F} : \mathbb{P}(N) = 0\}$ be the collection of $\mathbb{P}$-null sets. The *augmented filtration* of Brownian motion is the family $(\mathcal{F}_t)_{t\geq 0}$ defined by

$$\mathcal{F}_t := \sigma\big(\mathcal{F}_t^W \cup \mathcal{N}\big),$$

that is, the smallest σ-algebra containing $\mathcal{F}_t^W$ and all the $\mathbb{P}$-null sets.

This definition resolves the first limitation by construction. Moreover, it also resolves the second one automatically: for Brownian motion, the augmented filtration by itself satisfies the right-continuity property. The result can be found in [5], Proposition 2.7.7, and rests on Blumenthal's zero-one law.

Proposition 3.10 (Right-continuity of the augmented filtration)**.** *The filtration $(\mathcal{F}_t)_{t\geq 0}$ of Definition 3.9 is right-continuous: $\mathcal{F}_t = \bigcap_{s>t} \mathcal{F}_s$ for all $t \geq 0$.*

Consequently, $(\mathcal{F}_t)$ satisfies the so-called *usual conditions*, completeness and right-continuity, the standard framework on which the general theory of continuous processes is developed. **From now on, and throughout the rest of this work, $(\mathcal{F}_t)$ will without exception denote the augmented filtration of Brownian motion.** The distinction from the natural filtration $\mathcal{F}_t^W$ will not be made explicit again, since for informational purposes both encode the same events: any set in $\mathcal{F}_t$ differs from one in $\mathcal{F}_t^W$ by at most a $\mathbb{P}$-null set.

Definition 3.11 (Continuous-time martingale)**.** Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space. A stochastic process $\{M_t\}_{t\geq 0}$ is a *martingale* with respect to $(\mathcal{F}_t)$ if:

(i) M_t is $\mathcal{F}_t$-measurable for all $t \geq 0$ (*adaptedness*);

(ii) $\mathbb{E}[|M_t|] < \infty$ for all $t \geq 0$ (*integrability*);

(iii) $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ almost surely, for all $0 \leq s \leq t$.

This is the literal translation to the continuous setting of Definition 2.18; the elementary properties (linearity, expectation constant in time) are preserved, with proofs identical to the discrete case.

Canonical examples. We present two continuous martingales associated with Brownian motion. The filtration considered is the Brownian $(\mathcal{F}_t)$ fixed in Definition 3.9.

Proposition 3.12 (Brownian martingales)**.** *Let $\{W_t\}_{t\geq 0}$ be a standard Brownian motion. The following processes are martingales with respect to $(\mathcal{F}_t)$:*

(a) *Brownian motion itself, $\{W_t\}_{t\geq 0}$;*

(b) *for each $\lambda \in \mathbb{R}$, the* exponential martingale

$$\mathcal{E}_t^\lambda := \exp\Big(\lambda W_t - \tfrac{1}{2}\lambda^2 t\Big). \tag{3.5}$$

Proof. The adaptedness of both processes to $(\mathcal{F}_t)$ is immediate, since they are functions of W_t. Integrability follows from the Gaussian properties of Brownian motion. It remains to verify property (iii) of Definition 3.11.

(a) For $0 \leq s \leq t$, we write $W_t = W_s + (W_t - W_s)$, and by condition (iii) of Definition 3.1, the increment $W_t - W_s$ is independent of $\mathcal{F}_s$. Consequently,

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s + \mathbb{E}[W_t - W_s \mid \mathcal{F}_s] = W_s + \mathbb{E}[W_t - W_s] = W_s,$$

since $\mathbb{E}[W_t - W_s] = 0$ by condition (ii) of Definition 3.1.

(b) For the exponential martingale,

$$\mathcal{E}_t^\lambda = \exp\big(\lambda W_s - \tfrac{1}{2}\lambda^2 s\big) \cdot \exp\big(\lambda(W_t - W_s) - \tfrac{1}{2}\lambda^2(t-s)\big) = \mathcal{E}_s^\lambda \cdot \exp\big(\lambda(W_t - W_s) - \tfrac{1}{2}\lambda^2(t-s)\big).$$

The first factor is $\mathcal{F}_s$-measurable and the second is independent of $\mathcal{F}_s$. By the property "taking out what is known" and independence,

$$\mathbb{E}\big[\mathcal{E}_t^\lambda \mid \mathcal{F}_s\big] = \mathcal{E}_s^\lambda \, \mathbb{E}\big[\exp\big(\lambda(W_t - W_s)\big)\big] \cdot e^{-\frac{1}{2}\lambda^2(t-s)}.$$

Since $W_t - W_s \sim \mathcal{N}(0, t-s)$, its moment generating function gives $\mathbb{E}[\exp(\lambda(W_t - W_s))] = e^{\frac{1}{2}\lambda^2(t-s)}$, and the exponential factor cancels exactly with the last term, so that $\mathbb{E}[\mathcal{E}_t^\lambda \mid \mathcal{F}_s] = \mathcal{E}_s^\lambda$. □

Comment on the exponential martingale. The process (3.5), also known as the *stochastic exponential* of W, will play an important role in Chapter 4: a generalized version of formula (3.5), in which λ is replaced by an adapted process θ_t, is precisely the Radon-Nikodym derivative defining the change of measure in Girsanov's theorem. The quadratic form of the exponent, $-\frac{1}{2}\lambda^2 t$, is no accident: it is exactly the correction that makes the exponential have unit expectation and, therefore, that preserves the martingale property.

Having completed the study of Brownian motion and its associated martingales, we are in a position to develop the stochastic calculus that will occupy the next chapter. The tools to be introduced there (Itô integral, Itô's formula, Girsanov's theorem, Brownian martingale representation theorem) rest essentially on the properties studied so far.

Chapter 4

Stochastic calculus

This chapter develops the mathematical foundations of stochastic calculus on which the continuous-time pricing theory of Chapter 5 will be built. Its fundamental pillars are four: the *Itô stochastic integral*, which generalizes to the Brownian setting the discrete stochastic integral of Chapter 2; *Itô's formula*, which replaces the classical chain rule for Brownian processes; *Girsanov's theorem*, which formalizes the change of measure in continuous time; and the *Brownian martingale representation theorem*, which is the continuous analogue of the binomial representation theorem of Section 2.3.6. Each of these pillars thus has its counterpart in the discrete case of Chapter 2: the machinery we develop here introduces no essentially new concepts, but translates already familiar ideas to the continuous setting, where the irregularity of Brownian paths (Chapter 3) demands a considerably more careful technical construction.

4.1 Motivation: the need for a new integral

Before undertaking the construction of the stochastic integral, it is essential to articulate precisely why an *ad hoc* construction is necessary and why the classical integration methods do not suffice. The motivation unfolds in three steps: first, showing that the price process of the underlying must necessarily exhibit unbounded variation for the model to be economically sensible; second, checking that this condition makes the classical Riemann-Stieltjes procedure break down; and third, identifying the choice that will lead to the correct definition.

The price of the underlying cannot have smooth paths. The first observation, anchored in the no-arbitrage principle, is the following: in an arbitrage-free continuous model, the paths of the discounted price of the underlying cannot be of bounded variation. The argument, which we reproduce following [3], is constructive: we shall explicitly exhibit the arbitrage opportunity that would otherwise arise. We work with the discounted price $\widetilde{S}_t = e^{-rt}S_t$, on which the self-financing condition takes a particularly simple form; the variation properties of S and $\widetilde{S}$ are equivalent under $B_t = e^{rt}$.

The rigorous definition of a self-financing portfolio in continuous time will be formalized in Chapter 5; here we only need one of its consequences, the direct analogue of (2.20) in the discrete case: the discounted value at T of a self-financing portfolio starting from initial capital V_0 with

an adapted strategy ϕ having continuous paths is

$$\widetilde{V}_T = V_0 + \int_0^T \phi_t \, d\widetilde{S}_t, \tag{4.1}$$

where the integral, under the assumption of bounded variation of the integrator, is interpreted pathwise as a classical Riemann-Stieltjes integral.

Let $F : \mathbb{R} \to \mathbb{R}$ be of class C^1 and consider the strategy $\phi_t = F'(\widetilde{S}_t)$, starting from $V_0 = F(\widetilde{S}_0)$. By the fundamental theorem of calculus applied to the continuous integrator of bounded variation $\widetilde{S}$,

$$\int_0^T F'(\widetilde{S}_t) \, d\widetilde{S}_t = F(\widetilde{S}_T) - F(\widetilde{S}_0),$$

and substituting in (4.1) we obtain $\widetilde{V}_T = F(\widetilde{S}_T)$. Under the bounded variation assumption, any quantity of the form $F(\widetilde{S}_T)$ with $F \in C^1$ is thus replicable from initial capital $F(\widetilde{S}_0)$.

Let us specialize to $F(x) = (x - \widetilde{S}_0)^2$: the strategy $\phi_t = 2\,(\widetilde{S}_t - \widetilde{S}_0)$, starting from $V_0 = 0$, generates the nominal value at T

$$V_T = e^{rT} \left(\widetilde{S}_T - \widetilde{S}_0\right)^2,$$

which is nonnegative almost surely and strictly positive on the event $\{\widetilde{S}_T \neq \widetilde{S}_0\}$, of strictly positive probability in any nontrivial model. That is, it generates an arbitrage opportunity starting from zero capital, in contradiction with the fundamental assumption of the model.

The conclusion, therefore, is that any realistic model of the price in continuous time must employ processes whose paths have unbounded variation. The Brownian motion introduced in Chapter 3, far from being a mathematical eccentricity, is exactly the kind of process the theory demands.

Why Riemann-Stieltjes does not work. Once we accept that the integrator in an expression of the type $\int_0^T \phi_s \, dW_s$ will have unbounded variation, the technical problem arises immediately: the classical Riemann-Stieltjes construction ceases to work. Let us briefly recall why.

In the Riemann-Stieltjes integral, given a partition $\pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ and points $\tau_k \in [t_k, t_{k+1}]$, one considers the sum

$$S_\pi(\phi, f; \tau) := \sum_{k=0}^{n-1} \phi(\tau_k) \left(f(t_{k+1}) - f(t_k)\right).$$

If f is of bounded variation and ϕ is continuous, this sum converges to the same limit as $|\pi| \to 0$, regardless of the choice of the points τ_k. This common limit is, by definition, $\int_0^T \phi \, df$.

When f has unbounded variation, on the other hand, the result collapses: the limit, if it exists, depends on the choice of the points τ_k. The most transparent example is that of Brownian motion itself with $\phi = W$, that is, the sums that would approximate $\int_0^T W_s \, dW_s$. Consider the uniform partition $t_k = kT/n$ and the left-endpoint choice ($\tau_k = t_k$), and observe the algebraic identity

$$W_{t_k}(W_{t_{k+1}} - W_{t_k}) = \tfrac{1}{2}\left(W_{t_{k+1}}^2 - W_{t_k}^2\right) - \tfrac{1}{2}(W_{t_{k+1}} - W_{t_k})^2.$$

Summing over $k = 0, 1, \ldots, n-1$, the first part telescopes and yields $\frac{1}{2}(W_T^2 - W_0^2) = \frac{1}{2}W_T^2$, while

the second part tends, by Theorem 3.7, to $\frac{1}{2}T$ in the sense of L^2 convergence. Consequently,

$$\sum_{k=0}^{n-1} W_{t_k}(W_{t_{k+1}} - W_{t_k}) \xrightarrow{L^2} \tfrac{1}{2}\big(W_T^2 - T\big). \tag{4.2}$$

The analogous right-endpoint choice ($\tau_k = t_{k+1}$) produces, by an entirely parallel computation in which the sign of the quadratic term changes, the limit $\frac{1}{2}(W_T^2 + T)$. The two approximating sums therefore tend to different limits: they differ exactly by T, which is the quadratic variation of Brownian motion on $[0, T]$. The cause is now evident: the difference between the two sums is $\sum_k (W_{t_{k+1}} - W_{t_k})^2$, which for a function of bounded variation would tend to zero, but which for Brownian motion converges to T. The pathology that breaks Riemann-Stieltjes is exactly the property that distinguishes Brownian motion from classical processes.

Itô's choice. The conclusion of the preceding analysis is clear: to construct a well-defined integral $\int_0^T \phi_s \, dW_s$, we cannot simply take the limit of the partial sums, but must fix a specific convention for the evaluation points. The choice we shall adopt, *Itô's choice*, consists in always evaluating the integrand at the *left* endpoint of each subinterval,

$$\sum_{k=0}^{n-1} \phi_{t_k} \, (W_{t_{k+1}} - W_{t_k}).$$

This choice is justified for two reasons. The first, mathematical in nature, is that the resulting integral will preserve the martingale property (Theorem 4.10). The second, financial in nature, is that the left endpoint formalizes the notion of *predictability*: the quantity of stock held during the period $[t_k, t_{k+1})$ is chosen at the beginning of the period, at $t = t_k$, before knowing how the market will move during the subinterval. It is exactly the same idea we articulated in the discrete case in Section 2.3.5, now carried over to the continuous setting.

4.2 Construction of the Itô stochastic integral

Having presented the motivation, we turn to the rigorous construction of the stochastic integral $\int_0^T \phi_s \, dW_s$. The general strategy follows a standard scheme: we define the integral on a class of elementary integrands where the object reduces to a finite sum; we prove on that class a fundamental identity controlling the L^2 norm of the result; and we extend the integral by density to a sufficiently broad class of integrands, exploiting the completeness of L^2. Throughout the section we fix a time horizon $T > 0$ and a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which a standard Brownian motion $\{W_t\}_{t \in [0,T]}$ is considered together with its augmented filtration $(\mathcal{F}_t)_{0 \le t \le T}$.

4.2.1 Simple processes and the elementary integral

We begin by defining the class of elementary integrands on which the stochastic integral admits a direct definition, free of technical difficulties.

Definition 4.1 (Simple predictable process)**.** A stochastic process $\phi : [0, T] \times \Omega \to \mathbb{R}$ is *simple predictable*—or, for short, *simple*—if there exist a partition $0 = t_0 < t_1 < \cdots < t_n = T$ and random variables $\xi_0, \xi_1, \ldots, \xi_{n-1}$ such that:

(i) for each $i = 0, 1, \ldots, n-1$, ξ_i is bounded and $\mathcal{F}_{t_i}$-measurable;

(ii) the process ϕ admits the representation

$$\phi_t(\omega) = \sum_{i=0}^{n-1} \xi_i(\omega)\, \mathbf{1}_{(t_i, t_{i+1}]}(t). \tag{4.3}$$

We denote by $\mathcal{S}$ the class of all simple processes on $[0, T]$.

Let us pause briefly on condition (i). The measurability of ξ_i with respect to the σ-algebra $\mathcal{F}_{t_i}$ (the information available at the beginning of the subinterval $(t_i, t_{i+1}]$, not at its end) is the exact translation to the continuous setting of the notion of *predictability*[1] introduced in Section 2.3.5 for the discrete case. The value of the integrand during the subinterval is determined with the information available at its beginning, before observing the increment of Brownian motion.

Definition 4.2 (Stochastic integral of a simple process)**.** Let $\phi \in \mathcal{S}$ be a simple process with the representation (4.3). Its *stochastic integral* (or *Itô integral*) over $[0, T]$ is the random variable

$$I(\phi) \coloneqq \int_0^T \phi_s \, dW_s \coloneqq \sum_{i=0}^{n-1} \xi_i \left(W_{t_{i+1}} - W_{t_i} \right). \tag{4.4}$$

Formula (4.4) is the continuous version of the discrete stochastic integral $(H \cdot M)_n$ defined in (2.20): each increment of Brownian motion over a subinterval is weighted by the value the integrand takes at the beginning of that subinterval. Since the ξ_i are random variables, $I(\phi)$ is also a random variable, and by construction it is $\mathcal{F}_T$-measurable.

Well-definedness and linearity. Before proceeding, it is worth noting that the integral (4.4) does not depend on the particular representation used for ϕ: refining the partition, $\xi_i \left(W_{t_{i+1}} - W_{t_i}\right) = \xi_i \left(W_s - W_{t_i}\right) + \xi_i \left(W_{t_{i+1}} - W_s\right)$, a trivial identity by the additivity of Brownian increments, which implies that any two representations (for which a common refinement, their union, always exists) produce the same integral. Likewise, the integral is linear: for $\phi, \psi \in \mathcal{S}$ and $a, b \in \mathbb{R}$, $I(a\phi + b\psi) = a\, I(\phi) + b\, I(\psi)$ almost surely, which follows immediately from formula (4.4) after taking a common refinement of the partitions.

4.2.2 The Itô isometry

We next establish the *Itô isometry*: an identity that controls the L^2 norm of the integral in terms of the integrand.

Definition 4.3 (The space $\mathcal{H}^2$)**.** We denote by $\mathcal{H}^2[0, T]$, or simply $\mathcal{H}^2$ when the time horizon is clear from the context, the space of stochastic processes $\phi : [0, T] \times \Omega \to \mathbb{R}$ satisfying:

[1] The rigorous notion of predictable process in continuous time is formalized via the σ-algebra generated by the left-continuous adapted processes on $[0, T] \times \Omega$, and coincides for the class of simple processes with condition (i). It will not be necessary to work with it explicitly, since all the subsequent theory is extended by density from $\mathcal{S}$; see [5].

(i) ϕ is progressively measurable[2] with respect to the filtration $(\mathcal{F}_t)$;

(ii) $\mathbb{E}\Big[\int_0^T \phi_s^2\, ds\Big] < \infty$.

On $\mathcal{H}^2$ we consider the norm

$$\|\phi\|_{\mathcal{H}^2} := \mathbb{E}\Big[\int_0^T \phi_s^2\, ds\Big]^{1/2}. \tag{4.5}$$

The progressive measurability condition guarantees joint measurability in (t,ω) and, in particular, that the integral $\int_0^T \phi_s^2\, ds$ is well defined. For simple processes (Definition 4.1) both conditions hold trivially, and therefore $\mathcal{S} \subseteq \mathcal{H}^2$.

The isometry.

Theorem 4.4 (Itô isometry for simple processes)**.** *Let $\phi \in \mathcal{S}$ be a simple process. Then*

$$\mathbb{E}\Big[\big(I(\phi)\big)^2\Big] = \mathbb{E}\Big[\int_0^T \phi_s^2\, ds\Big] = \|\phi\|_{\mathcal{H}^2}^2. \tag{4.6}$$

As an immediate consequence, $\mathbb{E}[I(\phi)] = 0$.

Proof. Let $\phi \in \mathcal{S}$ with the representation (4.3), that is, $\phi_t = \sum_{i=0}^{n-1} \xi_i\, \mathbf{1}_{(t_i, t_{i+1}]}(t)$, with ξ_i bounded and $\mathcal{F}_{t_i}$-measurable. By Definition 4.2, $I(\phi) = \sum_{i=0}^{n-1} \xi_i\, (W_{t_{i+1}} - W_{t_i})$.

To lighten the notation, write $\Delta W_i := W_{t_{i+1}} - W_{t_i}$ and $\Delta_i := t_{i+1} - t_i$.

Step 1: computation of $\mathbb{E}[I(\phi)^2]$. Expanding the square of the sum,

$$I(\phi)^2 = \sum_{i=0}^{n-1} \xi_i^2\, (\Delta W_i)^2 + 2 \sum_{j<i} \xi_j\, \xi_i\, \Delta W_j\, \Delta W_i.$$

Taking expectations,

$$\mathbb{E}[I(\phi)^2] = \sum_{i=0}^{n-1} \mathbb{E}\big[\xi_i^2\, (\Delta W_i)^2\big] + 2 \sum_{j<i} \mathbb{E}\big[\xi_j\, \xi_i\, \Delta W_j\, \Delta W_i\big]. \tag{4.7}$$

Step 2: vanishing of the cross terms. Let $j < i$. The variables ξ_j, ξ_i and ΔW_j are all $\mathcal{F}_{t_i}$-measurable—indeed, ξ_j is $\mathcal{F}_{t_j}$-measurable and $\mathcal{F}_{t_j} \subseteq \mathcal{F}_{t_i}$, ξ_i is $\mathcal{F}_{t_i}$-measurable by assumption, and $\Delta W_j = W_{t_{j+1}} - W_{t_j}$ is $\mathcal{F}_{t_i}$-measurable since $t_{j+1} \le t_i$. By the property "taking out what is known" of conditional expectation (Proposition 2.17, property (iii)),

$$\mathbb{E}\big[\xi_j\, \xi_i\, \Delta W_j\, \Delta W_i \,\big|\, \mathcal{F}_{t_i}\big] = \xi_j\, \xi_i\, \Delta W_j\, \mathbb{E}[\Delta W_i \mid \mathcal{F}_{t_i}].$$

But ΔW_i is independent of $\mathcal{F}_{t_i}$ by condition (iii) of Definition 3.1, and has zero mean, hence $\mathbb{E}[\Delta W_i \mid \mathcal{F}_{t_i}] = 0$. The conditional expectation of the product is zero, and taking unconditional expectations, so is the expectation:

$$\mathbb{E}\big[\xi_j\, \xi_i\, \Delta W_j\, \Delta W_i\big] = 0, \qquad \text{for all } j < i.$$

[2] That is, for each $t \in [0,T]$, the restriction of ϕ to $[0,t] \times \Omega$ is $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$-measurable. The property implies adaptedness and holds automatically for adapted processes with left-continuous paths, in particular for the simple processes of Definition 4.1.

The cross terms of (4.7) vanish.

Step 3: computation of the diagonal terms. For each i, the variable ξ_i^2 is $\mathcal{F}_{t_i}$-measurable and $(\Delta W_i)^2$ is independent of $\mathcal{F}_{t_i}$. Applying again "taking out what is known" and independence,

$$\mathbb{E}\big[\xi_i^2\,(\Delta W_i)^2\big] = \mathbb{E}\big[\xi_i^2\,\mathbb{E}[(\Delta W_i)^2 \mid \mathcal{F}_{t_i}]\big] = \mathbb{E}[\xi_i^2]\,\mathbb{E}[(\Delta W_i)^2] = \mathbb{E}[\xi_i^2]\,\Delta_i,$$

where the last equality uses that $\Delta W_i \sim \mathcal{N}(0, \Delta_i)$.

Step 4: identification with the $\mathcal{H}^2$ norm. By (4.7) and the previous steps,

$$\mathbb{E}[I(\phi)^2] = \sum_{i=0}^{n-1} \mathbb{E}[\xi_i^2]\,\Delta_i.$$

On the other hand, on each subinterval $(t_i, t_{i+1}]$ the process ϕ equals ξ_i, hence $\phi_s^2 = \xi_i^2$ and

$$\int_0^T \phi_s^2\,ds = \sum_{i=0}^{n-1} \xi_i^2\,\Delta_i.$$

Taking expectations,

$$\mathbb{E}\left[\int_0^T \phi_s^2\,ds\right] = \sum_{i=0}^{n-1} \mathbb{E}[\xi_i^2]\,\Delta_i,$$

which coincides with $\mathbb{E}[I(\phi)^2]$. Identity (4.6) is proved.

For the consequence $\mathbb{E}[I(\phi)] = 0$, it suffices to take expectations in $I(\phi) = \sum_i \xi_i\,\Delta W_i$ and apply the same argument of Step 2 to each individual summand: $\mathbb{E}[\xi_i\,\Delta W_i] = \mathbb{E}[\xi_i\,\mathbb{E}[\Delta W_i \mid \mathcal{F}_{t_i}]] = 0$. □

Comment on the role of predictability. The preceding proof brings out with remarkable clarity the mathematical reason why Itô's choice, evaluating the integrand at the left endpoint of each subinterval, is the correct one. The decisive step is *Step 2*: the vanishing of the cross terms is obtained precisely because ξ_i is $\mathcal{F}_{t_i}$-measurable while the increment ΔW_i is independent of $\mathcal{F}_{t_i}$. Had we chosen to evaluate the integrand at the right endpoint, that is, if the variables were $\mathcal{F}_{t_{i+1}}$-measurable, the independence of the increment would no longer hold and the cross terms would no longer vanish. The isometry is thus a property specific to the Itô integral, not shared by integrals constructed with other conventions.

Reading as an isometry. Identity (4.6) admits a natural geometric reading. Regarding $\mathcal{S}$ as a subset of the Hilbert space $\mathcal{H}^2$ with the norm $\|\cdot\|_{\mathcal{H}^2}$ defined in (4.5), and $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ as a Hilbert space with its usual norm, identity (4.6) says exactly that the operator $I : (\mathcal{S}, \|\cdot\|_{\mathcal{H}^2}) \to (L^2(\Omega), \|\cdot\|_{L^2})$ is linear and norm-preserving. This is exactly the notion of *linear isometry* in the sense of functional analysis.

4.2.3 Extension to $\mathcal{H}^2$ by density

The isometry established in Theorem 4.4 allows us to extend the integral from $\mathcal{S}$ to $\mathcal{H}^2$ by the classical density argument: we approximate the integrand by simple processes and use the completeness of $L^2(\Omega)$ to pass to the limit.

Density of simple processes.

Lemma 4.5 (Density of $\mathcal{S}$ in $\mathcal{H}^2$)**.** *The class $\mathcal{S}$ of simple processes is dense in $\mathcal{H}^2$ with respect to the norm $\|\cdot\|_{\mathcal{H}^2}$. That is, for every process $\phi \in \mathcal{H}^2$ there exists a sequence $(\phi^{(n)}) \subset \mathcal{S}$ such that*

$$\|\phi^{(n)} - \phi\|_{\mathcal{H}^2} \xrightarrow{n\to\infty} 0.$$

The proof of the lemma, which we omit, proceeds in two stages. In the first, the problem is reduced to the case of bounded processes ϕ, via truncation. In the second, a given bounded process is approximated by constructing time regularizations of the form

$$\phi_t^{(n)} \coloneqq \sum_{i=0}^{n-1} \left(\frac{n}{T} \int_{(i-1)T/n}^{iT/n} \phi_s \, ds \right) \mathbf{1}_{(iT/n,\,(i+1)T/n]}(t),$$

which are simple processes and converge to ϕ in $\mathcal{H}^2$ by the continuity of translation in L^2. The detailed argument can be found in [5], Section 3.2.

Construction of the integral on $\mathcal{H}^2$. Let $\phi \in \mathcal{H}^2$. By Lemma 4.5 there exists a sequence $(\phi^{(n)}) \subset \mathcal{S}$ with $\|\phi^{(n)} - \phi\|_{\mathcal{H}^2} \to 0$. In particular, $(\phi^{(n)})$ is Cauchy in $\mathcal{H}^2$. Applying the Itô isometry (Theorem 4.4) and linearity to the difference $\phi^{(n)} - \phi^{(m)} \in \mathcal{S}$,

$$\mathbb{E}\Big[\big(I(\phi^{(n)}) - I(\phi^{(m)})\big)^2\Big] = \mathbb{E}\Big[\big(I(\phi^{(n)} - \phi^{(m)})\big)^2\Big] = \|\phi^{(n)} - \phi^{(m)}\|_{\mathcal{H}^2}^2 \xrightarrow{n,m\to\infty} 0. \tag{4.8}$$

The sequence $(I(\phi^{(n)}))$ is therefore Cauchy in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$. Since this space is complete, there exists a unique element $L \in L^2(\Omega)$ such that $I(\phi^{(n)}) \to L$ in L^2.

It remains to verify that the limit L does not depend on the particular approximating sequence. Let $(\phi^{(n)}), (\widetilde{\phi}^{(n)}) \subset \mathcal{S}$ be two sequences both converging to ϕ in $\mathcal{H}^2$, with respective limits $L, \widetilde{L} \in L^2(\Omega)$. Considering the interleaved sequence $\phi^{(1)}, \widetilde{\phi}^{(1)}, \phi^{(2)}, \widetilde{\phi}^{(2)}, \ldots$, which also converges to ϕ in $\mathcal{H}^2$, the images $I(\phi^{(1)}), I(\widetilde{\phi}^{(1)}), I(\phi^{(2)}), \ldots$ form a Cauchy sequence in $L^2(\Omega)$ by the same argument as (4.8), and hence converge to a unique limit. But the two subsequences $(I(\phi^{(n)}))$ and $(I(\widetilde{\phi}^{(n)}))$ must converge to that same limit, which forces $L = \widetilde{L}$ almost surely.

Definition 4.6 (Stochastic integral on $\mathcal{H}^2$)**.** Let $\phi \in \mathcal{H}^2$ and let $(\phi^{(n)}) \subset \mathcal{S}$ be any sequence of simple processes with $\|\phi^{(n)} - \phi\|_{\mathcal{H}^2} \to 0$. The *Itô stochastic integral* of ϕ over $[0, T]$ is the random variable

$$I(\phi) \coloneqq \int_0^T \phi_s \, dW_s \coloneqq L^2\text{-}\lim_{n\to\infty} I(\phi^{(n)}),$$

an element of $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ defined up to modifications on sets of measure zero.

The definition is consistent: for $\phi \in \mathcal{S}$, taking the constant sequence $\phi^{(n)} = \phi$ recovers Definition 4.2.

Theorem 4.7 (Properties of the stochastic integral on $\mathcal{H}^2$)**.** *The operator $I : \mathcal{H}^2 \to L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ given by Definition 4.6 satisfies the following properties:*

(i) Linearity: *for $\phi, \psi \in \mathcal{H}^2$ and $a, b \in \mathbb{R}$,*

$$I(a\phi + b\psi) = a\, I(\phi) + b\, I(\psi) \quad \text{a.s.}$$

(ii) Itô isometry: *for* $\phi \in \mathcal{H}^2$,

$$\mathbb{E}\Big[\big(I(\phi)\big)^2\Big] = \mathbb{E}\bigg[\int_0^T \phi_s^2 \, ds\bigg] = \|\phi\|_{\mathcal{H}^2}^2. \tag{4.9}$$

(iii) Zero expectation: *for* $\phi \in \mathcal{H}^2$, $\mathbb{E}[I(\phi)] = 0$.

Proof. The three properties follow from the corresponding properties on $\mathcal{S}$ by passing to the limit, thanks to the fact that I is continuous with respect to the norms $\|\cdot\|_{\mathcal{H}^2}$ and $\|\cdot\|_{L^2}$ (a continuity which is in turn a consequence of the isometry on $\mathcal{S}$).

By way of illustration, we prove (ii). Let $\phi \in \mathcal{H}^2$ and $(\phi^{(n)}) \subset \mathcal{S}$ with $\phi^{(n)} \to \phi$ in $\mathcal{H}^2$. Then $I(\phi^{(n)}) \to I(\phi)$ in $L^2(\Omega)$ by Definition 4.6. The continuity of the norm $\|\cdot\|_{L^2}$ gives

$$\mathbb{E}[I(\phi)^2] = \|I(\phi)\|_{L^2}^2 = \lim_{n\to\infty} \|I(\phi^{(n)})\|_{L^2}^2 = \lim_{n\to\infty} \mathbb{E}[I(\phi^{(n)})^2].$$

By Theorem 4.4 applied to each $\phi^{(n)} \in \mathcal{S}$,

$$\mathbb{E}[I(\phi^{(n)})^2] = \|\phi^{(n)}\|_{\mathcal{H}^2}^2,$$

and by the continuity of the norm $\|\cdot\|_{\mathcal{H}^2}$, $\|\phi^{(n)}\|_{\mathcal{H}^2}^2 \to \|\phi\|_{\mathcal{H}^2}^2$. Identity (4.9) is obtained by passing to the limit. The proofs of (i) and (iii) are entirely analogous. □

The integral so defined is, however, a *global* object on $[0,T]$. In many contexts we shall need the integral over $[0,t]$ with t variable: this motivates the study of the process $\{M_t\}$ in the next subsection.

4.2.4 The indefinite integral as a process

For each $t \in [0,T]$, the truncated process $\phi\,\mathbf{1}_{[0,t]}$ belongs to $\mathcal{H}^2$ (the restriction of a square-integrable process is square-integrable), which allows us to apply the integral operator I to it and thus obtain a family $\{M_t\}_{t\in[0,T]}$ that we shall study as a process indexed by t.

Definition 4.8 (Indefinite integral)**.** Let $\phi \in \mathcal{H}^2$. For each $t \in [0,T]$, we define

$$M_t \coloneqq \int_0^t \phi_s \, dW_s \coloneqq I\big(\phi\,\mathbf{1}_{[0,t]}\big)\,. \tag{4.10}$$

The family $\{M_t\}_{t\in[0,T]}$ so obtained is called the *indefinite integral* of ϕ with respect to the Brownian motion W.

Note that for $t = T$ we recover $M_T = I(\phi)$, the global operator of the previous subsection. Likewise, $M_0 = I(\phi\,\mathbf{1}_{\{0\}}) = 0$ almost surely, since the process $\phi\,\mathbf{1}_{\{0\}}$ has zero $\mathcal{H}^2$ norm.

Continuity of the paths. As we have constructed the process, each M_t is defined as an element of $L^2(\Omega)$, that is, up to modifications on sets of measure zero. This form of definition does not by itself guarantee that the paths $t \mapsto M_t(\omega)$ are continuous functions: the family $\{M_t\}_{t\in[0,T]}$ is indexed by a continuous parameter, and the uncountable union of the null sets associated with each M_t might, *a priori*, fail to be a null set. It is therefore necessary to verify the existence of

a continuous *version* of the process, that is, a choice of representatives in each equivalence class that yields continuous paths almost surely.

Theorem 4.9 (Continuity of the indefinite integral)**.** *Let $\phi \in \mathcal{H}^2$ and let $\{M_t\}_{t\in[0,T]}$ be the process defined in* (4.10)*. There exists a version of this process whose paths are continuous almost surely.*

The rigorous proof, which we omit, rests essentially on Doob's maximal inequality for L^2 martingales, together with the approximation argument by simple processes already employed. For simple processes, the paths of the process M_t are continuous by construction (they are piecewise linear in t); the passage to the limit by density, controlled by means of the maximal inequality applied to the difference of two approximations, guarantees almost sure uniform convergence on $[0,T]$, which preserves continuity. The detailed construction can be found in [5], Section 3.2.

From here on, we shall always identify $\{M_t\}$ with its continuous version, without explicit mention.

The martingale property.

Theorem 4.10 (Martingale property of the indefinite integral)**.** *Let $\phi \in \mathcal{H}^2$ and let $\{M_t\}_{t\in[0,T]}$ be the indefinite integral of ϕ. Then $\{M_t\}$ is a martingale with respect to the filtration $(\mathcal{F}_t)$, that is:*

(i) *M_t is $\mathcal{F}_t$-measurable for all $t \in [0,T]$;*

(ii) *$\mathbb{E}[|M_t|] < \infty$ for all $t \in [0,T]$;*

(iii) *$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ almost surely, for all $0 \le s \le t \le T$.*

Proof. Adaptedness and integrability. For simple processes, M_t is a finite sum of products of $\mathcal{F}_t$-measurable variables and hence $\mathcal{F}_t$-measurable. For general $\phi \in \mathcal{H}^2$, the L^2 convergence of the approximations $M_t^{(n)}$ to M_t (Definition 4.6) yields an almost surely convergent subsequence, and $\mathcal{F}_t$-measurability is preserved under almost sure limits. Integrability follows from Cauchy-Schwarz: $\mathbb{E}[|M_t|] \le \mathbb{E}[M_t^2]^{1/2} = \|\phi\,\mathbf{1}_{[0,t]}\|_{\mathcal{H}^2} \le \|\phi\|_{\mathcal{H}^2} < \infty$, where the middle equality uses the Itô isometry.

Martingale property. Let $0 \le s \le t \le T$. We begin with the case of a simple process $\phi \in \mathcal{S}$ with representation $\phi_r = \sum_i \xi_i\, \mathbf{1}_{(t_i,t_{i+1}]}(r)$. Taking a refinement of the partition that includes the points s and t—which alters neither the process ϕ nor the indefinite integral—we may assume without loss of generality that $s = t_p$ and $t = t_q$ for some $p \le q$. Then

$$M_t - M_s = \int_s^t \phi_r\, dW_r = \sum_{i=p}^{q-1} \xi_i \left(W_{t_{i+1}} - W_{t_i}\right).$$

Taking conditional expectations with respect to $\mathcal{F}_s = \mathcal{F}_{t_p}$ and using linearity,

$$\mathbb{E}[M_t - M_s \mid \mathcal{F}_s] = \sum_{i=p}^{q-1} \mathbb{E}\left[\xi_i \left(W_{t_{i+1}} - W_{t_i}\right) \,\middle|\, \mathcal{F}_{t_p}\right].$$

For each summand, observe that $\mathcal{F}_{t_p} \subseteq \mathcal{F}_{t_i}$ since $i \geq p$. Applying the tower property (Proposition 2.17, property (iv)) with $\mathcal{F}_{t_p} \subseteq \mathcal{F}_{t_i}$,

$$\mathbb{E}\big[\xi_i \, (W_{t_{i+1}} - W_{t_i}) \mid \mathcal{F}_{t_p}\big] = \mathbb{E}\Big[\mathbb{E}\big[\xi_i \, (W_{t_{i+1}} - W_{t_i}) \mid \mathcal{F}_{t_i}\big] \,\Big|\, \mathcal{F}_{t_p}\Big].$$

The inner conditional expectation is zero: ξ_i is $\mathcal{F}_{t_i}$-measurable and $W_{t_{i+1}} - W_{t_i}$ is independent of $\mathcal{F}_{t_i}$ with zero mean, hence taking out what is known and using independence, $\mathbb{E}[\xi_i \, (W_{t_{i+1}} - W_{t_i}) \mid \mathcal{F}_{t_i}] = \xi_i \, \mathbb{E}[W_{t_{i+1}} - W_{t_i} \mid \mathcal{F}_{t_i}] = 0$. Consequently, $\mathbb{E}[M_t - M_s \mid \mathcal{F}_s] = 0$, that is, $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ almost surely. The martingale property is established for simple processes.

For general $\phi \in \mathcal{H}^2$, take a sequence $(\phi^{(n)}) \subset \mathcal{S}$ with $\phi^{(n)} \to \phi$ in $\mathcal{H}^2$, and let $M_t^{(n)}$ be the corresponding indefinite integrals. By what has just been proved, $\mathbb{E}[M_t^{(n)} \mid \mathcal{F}_s] = M_s^{(n)}$ almost surely. Taking the limit in $L^2(\Omega)$ as $n \to \infty$, $M_t^{(n)} \to M_t$ and $M_s^{(n)} \to M_s$ by the isometry applied to the restrictions to $[0,t]$ and $[0,s]$ respectively. Since conditional expectation with respect to $\mathcal{F}_s$ is a continuous operator on L^2—it is the orthogonal projection onto the subspace $L^2(\Omega, \mathcal{F}_s, \mathbb{P})$—the equality passes to the limit: $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ almost surely. □

The isometry in process form. As an immediate consequence of the construction, the Itô isometry generalizes to the indefinite process: for all $t \in [0,T]$,

$$\mathbb{E}[M_t^2] = \mathbb{E}\left[\int_0^t \phi_s^2 \, ds\right]. \tag{4.11}$$

The identity follows by applying Theorem 4.7 to the restricted process $\phi \mathbf{1}_{[0,t]} \in \mathcal{H}^2$.

4.3 Itô's formula

This section develops *Itô's formula*, the continuous-time substitute for the classical chain rule and, together with Girsanov's theorem, one of the fundamental pieces of Chapter 5.

4.3.1 Itô processes and Itô's formula for $f(W_t)$

Definition 4.11 (Itô process)**.** A stochastic process $\{X_t\}_{t \in [0,T]}$ is an *Itô process* if it admits a representation of the form

$$X_t = X_0 + \int_0^t a_s \, ds + \int_0^t b_s \, dW_s, \qquad t \in [0,T], \tag{4.12}$$

where X_0 is an $\mathcal{F}_0$-measurable random variable, and the processes $\{a_t\}$ and $\{b_t\}$ are adapted to $(\mathcal{F}_t)$ and satisfy the integrability conditions

$$\int_0^T |a_s| \, ds < \infty \text{ almost surely}, \qquad b \in \mathcal{H}^2[0,T].$$

The processes a and b are called, respectively, the *drift coefficient* and the *diffusion coefficient* of the process X.

The condition $b \in \mathcal{H}^2[0,T]$ guarantees that the stochastic integral $\int_0^t b_s\,dW_s$ is well defined in the sense of Theorem 4.7, while the condition on a ensures that the integral $\int_0^t a_s\,ds$ is well defined pathwise as a Lebesgue integral.

Differential notation. The integral equality (4.12) is usually written, in abbreviated form, as

$$dX_t = a_t\,dt + b_t\,dW_t. \tag{4.13}$$

This notation is very convenient and will be used frequently from now on, but it should be stressed that it has no mathematical meaning of its own: it is nothing more than a compact way of expressing (4.12). Identities of the form (4.13) must always be understood as shorthand for the corresponding integral identities.

Statement of Itô's formula.

Theorem 4.12 (Itô's formula for $f(W_t)$)**.** *Let $f:\mathbb{R}\to\mathbb{R}$ be a function of class C^2 with f, f' and f'' bounded. Then, for all $t\in[0,T]$, almost surely,*

$$f(W_t) = f(0) + \int_0^t f'(W_s)\,dW_s + \frac{1}{2}\int_0^t f''(W_s)\,ds. \tag{4.14}$$

Equivalently, in differential notation,

$$d\big(f(W_t)\big) = f'(W_t)\,dW_t + \tfrac{1}{2}f''(W_t)\,dt.$$

Reading of the result. The differential of $f(W_t)$ is the sum of a term in dW_t, analogous to that of the classical chain rule, and an extra term in dt, the *Itô term*, absent in ordinary calculus. This correction is a consequence of the heuristic rule $(dW_t)^2 = dt$ anticipated in (3.4): when f is expanded in a Taylor series, the second-order term does not vanish and contributes exactly $\frac{1}{2}f''(W_t)\,dt$. The boundedness assumptions are the technically most convenient ones; the formula holds under weaker conditions via localization.

Proof. For each partition $\pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ of $[0,t]$, we write the telescoping expansion

$$f(W_t) - f(0) = \sum_{k=0}^{n-1}\big[f(W_{t_{k+1}}) - f(W_{t_k})\big]; \tag{4.15}$$

the resulting terms will produce the two integrals on the right-hand side of (4.14), plus a negligible remainder.

Step 1: Taylor expansion. By the assumption $f\in C^2$, for each k there exists $\theta_k \in [W_{t_k}, W_{t_{k+1}}]$ (or the corresponding interval with reversed endpoints) such that

$$f(W_{t_{k+1}}) - f(W_{t_k}) = f'(W_{t_k})\,\Delta W_k + \tfrac{1}{2}f''(W_{t_k})\,(\Delta W_k)^2 + R_k,$$

where $\Delta W_k \coloneqq W_{t_{k+1}} - W_{t_k}$ and the remainder is

$$R_k = \tfrac{1}{2}\big[f''(\theta_k) - f''(W_{t_k})\big]\,(\Delta W_k)^2.$$

Summing over k and comparing with (4.15),

$$f(W_t) - f(0) = \underbrace{\sum_{k=0}^{n-1} f'(W_{t_k})\,\Delta W_k}_{S_1^\pi} + \underbrace{\tfrac{1}{2}\sum_{k=0}^{n-1} f''(W_{t_k})\,(\Delta W_k)^2}_{S_2^\pi} + \underbrace{\sum_{k=0}^{n-1} R_k}_{R^\pi}. \tag{4.16}$$

Step 2: convergence of S_1^π. The sum S_1^π is exactly the stochastic integral of the simple process $\phi_s^\pi := \sum_k f'(W_{t_k})\,\mathbf{1}_{(t_k,t_{k+1}]}(s)$ with respect to W over $[0,t]$. Let us show that ϕ^π converges to $f'(W_s)$ in $\mathcal{H}^2[0,t]$ as $|\pi| \to 0$. By the definition of the norm,

$$\|\phi^\pi - f'(W)\|_{\mathcal{H}^2}^2 = \mathbb{E}\left[\int_0^t \big(\phi_s^\pi - f'(W_s)\big)^2\,ds\right] = \mathbb{E}\left[\sum_{k=0}^{n-1}\int_{t_k}^{t_{k+1}} \big(f'(W_{t_k}) - f'(W_s)\big)^2\,ds\right].$$

By the uniform continuity of f' on any compact set and the almost sure continuity of Brownian paths, the pointwise integrand tends to zero as $|\pi| \to 0$; by the boundedness of f', it is dominated by a constant. The dominated convergence theorem guarantees that the integral converges to zero, that is, $\phi^\pi \to f'(W)$ in $\mathcal{H}^2$. By the Itô isometry,

$$S_1^\pi \xrightarrow{L^2} \int_0^t f'(W_s)\,dW_s.$$

Step 3: convergence of S_2^π. We shall prove that

$$S_2^\pi \xrightarrow{L^2} \int_0^t f''(W_s)\,ds \quad \text{as } |\pi| \to 0.$$

To this end, we decompose

$$S_2^\pi - \int_0^t f''(W_s)\,ds = \underbrace{\sum_{k=0}^{n-1} f''(W_{t_k})\,\big[(\Delta W_k)^2 - \Delta_k\big]}_{A^\pi} + \underbrace{\sum_{k=0}^{n-1} f''(W_{t_k})\,\Delta_k - \int_0^t f''(W_s)\,ds}_{B^\pi},$$

where $\Delta_k := t_{k+1} - t_k$, and we shall prove that both A^π and B^π tend to zero in L^2.

For B^π: the quantity $\sum_k f''(W_{t_k})\,\Delta_k$ is the Riemann sum of the continuous function $s \mapsto f''(W_s)$ associated with the partition π and the choice $\tau_k = t_k$. By the almost sure pathwise continuity, it converges pathwise to $\int_0^t f''(W_s)\,ds$ as $|\pi| \to 0$, and the uniform bound on f'' allows us to apply dominated convergence to obtain convergence in L^2.

For A^π: by an argument analogous to that of the Itô isometry (Theorem 4.4, Step 2), using that $f''(W_{t_k})$ is $\mathcal{F}_{t_k}$-measurable and that $(\Delta W_k)^2 - \Delta_k$ has zero conditional expectation given $\mathcal{F}_{t_k}$, the cross terms vanish when squaring and taking expectations. Applying the uniform bound $|f''| \le M$ and the Gaussian fourth-moment computation $\mathbb{E}[((\Delta W_k)^2 - \Delta_k)^2] = 2\,\Delta_k^2$,

$$\mathbb{E}[(A^\pi)^2] = \sum_{k=0}^{n-1} \mathbb{E}\big[(f''(W_{t_k}))^2\big]\,\mathbb{E}\big[((\Delta W_k)^2 - \Delta_k)^2\big] \le 2M^2 \sum_{k=0}^{n-1} \Delta_k^2 \le 2M^2 \cdot |\pi| \cdot t \xrightarrow{|\pi|\to 0} 0.$$

Step 4: the remainder R^π is negligible. Recall that

$$R^\pi = \tfrac{1}{2}\sum_{k=0}^{n-1}\big[f''(\theta_k) - f''(W_{t_k})\big]\,(\Delta W_k)^2,$$

with θ_k between W_{t_k} and $W_{t_{k+1}}$. Since f'' is uniformly continuous on any compact set and Brownian paths are continuous almost surely, the factor $|f''(\theta_k) - f''(W_{t_k})|$ tends uniformly to zero as $|\pi| \to 0$, that is,

$$\varepsilon^\pi := \max_{0\le k\le n-1} |f''(\theta_k) - f''(W_{t_k})| \xrightarrow{|\pi|\to 0} 0 \quad \text{almost surely.}$$

Therefore,

$$|R^\pi| \le \tfrac{1}{2}\,\varepsilon^\pi \sum_{k=0}^{n-1}(\Delta W_k)^2.$$

The sum on the right-hand side converges to t in L^2 by Theorem 3.7, and $\varepsilon^\pi \to 0$ almost surely with uniform bound $2M$. Combining both facts (via dominated convergence) we conclude that $R^\pi \to 0$ in L^1, and hence in probability. Passing to a subsequence if necessary, $R^\pi \to 0$ almost surely.

Step 5: conclusion. Returning to the decomposition (4.16) and combining the limits obtained in the previous steps (after passing to a common subsequence ensuring that all convergences hold almost surely),

$$f(W_t) - f(0) = \int_0^t f'(W_s)\,dW_s + \tfrac{1}{2}\int_0^t f''(W_s)\,ds + 0,$$

almost surely, which is identity (4.14). □

4.3.2 General version and examples of application

Theorem 4.13 (Itô's formula for Itô processes)**.** *Let $\{X_t\}$ be an Itô process with dynamics $dX_t = a_t\,dt + b_t\,dW_t$, and let $f : [0,T]\times\mathbb{R} \to \mathbb{R}$ be a function of class $C^{1,2}$, that is, continuously differentiable in its first variable and twice continuously differentiable in the second, with the relevant derivatives bounded on the range of X. Then the process $\{f(t,X_t)\}_{t\in[0,T]}$ is also an Itô process, and it satisfies*

$$d\big(f(t,X_t)\big) = \Big[\partial_t f(t,X_t) + a_t\,\partial_x f(t,X_t) + \tfrac{1}{2}\,b_t^2\,\partial_{xx} f(t,X_t)\Big]\,dt + b_t\,\partial_x f(t,X_t)\,dW_t. \tag{4.17}$$

Equivalently, in integral form,

$$f(t,X_t) = f(0,X_0) + \int_0^t\Big[\partial_t f(s,X_s) + a_s\,\partial_x f(s,X_s) + \tfrac{1}{2}\,b_s^2\,\partial_{xx} f(s,X_s)\Big]ds + \int_0^t b_s\,\partial_x f(s,X_s)\,dW_s.$$

The proof of this theorem, which we omit, follows the same strategy as that of Theorem 4.12: a telescoping expansion followed by a second-order Taylor expansion in the space variable, now

with an additional first-order term in the time variable. The identification of the limits is entirely analogous; the details can be found in [5], Section 3.3.

Itô's mnemonic rules. The structure of (4.17) is easily memorized by operating with $dX_t = a_t\,dt + b_t\,dW_t$ as with an ordinary differential object, except for the multiplication rules

$$(dt)^2 = 0, \qquad dt\,dW_t = 0, \qquad (dW_t)^2 = dt, \tag{4.18}$$

which encode, respectively, that dt is of higher order than $\sqrt{dt}$ and the quadratic variation of Brownian motion of Theorem 3.7. Applying a formal Taylor expansion to $f(t, X_t)$ and using these rules to discard negligible terms, one recovers exactly (4.17).

Three examples. We illustrate the use of the formula with three examples that will reappear later on.

Example 1: the computation of $\int_0^t W_s\,dW_s$. We apply Theorem 4.12 to the function $f(x) = \frac{1}{2}x^2$, whose derivatives are $f'(x) = x$ and $f''(x) = 1$. Strictly speaking this function does not satisfy the boundedness assumptions, but the result still holds by localization arguments. The formula gives

$$\tfrac{1}{2}W_t^2 = 0 + \int_0^t W_s\,dW_s + \tfrac{1}{2}\int_0^t 1\,ds,$$

that is,

$$\int_0^t W_s\,dW_s = \tfrac{1}{2}\big(W_t^2 - t\big). \tag{4.19}$$

We thus recover, by the systematic route of Itô calculus, the result that in the motivation (Section 4.1) we had obtained by direct computation of the sums. Formula (4.19) also shows, via Theorem 4.10, that $\{W_t^2 - t\}$ is a martingale (a result that states the role of this quantity already anticipated in Proposition 3.12).

Example 2: geometric Brownian motion. Consider the process

$$S_t := S_0\,\exp\big(\nu t + \sigma\,W_t\big),$$

where $S_0 > 0$, $\nu \in \mathbb{R}$ and $\sigma > 0$. We apply Theorem 4.13 to the function $f(t,x) = S_0\,\exp(\nu t + \sigma x)$ and the process $X_t = W_t$ (which is an Itô process with $a \equiv 0$, $b \equiv 1$). The derivatives of f are

$$\partial_t f = \nu\,f, \qquad \partial_x f = \sigma\,f, \qquad \partial_{xx} f = \sigma^2\,f.$$

Substituting in the formula,

$$dS_t = \big[\nu\,S_t + 0 + \tfrac{1}{2}\sigma^2\,S_t\big]\,dt + \sigma\,S_t\,dW_t,$$

that is,

$$\boxed{dS_t = \mu\,S_t\,dt + \sigma\,S_t\,dW_t, \qquad \mu := \nu + \tfrac{1}{2}\sigma^2.} \tag{4.20}$$

Identity (4.20) is the *stochastic differential equation of geometric Brownian motion*, the canonical model for the price of the underlying in Black-Scholes (Chapter 5).

Example 3: the generalized exponential martingale. Let $\{\theta_t\}_{t\in[0,T]}$ be an adapted process with $\theta \in \mathcal{H}^2$. Consider the process $\{L_t\}$ defined by

$$L_t \coloneqq \exp\Big(-\int_0^t \theta_s\, dW_s - \tfrac{1}{2}\int_0^t \theta_s^2\, ds\Big). \tag{4.21}$$

We apply Itô's formula to identify its SDE. Define $Y_t \coloneqq -\int_0^t \theta_s\, dW_s - \frac{1}{2}\int_0^t \theta_s^2\, ds$, which is an Itô process with $dY_t = -\frac{1}{2}\theta_t^2\, dt - \theta_t\, dW_t$. The process $L_t = \exp(Y_t)$ is a function of the Itô process Y via $f(y) = e^y$, with $f'(y) = f''(y) = e^y = L_t$. By Theorem 4.13 applied to $f(Y_t)$, where the drift coefficient of Y is $-\frac{1}{2}\theta_t^2$ and the diffusion coefficient is $-\theta_t$,

$$dL_t = L_t\left[-\tfrac{1}{2}\theta_t^2 + \tfrac{1}{2}\,(-\theta_t)^2\right] dt + L_t\,(-\theta_t)\, dW_t.$$

The two terms in the bracket cancel exactly, leaving

$$dL_t = -\theta_t\, L_t\, dW_t. \tag{4.22}$$

The cancellation that produces the zero drift term is not accidental: the correction $-\frac{1}{2}\int \theta_s^2\, ds$ appearing in the exponent of (4.21) is calibrated precisely so that this happens. The consequence is that $\{L_t\}$ is a driftless process, expressible as the stochastic integral of $-\theta_t\, L_t$ with respect to W. By Theorem 4.10, if $\theta\, L \in \mathcal{H}^2$, then $\{L_t\}$ is a martingale.

The process (4.21) generalizes the exponential martingale $\mathcal{E}_t^\lambda$ of Chapter 3, replacing the parameter λ by an adapted process θ_t. It will reappear in §4.4 as the Radon-Nikodym derivative of Girsanov's change of measure.

4.4 Girsanov's theorem

In the binomial model of Chapter 2, the change $\mathbb{P} \to \mathbb{Q}$ was constructed explicitly by modifying the probabilities of the random walk. In the continuous setting, what is modified is the measure on the space of paths, and the question is under which measure $\mathbb{Q}$ a process with drift θ_t relative to W becomes a martingale. The answer is provided by *Girsanov's theorem.*

4.4.1 Change of measure in continuous time

Before stating the theorem, it is worth introducing the language and concepts on which it will rest: equivalence of measures and the Radon-Nikodym derivative in the continuous setting.

Definition 4.14 (Equivalent measures)**.** Two probability measures $\mathbb{P}$ and $\mathbb{Q}$ on a measurable space $(\Omega, \mathcal{F})$ are said to be *equivalent*, written $\mathbb{Q} \sim \mathbb{P}$, if for all $A \in \mathcal{F}$,

$$\mathbb{P}(A) = 0 \iff \mathbb{Q}(A) = 0.$$

Equivalence of measures is the minimal degree of compatibility we shall need: two equivalent measures assign zero probability to exactly the same events, although they may weigh the re-

maining ones differently. In particular, statements of the type "almost surely" are interchangeable between the two.

Radon-Nikodym derivative.

Theorem 4.15 (Radon-Nikodym, equivalent version)**.** *Let $\mathbb{P}$ and $\mathbb{Q}$ be two equivalent probability measures on $(\Omega, \mathcal{F})$. There exists a unique (up to almost sure equality) random variable $L : \Omega \to (0, \infty)$ which is $\mathcal{F}$-measurable, $\mathbb{P}$-integrable, and satisfies*

$$\mathbb{Q}(A) = \int_A L \, d\mathbb{P}, \qquad A \in \mathcal{F}. \tag{4.23}$$

The variable L is called the Radon-Nikodym derivative of $\mathbb{Q}$ with respect to $\mathbb{P}$*, and is denoted $L = d\mathbb{Q}/d\mathbb{P}$.*

The proof can be found in [9]. Identity (4.23) can be reformulated, equivalently, in terms of expectations: for any random variable X such that LX is $\mathbb{P}$-integrable,

$$\mathbb{E}_{\mathbb{Q}}[X] = \mathbb{E}_{\mathbb{P}}[LX], \tag{4.24}$$

a formula that determines how expectations under $\mathbb{Q}$ are computed from expectations under $\mathbb{P}$.

Remark 4.16 (Change of measure as a general method). Identity (4.24) reveals an idea that goes beyond the pricing framework: it allows one to compute expectations under a measure $\mathbb{Q}$ by operating under another measure $\mathbb{P}$, simply by reweighting through the density L. Replacing a measure by an equivalent one in order to turn a difficult computation into a tractable one is a device of general use in probability; in simulation, for instance, it is the foundation of *importance sampling*, which reduces the variance of Monte Carlo estimates of rare events (see [4], Section 4.6). Girsanov's theorem, which we address next, is the version of this method tailored to Brownian motion.

The filtered case: the density as a process. In the setting of interest to us, recall that $(\Omega, \mathcal{F})$ is moreover endowed with a filtration $(\mathcal{F}_t)_{t\in[0,T]}$, the augmented Brownian filtration. If $\mathbb{P}$ and $\mathbb{Q}$ are equivalent on $(\Omega, \mathcal{F}_T)$, we can restrict them to each $\mathcal{F}_t$ with $t < T$, and thus obtain a family of Radon-Nikodym derivatives

$$L_t := \frac{d(\mathbb{Q}\restriction_{\mathcal{F}_t})}{d(\mathbb{P}\restriction_{\mathcal{F}_t})}, \qquad t \in [0, T]. \tag{4.25}$$

The Radon-Nikodym derivative thus becomes a *process* rather than a random variable. The fundamental structural property of this process is the following.

Proposition 4.17 (Martingale property of the density process)**.** *Let $\mathbb{P}$ and $\mathbb{Q}$ be equivalent measures on $(\Omega, \mathcal{F}_T)$, and let $\{L_t\}_{t\in[0,T]}$ be the process defined in* (4.25)*. Then $\{L_t\}$ is a $\mathbb{P}$-martingale with respect to $(\mathcal{F}_t)$ with $\mathbb{E}_{\mathbb{P}}[L_t] = 1$ for all t.*

Proof. Adaptedness and integrability are immediate: L_t is $\mathcal{F}_t$-measurable by construction, and $\mathbb{E}_{\mathbb{P}}[L_t] = \mathbb{Q}(\Omega) = 1$ by (4.23) applied with $A = \Omega$.

For the martingale property, let $0 \le s < t \le T$ and $A \in \mathcal{F}_s$. Since $\mathcal{F}_s \subseteq \mathcal{F}_t$, also $A \in \mathcal{F}_t$. Applying (4.23) twice,

$$\int_A L_t \, d\mathbb{P} = \mathbb{Q}(A) = \int_A L_s \, d\mathbb{P}.$$

By the uniqueness of conditional expectation (Definition 2.15), $\mathbb{E}_{\mathbb{P}}[L_t \mid \mathcal{F}_s] = L_s$ almost surely. □

Construction of equivalent measures via exponential martingales. We shall follow the reverse path: given a positive $\mathbb{P}$-martingale $\{L_t\}$ with $\mathbb{E}_{\mathbb{P}}[L_T] = 1$, we can *define* a new measure $\mathbb{Q}$ on $(\Omega, \mathcal{F}_T)$ by

$$\mathbb{Q}(A) \coloneqq \int_A L_T \, d\mathbb{P}, \qquad A \in \mathcal{F}_T. \tag{4.26}$$

Integrability and the condition $\mathbb{E}_{\mathbb{P}}[L_T] = 1$ guarantee that $\mathbb{Q}$ is a probability measure, and the strict positivity of L_T ensures $\mathbb{Q} \sim \mathbb{P}$. The Radon-Nikodym derivative is then $d\mathbb{Q}/d\mathbb{P} = L_T$, and the density process recovers, by Proposition 4.17, the original martingale $\{L_t\}$.

The question, therefore, is: what kind of positive martingales with value one at $t = 0$ can we construct? Example 3 of Section 4.3.2 offers the natural answer: the exponential processes

$$L_t = \exp\!\Big(- \int_0^t \theta_s \, dW_s - \tfrac{1}{2} \int_0^t \theta_s^2 \, ds \Big), \tag{4.27}$$

associated with an adapted process θ, are canonical candidates. By construction they are positive and satisfy $L_0 = 1$. We have also verified that they satisfy the SDE $dL_t = -\theta_t \, L_t \, dW_t$, that is, they have no drift. By Theorem 4.10, if $\theta \, L \in \mathcal{H}^2$, then $\{L_t\}$ is a $\mathbb{P}$-martingale. And in that case $\mathbb{E}_{\mathbb{P}}[L_T] = \mathbb{E}_{\mathbb{P}}[L_0] = 1$, as required by (4.26).

Novikov's condition. The condition $\theta \, L \in \mathcal{H}^2$ is delicate to verify in practice, since it involves the very process L we are defining. There is a more manageable standard criterion, known as *Novikov's condition*, which guarantees that the process (4.27) is indeed a martingale.

Theorem 4.18 (Novikov's condition)**.** *Let $\{\theta_t\}_{t \in [0,T]}$ be an adapted process such that*

$$\mathbb{E}_{\mathbb{P}}\left[\exp\!\Big(\tfrac{1}{2} \int_0^T \theta_s^2 \, ds \Big) \right] < \infty. \tag{4.28}$$

Then the process $\{L_t\}$ defined in (4.27) is a $\mathbb{P}$-martingale.

The proof of this result lies beyond the scope of the present work and can be found in [5], Section 3.5. We shall use it as a tool to guarantee, in practice, the martingale condition. In the applications we address in Chapter 5, the process θ will turn out to be deterministic (in fact constant), and Novikov's condition will be trivially satisfied.

4.4.2 Girsanov's theorem

Lévy's characterization.

Theorem 4.19 (Lévy's characterization of Brownian motion)**.** *Let $\{X_t\}_{t \in [0,T]}$ be a continuous stochastic process adapted to a filtration $(\mathcal{F}_t)$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with $X_0 = 0$ almost surely. If:*

(i) $\{X_t\}$ *is a martingale with respect to* $(\mathcal{F}_t)$ *under* $\mathbb{P}$*;*

(ii) *the process* $\{X_t^2 - t\}$ *is also a martingale with respect to* $(\mathcal{F}_t)$ *under* $\mathbb{P}$*;*

then $\{X_t\}$ *is a standard Brownian motion under* $\mathbb{P}$*.*

The proof of this theorem, as formulated here, can be found in [3], Section 4.3.

Condition (ii) is equivalent, for continuous martingales, to $[X]_t = t$. The characterization therefore says: a continuous martingale with $X_0 = 0$ and quadratic variation t is necessarily a Brownian motion.

Girsanov's theorem. We turn to the statement of the central result.

Theorem 4.20 (Girsanov)**.** *Let* $\{W_t\}_{t\in[0,T]}$ *be a standard Brownian motion on* $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$*, and let* $\{\theta_t\}_{t\in[0,T]}$ *be an adapted process satisfying Novikov's condition* (4.28)*. Define* $\{L_t\}$ *by* (4.27) *and the measure* $\mathbb{Q}$ *on* $(\Omega, \mathcal{F}_T)$ *by* $\mathbb{Q}(A) = \int_A L_T \, d\mathbb{P}$*. Then:*

(i) $\mathbb{Q}$ *is a probability measure equivalent to* $\mathbb{P}$*, with* $d\mathbb{Q}/d\mathbb{P} = L_T$*;*

(ii) *the process defined by*

$$\widetilde{W}_t := W_t + \int_0^t \theta_s \, ds, \qquad t \in [0, T], \tag{4.29}$$

is a standard Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{Q})$*.*

Proof. Claim (i) is essentially proved in the preceding subsection. By Novikov's condition and Theorem 4.18, $\{L_t\}$ is a strictly positive $\mathbb{P}$-martingale with $L_0 = 1$, and therefore $\mathbb{E}_{\mathbb{P}}[L_T] = 1$. The map $A \mapsto \int_A L_T \, d\mathbb{P}$ then defines a probability measure on $(\Omega, \mathcal{F}_T)$, equivalent to $\mathbb{P}$ by the strict positivity of L_T, and with Radon-Nikodym derivative L_T by construction.

For claim (ii) we apply Lévy's characterization (Theorem 4.19) under the measure $\mathbb{Q}$. We must verify that $\widetilde{W}$ satisfies the hypotheses of the theorem: pathwise continuity, $\widetilde{W}_0 = 0$, the martingale property under $\mathbb{Q}$, and the property that $\{\widetilde{W}_t^2 - t\}$ is a $\mathbb{Q}$-martingale.

Step 1: continuity and initial value. The continuity of $\{\widetilde{W}_t\}$ is immediate: W_t is continuous almost surely (condition (iv) of Definition 3.1) and the integral $\int_0^t \theta_s \, ds$ is continuous in t by construction. The sum is therefore continuous. Likewise, $\widetilde{W}_0 = W_0 + 0 = 0$.

Step 2: $\{\widetilde{W}_t\}$ *is a* $\mathbb{Q}$*-martingale.* The strategy is the following: by a classical lemma that we now detail, a process $\{Y_t\}$ is a $\mathbb{Q}$-martingale if and only if $\{Y_t \, L_t\}$ is a $\mathbb{P}$-martingale. We shall apply this characterization to $Y_t = \widetilde{W}_t$.
Let us first prove the lemma. If $\{Y_t L_t\}$ is a $\mathbb{P}$-martingale, then for $0 \le s \le t$ and $A \in \mathcal{F}_s$,

$$\int_A Y_t \, L_t \, d\mathbb{P} = \int_A Y_s \, L_s \, d\mathbb{P}.$$

The left-hand side, by (4.26) applied with restriction to $\mathcal{F}_t$ (recall that L_t is the Radon-Nikodym derivative of $\mathbb{Q}\restriction_{\mathcal{F}_t}$ with respect to $\mathbb{P}\restriction_{\mathcal{F}_t}$), equals $\int_A Y_t \, d\mathbb{Q}$. Similarly, the right-hand side equals $\int_A Y_s \, d\mathbb{Q}$. By the uniqueness of conditional expectation, $\mathbb{E}_{\mathbb{Q}}[Y_t \mid \mathcal{F}_s] = Y_s$, that is, $\{Y_t\}$ is a $\mathbb{Q}$-martingale. The converse implication is obtained by reversing the argument.

Having established the lemma, it suffices to prove that $\{\widetilde{W}_t \, L_t\}$ is a $\mathbb{P}$-martingale. Let us compute its SDE under $\mathbb{P}$ by applying the product rule, which is a particular case of the general

Itô formula (Theorem 4.13) applied to $f(x,y) = xy$ with $X_t = \widetilde{W}_t$ and $Y_t = L_t$ as arguments. The differentials of the processes are

$$d\widetilde{W}_t = \theta_t\,dt + dW_t, \qquad dL_t = -\theta_t\,L_t\,dW_t,$$

where the first identity follows from the definition (4.29), and the second was established in (4.22). The product rule yields

$$d(\widetilde{W}_t\,L_t) = L_t\,d\widetilde{W}_t + \widetilde{W}_t\,dL_t + d\widetilde{W}_t\,dL_t,$$

where the last term is the formal product of the two differentials, evaluated by means of Itô's rules (4.18). Let us compute it:

$$d\widetilde{W}_t\,dL_t = (\theta_t\,dt + dW_t)\cdot(-\theta_t\,L_t\,dW_t) = -\theta_t^2\,L_t\,dt\,dW_t - \theta_t\,L_t\,(dW_t)^2.$$

By Itô's rules, $dt\,dW_t = 0$ and $(dW_t)^2 = dt$, so that $d\widetilde{W}_t\,dL_t = -\theta_t\,L_t\,dt$. Substituting,

$$\begin{aligned} d(\widetilde{W}_t\,L_t) &= L_t\,(\theta_t\,dt + dW_t) + \widetilde{W}_t\,(-\theta_t\,L_t\,dW_t) - \theta_t\,L_t\,dt \\ &= \underbrace{(L_t\,\theta_t - \theta_t\,L_t)}_{=0}\,dt + L_t\,dW_t - \widetilde{W}_t\,\theta_t\,L_t\,dW_t \\ &= L_t\,(1 - \theta_t\,\widetilde{W}_t)\,dW_t. \end{aligned}$$

The cancellation of the dt term is the crucial computation of the proof: it shows that $\{\widetilde{W}_t\,L_t\}$ is a driftless process, expressible as the stochastic integral of the process $L_t(1-\theta_t\,\widetilde{W}_t)$ with respect to W. By Theorem 4.10, assuming sufficient integrability conditions on θ, ultimately guaranteed by Novikov's condition, $\{\widetilde{W}_t\,L_t\}$ is a $\mathbb{P}$-martingale. By the preceding lemma, $\{\widetilde{W}_t\}$ is a $\mathbb{Q}$-martingale.

Step 3: $\{\widetilde{W}_t^2 - t\}$ is a $\mathbb{Q}$-martingale. By the characterization of Step 2, it suffices to prove that $\{(\widetilde{W}_t^2 - t)\,L_t\}$ is a $\mathbb{P}$-martingale. Applying Itô's formula to $f(x) = x^2$ with $X_t = \widetilde{W}_t$ and using $(d\widetilde{W}_t)^2 = dt$ by Itô's rules, we obtain $d(\widetilde{W}_t^2 - t) = 2\,\widetilde{W}_t\,\theta_t\,dt + 2\,\widetilde{W}_t\,dW_t$. Multiplying by L_t and applying the product rule exactly as in Step 2, the dt terms cancel for the same underlying reason (the correction $-\frac{1}{2}\int\theta_s^2\,ds$ in the exponent of L_t is calibrated to produce this cancellation), leaving

$$d\big((\widetilde{W}_t^2 - t)\,L_t\big) = (\text{terms in } dW_t),$$

with no dt term. The process $(\widetilde{W}_t^2 - t)\,L_t$ is therefore a $\mathbb{P}$-martingale under the appropriate integrability conditions, and by the equivalence lemma $\{\widetilde{W}_t^2 - t\}$ is a $\mathbb{Q}$-martingale.

Conclusion. The process $\{\widetilde{W}_t\}$ is continuous, satisfies $\widetilde{W}_0 = 0$, is a $\mathbb{Q}$-martingale, and $\{\widetilde{W}_t^2 - t\}$ is also a $\mathbb{Q}$-martingale. By Lévy's characterization (Theorem 4.19), $\widetilde{W}$ is a standard Brownian motion under $\mathbb{Q}$. □

Reading of the theorem. Girsanov's theorem is a result of great interest for the purposes of our work: the change of measure encoded by the exponential martingale $\{L_t\}$ *absorbs the additional drift θ_t of the process $W_t + \int_0^t \theta_s\,ds$ until it disappears.* Under the new measure $\mathbb{Q}$,

what was a process with drift θ_t becomes a pure Brownian motion. The operation is reversible: applying the theorem with $-\theta$ recovers W from $\widetilde{W}$.

Reformulation in terms of Itô processes. The practical use of the theorem in Chapter 5 will go through the following reformulation. Let $\{X_t\}$ be an Itô process with dynamics

$$dX_t = a_t\,dt + b_t\,dW_t,$$

and suppose $b_t > 0$ almost surely. Define $\theta_t := a_t/b_t$, and suppose it satisfies Novikov's condition. Applying Theorem 4.20 with this θ, under the resulting measure $\mathbb{Q}$ the process $\widetilde{W}$ defined in (4.29) is a Brownian motion. Substituting $dW_t = d\widetilde{W}_t - \theta_t\,dt$ into the SDE of X,

$$dX_t = a_t\,dt + b_t\,(d\widetilde{W}_t - \theta_t\,dt) = (a_t - b_t\,\theta_t)\,dt + b_t\,d\widetilde{W}_t.$$

Substituting $\theta_t = a_t/b_t$, the coefficient of dt vanishes, and

$$\boxed{dX_t = b_t\,d\widetilde{W}_t.} \tag{4.30}$$

Under the measure $\mathbb{Q}$, the process X loses its drift and becomes a pure stochastic integral with respect to the new Brownian motion $\widetilde{W}$. In particular, by Theorem 4.10, $\{X_t\}$ is a $\mathbb{Q}$-martingale.

4.5 The Brownian martingale representation theorem

This section presents the generalization to the continuous setting of the binomial representation theorem (Subsection 2.3.6). Unlike the discrete case, where the integrand was constructed explicitly by solving a linear system node by node, the continuous version relies on functional-analytic arguments and *does not provide an explicit formula*: it guarantees the existence of the integrand, not its construction.

Throughout the section we work on the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$, where $(\mathcal{F}_t)$ is the filtration of the Brownian motion $\{W_t\}$ fixed in Definition 3.9. It is essential to stress that the theorem requires working with exactly this filtration (generated by the Brownian motion together with the null sets): if the underlying filtration contained strictly additional information, extraneous to the Brownian motion, the result would cease to hold. This restriction will not be a problem in the applications of Chapter 5, where the only source of randomness will be precisely the Brownian motion modeling the underlying.

Theorem 4.21 (Representation of L^2 random variables)**.** *Let X be an $\mathcal{F}_T$-measurable random variable with $\mathbb{E}[X^2] < \infty$. There exists a unique process $\phi \in \mathcal{H}^2[0,T]$ such that*

$$X = \mathbb{E}[X] + \int_0^T \phi_s\,dW_s \qquad \textit{almost surely.} \tag{4.31}$$

The proof lies beyond the scope of the present work, but we can describe the strategy in detail; it is the same one we already employed when constructing the stochastic integral in Section 4.2.3: identify a sufficiently large subspace of $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ where the representation is direct, and extend it by density and isometry to the whole space.

Specifically, one considers the subspace

$$K \coloneqq \left\{ M_0 + \int_0^T \phi_s \, dW_s : M_0 \in \mathbb{R},\ \phi \in \mathcal{H}^2[0,T] \right\} \subseteq L^2(\Omega, \mathcal{F}_T, \mathbb{P}),$$

consisting precisely of the random variables that admit the desired representation. From here the proof rests on two facts.

The first is that K is a *closed* subspace of $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$. The verification is direct: if $(Y_n) \subset K$ converges in L^2 to some Y, the constants $\mathbb{E}[Y_n]$ converge to $\mathbb{E}[Y]$ by continuity of expectation, and the associated integrands $\phi^{(n)}$ form a Cauchy sequence in $\mathcal{H}^2[0,T]$ by the *Itô isometry* (Theorem 4.7); their limit ϕ provides the representation of Y, so $Y \in K$.

The second is that the *stochastic exponentials* of the form

$$\mathcal{E}(f) \coloneqq \exp\!\left(\int_0^T f(s)\, dW_s - \tfrac{1}{2} \int_0^T f(s)^2 \, ds \right),$$

with $f : [0,T] \to \mathbb{R}$ deterministic and square-integrable, belong to K and, moreover, span a dense subspace of $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$. Membership in K is a direct computation with Itô's formula applied to the SDE $d\mathcal{E}(f)_t = \mathcal{E}(f)_t \, f(t)\, dW_t$, which in integral form gives $\mathcal{E}(f) = 1 + \int_0^T \mathcal{E}(f)_s \, f(s)\, dW_s$. Density, on the other hand, is the truly delicate part of the argument: it is proved by showing that no $Z \in L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ can be orthogonal to all the exponentials, which reduces, via an analytic change of parameter, to an injectivity argument for the Fourier transform of the finite-dimensional Gaussian distributions of Brownian motion.

Combining both facts, K is a closed subspace containing a dense subset, and therefore coincides with all of $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$. The uniqueness of the integrand ϕ follows, once again, from the Itô isometry. For a detailed exposition we refer to [3], Section 4.6.

Version for martingales. Theorem 4.21 admits the following general formulation, which is the one we shall work with in Chapter 5.

Theorem 4.22 (Brownian martingale representation)**.** *Let $\{M_t\}_{t \in [0,T]}$ be a martingale with respect to the Brownian filtration $(\mathcal{F}_t)$, with $M_t \in L^2$ for all $t \in [0,T]$. There exists a unique process $\phi \in \mathcal{H}^2[0,T]$ such that*

$$M_t = M_0 + \int_0^t \phi_s \, dW_s, \qquad t \in [0,T], \quad \textit{almost surely.} \tag{4.32}$$

Proof. Applying Theorem 4.21 to the variable M_T, there exists a unique $\phi \in \mathcal{H}^2$ with

$$M_T = M_0 + \int_0^T \phi_s \, dW_s$$

almost surely, where we have used $\mathbb{E}[M_T] = M_0$ by the martingale property. To extend the identity to $t < T$, we take conditional expectations with respect to $\mathcal{F}_t$ on both sides. The left-hand side gives $\mathbb{E}[M_T \mid \mathcal{F}_t] = M_t$ by the martingale property of M. The right-hand side gives $M_0 + \mathbb{E}[\int_0^T \phi_s \, dW_s \mid \mathcal{F}_t] = M_0 + \int_0^t \phi_s \, dW_s$, since the indefinite integral up to T is a martingale (Theorem 4.10) and therefore its conditional expectation given $\mathcal{F}_t$ is the integral up to t. Identity (4.32) is proved. Uniqueness follows from the uniqueness in Theorem 4.21. $\square$

Financial reading of the integrand. The financial interpretation of the integrand ϕ in Theorem 4.22 is entirely analogous to that of the discrete case. If $\{M_t\}$ is the martingale associated with the discounted value of a derivative, constructed in Chapter 5 as $M_t = \mathbb{E}_{\mathbb{Q}}[e^{-rT} C_T \mid \mathcal{F}_t]$ (following the same pattern as in (2.25)), then the process ϕ_t appearing in its representation is the *quantity of the underlying* that the replicating portfolio must hold at time t. It is the generalization to the continuous setting of the explicit formula (2.22) of the binomial model, with the caveat already noted: in the continuous case ϕ exists but is not constructed explicitly.

4.6 Synthesis: from the binomial model to stochastic calculus

We close the chapter by establishing the correspondence between the tools of the binomial model and those of continuous stochastic calculus.

The correspondence between the two settings. Each of the four pieces of continuous stochastic calculus has its counterpart in the discrete case. The *Itô stochastic integral*, $\int_0^t \phi_s \, dW_s$, is the continuous version of the discrete stochastic integral $(\phi \cdot \widetilde{S})_n$ of (2.20): both express the accumulated gain of a predictable strategy ϕ, and the predictability condition, formulated in discrete time as $\mathcal{F}_{k-1}$-measurability and in continuous time as measurability with respect to $\mathcal{F}_{t_i}$ (on simple processes), is what makes the integral preserve the martingale property (Theorems 2.22 and 4.10). *Itô's formula* is the continuous version of the classical chain rule, with the novelty of the extra term $\frac{1}{2} b_t^2 \, \partial_{xx} f \, dt$ in (4.17), which has no counterpart in the discrete case and reflects the nonzero quadratic variation of Brownian motion.

Girsanov's theorem is the continuous version of the explicit construction of the measure $\mathbb{Q}$ in the binomial model (2.19). In both cases, the change of measure is governed by a density $L = d\mathbb{Q}/d\mathbb{P}$. The difference is that in discrete time L is constructed combinatorially, assigning to each path a probability $q^j(1-q)^{N-j}$, whereas in continuous time L_T is the exponential martingale $\exp(-\int \theta \, dW - \frac{1}{2}\int \theta^2 \, ds)$. The *Brownian martingale representation theorem* (Theorem 4.22) is the continuous version of the binomial representation theorem (Theorem 2.23). In the discrete case, the proof was constructive and we have an explicit formula for the integrand; in the continuous case, by contrast, the proof appeals to abstract density and completeness arguments and provides no explicit expression: knowing how to construct it will be, in each specific problem, an additional question to be settled by means of Itô's formula.

The three-step recipe, continuous version. The four pieces above are the ingredients with which we shall obtain the solution of the pricing problem in the continuous setting. The strategy, faithful to the scheme of the discrete binomial model (Subsection 2.4.1), will follow the same three steps:

Step 1. *Change of measure.* Identify the measure $\mathbb{Q}$ under which the discounted price of the underlying is a martingale. This identification, carried out in the discrete case by explicit construction, will be carried out in the continuous case by applying *Girsanov's theorem.*

Step 2. *Construction of the discounted value process.* Define the process $\{M_t\}$ associated with the discounted *payoff* of the derivative via $M_t = \mathbb{E}_{\mathbb{Q}}[e^{-rT} C_T \mid \mathcal{F}_t]$. By the tower

property, $\{M_t\}$ is a $\mathbb{Q}$-martingale. This step is identical to that of the discrete case.

Step 3. *Application of the representation theorem.* Apply the *Brownian martingale representation theorem* to obtain a process ϕ such that $M_t = M_0 + \int_0^t \phi_s \, d\widetilde{W}_s$, where $\widetilde{W}$ is the Brownian motion under $\mathbb{Q}$ produced by Girsanov. The integrand ϕ encodes the quantity of the underlying in the replicating portfolio.

The structure is the same as in the discrete case: only the technical sophistication of the tools changes. Chapter 5 applies this recipe to geometric Brownian motion as the underlying and obtains, as its central result, the Black-Scholes formula for the European *call* option.

Chapter 5

The Black-Scholes model

We reach the ultimate goal of this work: to apply the apparatus of stochastic calculus to the *Black-Scholes model* and obtain, as its central result, the classical pricing formula for the European *call* option. The structure of the chapter follows the same pattern as Chapter 2: model assumptions, execution of the three-step recipe, specialization to the *call*, and recovery of the classical PDE.

5.1 The Black-Scholes model

5.1.1 Market assumptions and dynamics of the underlying

The model assumptions. The Black-Scholes model describes an idealized market built on the following structural assumptions:

(H1) The market consists of two assets: a risky *underlying* whose price is a stochastic process $\{S_t\}_{t\in[0,T]}$, and a *risk-free bond* with constant interest rate $r > 0$.

(H2) The underlying follows a geometric Brownian motion (GBM): its dynamics are governed by the stochastic differential equation

$$dS_t = \mu\, S_t\, dt + \sigma\, S_t\, dW_t, \qquad S_0 > 0, \tag{5.1}$$

where $\mu \in \mathbb{R}$ is the expected return and $\sigma > 0$ is the *volatility*, both constant, and $\{W_t\}$ is a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ with $(\mathcal{F}_t)$ the augmented Brownian filtration.

(H3) The market is *frictionless*: there are no transaction costs, assets are infinitely divisible, short selling is allowed, and lending and borrowing take place at the same rate r.

(H4) The market is *arbitrage-free*: there is no strategy which, starting from zero capital, produces with strictly positive probability a strictly positive terminal value and never a negative one.

Assumptions (H1) and (H3)–(H4) are the literal translation to the continuous setting of the assumptions of the binomial model of Chapter 2: the market structure, the idealization and the no-arbitrage condition remain intact. The novelty is (H2): the specific choice of the dynamics of

the underlying, which in the discrete case was the multiplicative binomial random walk and in the continuous case is the GBM.

Explicit form of the underlying. The SDE (5.1) admits a closed-form solution whose expression we already obtained as Example 2 of Section 4.3.2: the GBM with SDE (5.1) is exactly

$$S_t = S_0 \exp\Big(\Big(\mu - \tfrac{1}{2}\sigma^2\Big)t + \sigma\, W_t\Big), \qquad t \in [0,T]. \tag{5.2}$$

Identity (5.2) shows that S_t is strictly positive almost surely (an economically required property for the price of an asset), and that its logarithm $\log S_t$ has Gaussian distribution $\mathcal{N}\big(\log S_0 + (\mu - \frac{1}{2}\sigma^2)t,\, \sigma^2 t\big)$. This log-normality property will be the key ingredient in the explicit computation of the Black-Scholes formula in Section 5.3.

The risk-free bond. With constant interest rate r, the price of the risk-free bond evolves deterministically according to

$$B_t = e^{rt}, \qquad t \in [0,T], \tag{5.3}$$

which is the solution of the ordinary differential equation $dB_t = r\, B_t\, dt$ with $B_0 = 1$, with no stochastic term (the bond carries no risk).

5.1.2 Portfolios in continuous time

Definition 5.1 (Portfolio in continuous time)**.** A *portfolio* (or *strategy*) on the horizon $[0,T]$ is a pair of stochastic processes $(\phi_t, \psi_t)_{t\in[0,T]}$ where:

(i) ϕ_t represents the quantity of the underlying in the portfolio at time t;

(ii) ψ_t represents the quantity of the risk-free bond in the portfolio at time t;

(iii) both processes are adapted to the Brownian filtration $(\mathcal{F}_t)$ and satisfy the technical integrability conditions

$$\phi \in \mathcal{H}^2[0,T], \qquad \int_0^T |\psi_s|\, ds < \infty \text{ almost surely.}$$

The *value* of the portfolio at time t is denoted V_t and is given by

$$V_t \coloneqq \phi_t\, S_t + \psi_t\, B_t. \tag{5.4}$$

The integrability conditions on (ϕ, ψ) are those guaranteeing that the integrals that will appear next, the stochastic integral $\int \phi\, dS$ and the ordinary integral $\int \psi\, dB$, are well defined in the sense of Chapter 4. The adaptedness of ϕ and ψ to $(\mathcal{F}_t)$ is the translation to the continuous setting of discrete predictability (Section 2.3.5): the portfolio is decided with the information available before the market moves.

5.1.3 The *self-financing* condition

The most delicate structural piece of the continuous model is the translation to the continuous setting of the *self-financing* condition introduced in the discrete case in Section 2.4.1. Let us

recall its content: a binomial strategy was *self-financing* if at each node of the tree

$$\phi_{k+1}\, S_k + \psi_{k+1}\, B_k = \phi_k\, S_k + \psi_k\, B_k, \tag{5.5}$$

that is, if the amount of capital available at the beginning of period $k+1$, after reallocating the portfolio, coincides with that available at the end of period k. Rearranging terms, (5.5) is equivalent to

$$S_k\,(\phi_{k+1} - \phi_k) + B_k\,(\psi_{k+1} - \psi_k) = 0,$$

and consequently, summing over the periods up to time n,

$$V_n - V_0 = \sum_{k=0}^{n-1} \big[\phi_{k+1}\,(S_{k+1} - S_k) + \psi_{k+1}\,(B_{k+1} - B_k)\big].$$

The value of the portfolio changes exclusively due to market movements, not to injections or withdrawals of capital.

Definition 5.2 (*Self-financing* strategy)**.** A strategy $(\phi_t, \psi_t)_{t\in[0,T]}$ is said to be *self-financing* if its value $V_t = \phi_t\, S_t + \psi_t\, B_t$ satisfies

$$dV_t = \phi_t\, dS_t + \psi_t\, dB_t, \qquad t \in [0,T], \tag{5.6}$$

or, in integral form,

$$V_t = V_0 + \int_0^t \phi_s\, dS_s + \int_0^t \psi_s\, dB_s, \qquad t \in [0,T]. \tag{5.7}$$

Identity (5.6) is the exact translation to the continuous setting of the discrete condition: it states that the change in the value of the portfolio at time t comes exclusively from changes in the prices of the underlying and of the bond, weighted respectively by the quantities ϕ_t and ψ_t held at each instant.

Equivalence with the primitive formulation. Definition 5.2 is the operational formulation of the self-financing condition. It is worth verifying its equivalence with the original formulation, which requires zero cost in the infinitesimal reallocation of the portfolio.

Proposition 5.3 (Equivalence of the two formulations)**.** *Let (ϕ_t, ψ_t) be a strategy with $V_t = \phi_t\, S_t + \psi_t\, B_t$. The following conditions are equivalent:*

(i) $dV_t = \phi_t\, dS_t + \psi_t\, dB_t$;

(ii) $S_t\, d\phi_t + B_t\, d\psi_t + d\langle \phi, S\rangle_t = 0$,

where $\langle \phi, S\rangle_t$ denotes the quadratic covariation of the processes ϕ and S.

Proof. We apply the product rule (a particular case of Itô's formula, Theorem 4.13) to the product $\phi_t\, S_t$:

$$d(\phi_t\, S_t) = \phi_t\, dS_t + S_t\, d\phi_t + d\langle \phi, S\rangle_t.$$

Similarly, for $\psi_t\, B_t$, where B_t has bounded variation, so that $\langle \psi, B\rangle_t = 0$,

$$d(\psi_t\, B_t) = \psi_t\, dB_t + B_t\, d\psi_t.$$

Adding both identities, we obtain the differential of the value:

$$dV_t = d(\phi_t S_t + \psi_t B_t) = \phi_t\, dS_t + \psi_t\, dB_t + \big[S_t\, d\phi_t + B_t\, d\psi_t + d\langle \phi, S\rangle_t\big].$$

Condition (i) is therefore equivalent to the vanishing of the term in brackets, which is exactly condition (ii). □

5.2 Application of the three-step recipe

In this section we execute the three-step recipe set out in Subsection 2.4.1 for the discrete case, now in the continuous setting.

5.2.1 Statement of the problem

European derivatives.

Definition 5.4 (European derivative)**.** A *European derivative* with maturity T on the underlying $\{S_t\}$ is a contract that pays the holder, at time $t = T$, an amount $C_T = h(S_T)$, where $h : (0, \infty) \to \mathbb{R}$ is a measurable function, the *payoff* of the derivative, satisfying the integrability condition

$$\mathbb{E}_{\mathbb{P}}\big[h(S_T)^2\big] < \infty. \tag{5.8}$$

Condition (5.8) is the minimum required for the processes built from the *payoff* to lie in the L^2 spaces demanded by the Brownian martingale representation theorem. For standard *payoffs*, including the European *call* $h(S_T) = (S_T - K)^+$ that we address in Section 5.3, the condition holds automatically by the log-normality of S_T.

The pricing problem. The central problem is to determine, at each time $t \in [0, T]$, a fair price $V(t, S_t)$ for the derivative, that is, a price compatible with the no-arbitrage assumption (H4). The procedure to follow, as in the discrete scheme, goes through the notion of *replicating strategy*:

Definition 5.5 (Replicating strategy)**.** A *replicating strategy* for the European derivative $h(S_T)$ is a *self-financing* strategy $(\phi_t, \psi_t)_{t\in[0,T]}$ whose terminal value coincides with the *payoff*, that is,

$$V_T = \phi_T\, S_T + \psi_T\, B_T = h(S_T) \qquad \text{almost surely.}$$

By the no-arbitrage argument of the discrete case, if the replicating strategy exists, its value V_t at each time is necessarily the price of the derivative at t. The problem therefore reduces to constructing a replicating strategy.

5.2.2 Step 1: change of measure via Girsanov

We replace the market measure $\mathbb{P}$ by an equivalent measure $\mathbb{Q}$ under which the discounted price of the underlying is a martingale. The identification is carried out via Girsanov's theorem.[1]

The dynamics of the discounted price. To set up the change of measure, it is convenient first to compute the SDE of the discounted price process

$$\widetilde{S}_t := e^{-rt}\, S_t, \qquad t \in [0, T]. \tag{5.9}$$

We apply Itô's formula (Theorem 4.13) to the function $f(t, x) = e^{-rt}\, x$ and the process $X_t = S_t$. The relevant derivatives are

$$\partial_t f = -r\, e^{-rt}\, x, \qquad \partial_x f = e^{-rt}, \qquad \partial_{xx} f = 0.$$

With $a_t = \mu\, S_t$, $b_t = \sigma\, S_t$ the coefficients of the SDE (5.1), Itô's formula gives

$$\begin{aligned} d\widetilde{S}_t &= \left[-r\, e^{-rt}\, S_t + e^{-rt}\, \mu\, S_t + 0\right] dt + e^{-rt}\, \sigma\, S_t\, dW_t \\ &= (\mu - r)\, e^{-rt}\, S_t\, dt + \sigma\, e^{-rt}\, S_t\, dW_t. \end{aligned}$$

Since $e^{-rt} S_t = \widetilde{S}_t$, we obtain

$$\boxed{d\widetilde{S}_t = (\mu - r)\, \widetilde{S}_t\, dt + \sigma\, \widetilde{S}_t\, dW_t.} \tag{5.10}$$

The SDE (5.10) shows that the discounted price is a new Itô process with drift coefficient $(\mu - r)\, \widetilde{S}_t$ (proportional to the excess return of the underlying over the risk-free rate) and, therefore, $\{\widetilde{S}_t\}$ is not a martingale under $\mathbb{P}$. The goal of the change of measure is to remove that drift.

The choice of θ. The SDE (5.10) fits exactly into the scheme of Girsanov's theorem reformulated in terms of Itô processes, eq. (4.30): we have a process with dynamics $dX_t = a_t\, dt + b_t\, dW_t$ and we want to remove the drift a_t by means of the choice

$$\theta_t := \frac{a_t}{b_t} = \frac{(\mu - r)\, \widetilde{S}_t}{\sigma\, \widetilde{S}_t} = \frac{\mu - r}{\sigma}.$$

The quantity

$$\theta := \frac{\mu - r}{\sigma}, \tag{5.11}$$

constant in the Black-Scholes model, goes by the classical name of *market price of risk*: it is exactly the quantity that Girsanov's theorem requires in order to cancel the drift of the discounted price.

Verification of Novikov's condition. Since θ is constant, Novikov's condition (4.28) holds trivially:

$$\mathbb{E}_{\mathbb{P}}\left[\exp\left(\tfrac{1}{2}\int_0^T \theta^2\, ds\right)\right] = \exp\left(\tfrac{1}{2}\theta^2\, T\right) < \infty.$$

[1] Here reappears, in its continuous form, the idea anticipated in Remark 4.16: passing to an equivalent measure (the risk-neutral one) to simplify the problem. What in the binomial model was an explicit reweighting of the probabilities on the paths of the tree is now the change of measure established by Girsanov's theorem.

We are therefore in a position to apply Theorem 4.20 with $\theta_t \equiv \theta$.

Application of Girsanov's theorem. The density process associated with θ is, according to (4.27),

$$L_T = \exp\Big(-\theta\, W_T - \tfrac{1}{2}\,\theta^2\, T\Big). \tag{5.12}$$

We define the measure $\mathbb{Q}$ on $(\Omega, \mathcal{F}_T)$ by

$$\mathbb{Q}(A) \coloneqq \int_A L_T\, d\mathbb{P}, \qquad A \in \mathcal{F}_T.$$

By Theorem 4.20, $\mathbb{Q}$ is a probability measure equivalent to $\mathbb{P}$, with $d\mathbb{Q}/d\mathbb{P} = L_T$, and the process

$$\widetilde{W}_t \coloneqq W_t + \theta\, t, \qquad t \in [0, T], \tag{5.13}$$

is a standard Brownian motion under $\mathbb{Q}$.

Characterization of $\mathbb{Q}$ as a martingale measure. Substituting $dW_t = d\widetilde{W}_t - \theta\, dt$ into the SDE (5.10) of the discounted price:

$$d\widetilde{S}_t = (\mu - r)\, \widetilde{S}_t\, dt + \sigma\, \widetilde{S}_t\, (d\widetilde{W}_t - \theta\, dt).$$

Recalling $\theta = (\mu - r)/\sigma$, the dt terms cancel:

$$\boxed{d\widetilde{S}_t = \sigma\, \widetilde{S}_t\, d\widetilde{W}_t.} \tag{5.14}$$

Identity (5.14) shows that, under $\mathbb{Q}$, the discounted price of the underlying solves a driftless SDE: it is a pure stochastic integral with respect to the Brownian motion $\widetilde{W}$. Thus, by Theorem 4.10, $\{\widetilde{S}_t\}$ is a $\mathbb{Q}$-martingale. The measure $\mathbb{Q}$ is accordingly called the *equivalent martingale measure* or *risk-neutral measure*.

The dynamics of the underlying under $\mathbb{Q}$. It is also worth making explicit the dynamics of the underlying S_t itself (undiscounted) under the new measure. Substituting $dW_t = d\widetilde{W}_t - \theta\, dt$ into (5.1),

$$\boxed{dS_t = r\, S_t\, dt + \sigma\, S_t\, d\widetilde{W}_t.} \tag{5.15}$$

Under $\mathbb{Q}$, the underlying is still a GBM, but with *trend r instead of μ*. This is the translation to the continuous setting of the phenomenon we already saw in the binomial model: the change of measure modifies the dynamics of the underlying so that its expected return coincides with that of the risk-free bond, whence the name *risk-neutral*.
The explicit form of the underlying under $\mathbb{Q}$, analogous to (5.2), is obtained by solving the SDE (5.15) by the same technique as in Example 2 of Section 4.3.2:

$$S_t = S_0\, \exp\Big(\Big(r - \tfrac{1}{2}\sigma^2\Big)t + \sigma\, \widetilde{W}_t\Big)\,. \tag{5.16}$$

Identity (5.16) will be the key tool for the explicit computations of Section 5.3. Under $\mathbb{Q}$, $\log S_T$ has Gaussian distribution $\mathcal{N}\big(\log S_0 + (r - \tfrac{1}{2}\sigma^2)T,\, \sigma^2 T\big)$.

5.2.3 Step 2: the $\mathbb{Q}$-martingale associated with the discounted *payoff*

Having identified the risk-neutral measure $\mathbb{Q}$, the second step of the recipe, in analogy with the discrete case (Subsection 2.4.1), consists in constructing a $\mathbb{Q}$-martingale process associated with the *payoff*.

Definition of the process. Let $h(S_T)$ be the *payoff* of the derivative. We define the process $\{M_t\}_{t\in[0,T]}$ by

$$M_t := \mathbb{E}_{\mathbb{Q}}\big[e^{-rT}\, h(S_T) \,\big|\, \mathcal{F}_t\big], \qquad t \in [0,T]. \tag{5.17}$$

The financial interpretation of M_t is clear: it represents the best estimate, conditional on the information available at time t, of the discounted value of the terminal *payoff*, computed under the risk-neutral measure $\mathbb{Q}$.

Well-definedness and square integrability. For the *payoffs* of standard derivatives, including the European *call* $(S_T - K)^+$, the condition $h(S_T) \in L^2(\mathbb{Q})$ holds thanks to the log-normality of the underlying under $\mathbb{Q}$ (see (5.16)) and the equivalence between $\mathbb{P}$ and $\mathbb{Q}$ (see [5], Section 5.8, for the details). Consequently, $M_t \in L^2(\mathbb{Q})$ for all t, a necessary condition for applying the Brownian martingale representation theorem in the next step.

Martingale property.

Proposition 5.6 ($\{M_t\}$ is a $\mathbb{Q}$-martingale)**.** *The process $\{M_t\}$ defined in* (5.17) *is a $\mathbb{Q}$-martingale with respect to the filtration $(\mathcal{F}_t)$.*

Proof. The adaptedness of M_t to $\mathcal{F}_t$ is immediate from the very definition of conditional expectation. Square integrability follows from $h(S_T) \in L^2(\mathbb{Q})$ and Jensen's inequality applied to conditional expectation: $\mathbb{E}_{\mathbb{Q}}[M_t^2] \le \mathbb{E}_{\mathbb{Q}}\big[\mathbb{E}_{\mathbb{Q}}[(e^{-rT}h(S_T))^2 \mid \mathcal{F}_t]\big] = \mathbb{E}_{\mathbb{Q}}[(e^{-rT}h(S_T))^2] < \infty$.

For the martingale property, let $0 \le s \le t \le T$. Applying the tower property (Proposition 2.17, property (iv)) with $\mathcal{F}_s \subseteq \mathcal{F}_t$,

$$\mathbb{E}_{\mathbb{Q}}[M_t \mid \mathcal{F}_s] = \mathbb{E}_{\mathbb{Q}}\Big[\,\mathbb{E}_{\mathbb{Q}}\big[e^{-rT}\, h(S_T) \,\big|\, \mathcal{F}_t\big] \,\Big|\, \mathcal{F}_s\Big] = \mathbb{E}_{\mathbb{Q}}\big[e^{-rT}\, h(S_T) \,\big|\, \mathcal{F}_s\big] = M_s.$$

□

5.2.4 Step 3: representation, replication and *self-financing*

Application of the representation theorem. The $\mathbb{Q}$-martingale $\{M_t\}$ is adapted to the filtration of the Brownian motion $\widetilde{W}$ since, by (5.13), the filtration $(\mathcal{F}_t)$ generated by W and augmented with the null sets is the same as the one generated by $\widetilde{W}$; moreover, it satisfies $M_t \in L^2(\mathbb{Q})$ for all t by Proposition 5.6. We are therefore in a position to apply Theorem 4.22 under the measure $\mathbb{Q}$ and with respect to the Brownian motion $\widetilde{W}$: there exists a unique process $\widetilde{\phi} \in \mathcal{H}^2[0,T]$ (where $\mathcal{H}^2$ is now understood with respect to $\mathbb{Q}$) such that

$$M_t = M_0 + \int_0^t \widetilde{\phi}_s \, d\widetilde{W}_s, \qquad t \in [0,T]. \tag{5.18}$$

Identification of the portfolio. The integrand $\widetilde{\phi}_t$ provided by (5.18) is the piece that allows us to construct the replicating strategy. The idea is the following: we want to define (ϕ_t, ψ_t) so

that the discounted value of the portfolio, $\widetilde{V}_t := e^{-rt} V_t$, coincides with M_t at every time. To this end, observe that if such an identity holds, then by (5.18) the differential of $\widetilde{V}_t$ would be $d\widetilde{V}_t = \widetilde{\phi}_t \, d\widetilde{W}_t$. On the other hand, if the strategy is *self-financing* and $\widetilde{V}_t = e^{-rt} V_t$, a direct computation (which we shall make precise below) gives

$$d\widetilde{V}_t = \phi_t \, d\widetilde{S}_t = \phi_t \, \sigma \, \widetilde{S}_t \, d\widetilde{W}_t,$$

where the last equality uses (5.14). Comparing both expressions for $d\widetilde{V}_t$, we identify $\widetilde{\phi}_t = \phi_t \, \sigma \, \widetilde{S}_t$, that is,

$$\boxed{\phi_t := \frac{\widetilde{\phi}_t}{\sigma \, \widetilde{S}_t} = \frac{\widetilde{\phi}_t \, e^{rt}}{\sigma \, S_t}.} \tag{5.19}$$

The bond holding ψ_t is then defined by forcing the value of the portfolio to equal the discounted value given by the representation,

$$\boxed{\psi_t := \frac{V_t - \phi_t \, S_t}{B_t}, \qquad \text{with } V_t := e^{rt} \, M_t.} \tag{5.20}$$

Thus $V_t = \phi_t \, S_t + \psi_t \, B_t$ by construction and, in particular, $V_T = e^{rT} \, M_T = e^{rT} \, e^{-rT} \, h(S_T) = h(S_T)$: the strategy would therefore satisfy the replication condition (Definition 5.5). Consequently, it suffices to verify that (ϕ_t, ψ_t) is *self-financing* in the sense of Definition 5.2.

Verification of the *self-financing* condition.

Theorem 5.7 (The strategy is *self-financing*)**.** *The strategy (ϕ_t, ψ_t) defined by* (5.19) *and* (5.20) *is* self-financing*, that is, it satisfies*

$$dV_t = \phi_t \, dS_t + \psi_t \, dB_t.$$

Proof. The proof consists in computing both sides of the identity and checking that they coincide.

Step 1: computation of dV_t. By construction, $V_t = e^{rt} \, M_t$. We apply Itô's formula (Theorem 4.13) to the function $g(t, m) = e^{rt} \, m$ and the process $X_t = M_t$. The derivatives are $\partial_t g = r \, e^{rt} \, m$, $\partial_m g = e^{rt}$ and $\partial_{mm} g = 0$. Since $\{M_t\}$ has differential (5.18), with zero drift and diffusion coefficient $\widetilde{\phi}_t$ with respect to $\widetilde{W}$, the formula gives

$$dV_t = r \, e^{rt} \, M_t \, dt + e^{rt} \, \widetilde{\phi}_t \, d\widetilde{W}_t = r \, V_t \, dt + e^{rt} \, \widetilde{\phi}_t \, d\widetilde{W}_t. \tag{5.21}$$

Step 2: computation of $\phi_t \, dS_t + \psi_t \, dB_t$. For $\phi_t \, dS_t$, we substitute (5.15) into the definition (5.19):

$$\begin{aligned} \phi_t \, dS_t &= \frac{\widetilde{\phi}_t \, e^{rt}}{\sigma \, S_t} \left(r \, S_t \, dt + \sigma \, S_t \, d\widetilde{W}_t \right) \\ &= \frac{r \, \widetilde{\phi}_t \, e^{rt}}{\sigma} \, dt + \widetilde{\phi}_t \, e^{rt} \, d\widetilde{W}_t. \end{aligned}$$

For $\psi_t \, dB_t$, using $dB_t = r \, B_t \, dt$ and the definition (5.20) of ψ_t:

$$\psi_t \, dB_t = r \, \psi_t \, B_t \, dt = r \, (V_t - \phi_t \, S_t) \, dt.$$

Adding both contributions,

$$\phi_t\, dS_t + \psi_t\, dB_t = \frac{r\,\widetilde{\phi}_t\, e^{rt}}{\sigma}\, dt + \widetilde{\phi}_t\, e^{rt}\, d\widetilde{W}_t + r\, V_t\, dt - r\,\phi_t\, S_t\, dt.$$

Substituting $\phi_t\, S_t = \widetilde{\phi}_t\, e^{rt}/\sigma$ (solving from (5.19)), the terms $\frac{r\,\widetilde{\phi}_t\, e^{rt}}{\sigma}\, dt$ and $-r\,\phi_t\, S_t\, dt$ cancel, leaving

$$\phi_t\, dS_t + \psi_t\, dB_t = r\, V_t\, dt + \widetilde{\phi}_t\, e^{rt}\, d\widetilde{W}_t. \tag{5.22}$$

Step 3: comparison. The expressions (5.21) and (5.22) coincide term by term. The strategy (ϕ_t, ψ_t) is therefore *self-financing*. □

The central result. Combining the three steps, we are in a position to state the main result of the section.

Theorem 5.8 (Pricing of European derivatives in the Black-Scholes model)**.** *Let $h(S_T)$ be a European* payoff *satisfying the integrability condition* (5.8)*. Then there exists a unique* self-financing *replicating strategy (ϕ_t, ψ_t) for the derivative, with portfolio value at each time $t \in [0, T]$ given by*

$$\boxed{V(t, S_t) = \mathbb{E}_{\mathbb{Q}}\big[e^{-r(T-t)}\, h(S_T) \,\big|\, \mathcal{F}_t\big].} \tag{5.23}$$

By the no-arbitrage argument, $V(t, S_t)$ is the fair price of the derivative at time t.

Proof. Existence follows from the construction carried out in Steps 1, 2 and 3, together with the *self-financing* verification of Theorem 5.7. Formula (5.23) is obtained from $V_t = e^{rt}\, M_t = e^{rt}\, \mathbb{E}_{\mathbb{Q}}[e^{-rT}\, h(S_T) \mid \mathcal{F}_t] = \mathbb{E}_{\mathbb{Q}}[e^{-r(T-t)}\, h(S_T) \mid \mathcal{F}_t]$, using the definition (5.17) of M_t and the $\mathcal{F}_t$-measurability of the factor e^{rt}.

Uniqueness follows from the uniqueness of the integrand $\widetilde{\phi}$ in the representation theorem (Theorem 4.22): if two *self-financing* replicating strategies produced the same terminal value $h(S_T)$, their respective martingales $\mathbb{E}_{\mathbb{Q}}[e^{-rT}\, h(S_T) \mid \mathcal{F}_t]$ would coincide, their representation would be unique, and by (5.19)–(5.20) the two strategies would necessarily be equal. □

Reading of the result. Formula (5.23) is the continuous-time translation of Theorem 2.26: the price at each time is the conditional expectation, under the risk-neutral measure, of the discounted *payoff*.

We have thus solved the general pricing problem. Formula (5.23) expresses, for any European *payoff* h, the price of the derivative as a conditional expectation under $\mathbb{Q}$. The next section specializes this formula to the case of the European *call* option, $h(S_T) = (S_T - K)^+$, and obtains by direct computation the classical Black-Scholes formula.

5.3 The Black-Scholes formula for the European *call* option

We now specialize the formula of Theorem 5.8 to the *payoff* of the European *call*, $h(S_T) = (S_T - K)^+$, thereby obtaining the classical Black-Scholes formula (1973) with its coefficients d_1 and d_2.

The European *call* option. Let us briefly recall the contract. The European *call* option with *strike* $K > 0$ and maturity T is the European derivative whose *payoff* at T is

$$h(S_T) = (S_T - K)^+ \coloneqq \max(S_T - K, 0). \tag{5.24}$$

Its holder acquires the right (but not the obligation) to buy the underlying at the price K at T: if at that time $S_T > K$ they exercise the right and obtain a profit $S_T - K$; if $S_T \le K$ they do not exercise it and the *payoff* is zero. The integrability condition (5.8) holds automatically for this *payoff* since $0 \le (S_T - K)^+ \le S_T$ and the moments of the GBM are finite of all orders, so that Theorem 5.8 applies.

Strategy of the computation. We shall first compute the price of the *call* at $t = 0$, that is, $V(0, S_0) = \mathbb{E}_{\mathbb{Q}}[e^{-rT}\,(S_T - K)^+]$, and at the end obtain the generalization to the case $t > 0$ by an argument based on the Markov property of $\{S_t\}$. For $t = 0$, the strategy consists in decomposing the expectation by means of the exercise indicator function:

$$\mathbb{E}_{\mathbb{Q}}\big[(S_T - K)^+\big] = \mathbb{E}_{\mathbb{Q}}\big[(S_T - K)\,\mathbf{1}_{\{S_T > K\}}\big] = \underbrace{\mathbb{E}_{\mathbb{Q}}\big[S_T\,\mathbf{1}_{\{S_T > K\}}\big]}_{\text{(I)}} - K\,\underbrace{\mathbb{Q}(S_T > K)}_{\text{(II)}}. \tag{5.25}$$

Expression of S_T under $\mathbb{Q}$. By (5.16) with $t = T$,

$$S_T = S_0\,\exp\!\Big(\Big(r - \tfrac{1}{2}\sigma^2\Big)T + \sigma\,\widetilde{W}_T\Big)\,, \tag{5.26}$$

where $\widetilde{W}_T \sim \mathcal{N}(0, T)$ under $\mathbb{Q}$. Rewriting $\widetilde{W}_T = \sqrt{T}\,Z$ with $Z \sim \mathcal{N}(0,1)$ under $\mathbb{Q}$,

$$S_T = S_0\,\exp\!\Big(\Big(r - \tfrac{1}{2}\sigma^2\Big)T + \sigma\,\sqrt{T}\,Z\Big)\,.$$

The inequality $S_T > K$ is equivalent, taking logarithms, to

$$\log\!\Big(\frac{S_0}{K}\Big) + \Big(r - \tfrac{1}{2}\sigma^2\Big)T + \sigma\sqrt{T}\,Z > 0,$$

that is, $Z > -d_2$, where

$$d_2 \coloneqq \frac{\log(S_0/K) + \big(r - \tfrac{1}{2}\sigma^2\big)T}{\sigma\,\sqrt{T}}. \tag{5.27}$$

Computation of the term (II)**.** By the symmetry of the Gaussian distribution, (II) $= \mathbb{Q}(Z > -d_2) = N(d_2)$, where $N(x) \coloneqq (2\pi)^{-1/2}\int_{-\infty}^{x} e^{-y^2/2}\,dy$ is the standard normal distribution function.

Computation of the term (I)**.** We have

$$\text{(I)} = \mathbb{E}_{\mathbb{Q}}\big[S_T\,\mathbf{1}_{\{Z > -d_2\}}\big] = S_0\,e^{(r - \frac{1}{2}\sigma^2)T}\,\mathbb{E}_{\mathbb{Q}}\Big[e^{\sigma\sqrt{T}\,Z}\,\mathbf{1}_{\{Z > -d_2\}}\Big]\,.$$

To evaluate the expectation, we write the Gaussian integral explicitly, using that $Z \sim \mathcal{N}(0,1)$,

and complete the square in the exponent:

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}}\Big[e^{\sigma\sqrt{T}\,Z}\,\mathbf{1}_{\{Z>-d_2\}}\Big] &= \frac{1}{\sqrt{2\pi}}\int_{-d_2}^{\infty} e^{\sigma\sqrt{T}\,z}\,e^{-z^2/2}\,dz \\ &= \frac{1}{\sqrt{2\pi}}\int_{-d_2}^{\infty} \exp\Big(-\tfrac{1}{2}\big(z-\sigma\sqrt{T}\big)^2+\tfrac{1}{2}\sigma^2T\Big)\,dz \\ &= e^{\frac{1}{2}\sigma^2T}\cdot\frac{1}{\sqrt{2\pi}}\int_{-d_2}^{\infty}\exp\Big(-\tfrac{1}{2}\big(z-\sigma\sqrt{T}\big)^2\Big)\,dz.\end{aligned}$$

We make the change of variable $u \coloneqq z-\sigma\sqrt{T}$, with $du=dz$ and new limits $u>-d_2-\sigma\sqrt{T}$:

$$\mathbb{E}_{\mathbb{Q}}\Big[e^{\sigma\sqrt{T}\,Z}\,\mathbf{1}_{\{Z>-d_2\}}\Big] = e^{\frac{1}{2}\sigma^2T}\,\frac{1}{\sqrt{2\pi}}\int_{-d_2-\sigma\sqrt{T}}^{\infty} e^{-u^2/2}\,du = e^{\frac{1}{2}\sigma^2T}\,N(d_2+\sigma\sqrt{T}),$$

where the last equality again uses the symmetry of the Gaussian. Defining

$$d_1 \coloneqq d_2+\sigma\sqrt{T} = \frac{\log(S_0/K)+\big(r+\frac{1}{2}\sigma^2\big)T}{\sigma\sqrt{T}}, \tag{5.28}$$

and substituting into the expression for (I), the factor $e^{\frac{1}{2}\sigma^2T}$ cancels exactly with the corresponding term in the drift $r-\frac{1}{2}\sigma^2$ of the exponent of S_T, leaving

$$\text{(I)} = S_0\,e^{rT}\,N(d_1). \tag{5.29}$$

The Black-Scholes formula. Substituting (I) and (II) into (5.25) and multiplying by the discount factor e^{-rT}, we obtain the central formula:

Theorem 5.9 (Black-Scholes formula for the European *call*)**.** *The price at $t=0$ of the European* call *option with* strike K *and maturity T in the Black-Scholes model is*

$$\boxed{V(0,S_0) = S_0\,N(d_1)-K\,e^{-rT}\,N(d_2),} \tag{5.30}$$

where

$$d_1 = \frac{\log(S_0/K)+\big(r+\frac{1}{2}\sigma^2\big)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1-\sigma\sqrt{T} = \frac{\log(S_0/K)+\big(r-\frac{1}{2}\sigma^2\big)T}{\sigma\sqrt{T}},$$

and $N(\cdot)$ is the standard normal distribution function.

Generalization to the case $t>0$. Formula (5.30) has been derived at $t=0$. The generalization to the case $t\in(0,T)$ is obtained by taking into account the following observation: the process $\{S_t\}$ under $\mathbb{Q}$ is Markovian (since it solves the SDE (5.15), whose coefficients depend only on the current state) and, therefore, the conditional expectation $\mathbb{E}_{\mathbb{Q}}[(S_T-K)^+ \mid \mathcal{F}_t]$ depends exclusively on S_t, not on the whole past path. By the Markov property, computing this conditional expectation is equivalent to computing the unconditional expectation of the problem "started" at t with initial value S_t and duration $T-t$. Thus, the derivation we carried out for $t=0$, repeated with S_0 replaced by S_t and T replaced by $T-t$, gives

$$\boxed{V(t,S_t) = S_t\,N(d_1(t,S_t))-K\,e^{-r(T-t)}\,N(d_2(t,S_t)),} \tag{5.31}$$

where

$$d_1(t,S_t) = \frac{\log(S_t/K) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}},$$
$$d_2(t,S_t) = d_1(t,S_t) - \sigma\sqrt{T-t}.$$

Formula (5.31) expresses the price of the *call* at any time $t \in [0,T)$ as a function of the current price of the underlying S_t and the time to maturity $T-t$. It is the *value function* of the derivative, to which we shall return in the next section when deriving the Black-Scholes PDE.

5.4 The Black-Scholes PDE

We close the chapter with the derivation of the classical Black-Scholes PDE from the martingale calculus.

The derivation we have carried out in the preceding sections (via change of measure, construction of the $\mathbb{Q}$-martingale associated with the *payoff*, and application of the representation theorem) is the *martingale route* to the Black-Scholes theory. There is an alternative classical derivation, historically earlier, which starts from the idea of *instantaneous hedging* and leads to a partial differential equation. This section recovers this second formulation from formula (5.31), showing the consistency between the two routes.

Application of Itô to the value function. The value function of the European *call*, $V(t,S_t)$, given by (5.31), is of class $C^{1,2}$ on the domain $[0,T)\times(0,\infty)$. We apply Itô's formula (Theorem 4.13) to $V(t,S_t)$, with S_t an Itô process under $\mathbb{Q}$ with dynamics (5.15):

$$dV(t,S_t) = \left[\partial_t V + r\,S_t\,\partial_S V + \tfrac{1}{2}\,\sigma^2\,S_t^2\,\partial_{SS}V\right]dt + \sigma\,S_t\,\partial_S V\,d\widetilde{W}_t. \tag{5.32}$$

Comparison with the dynamics of the replicating portfolio. On the other hand, in Subsection 5.2.4 we proved that the value of the replicating portfolio satisfies (combining (5.21) and the identification (5.19)):

$$dV_t = r\,V_t\,dt + \sigma\,S_t\,\phi_t\,d\widetilde{W}_t, \tag{5.33}$$

where $\phi_t = \widetilde{\phi}_t\,e^{rt}/(\sigma\,S_t)$ is the quantity of the underlying. Since $V(t,S_t) = V_t$ by construction, expressions (5.32) and (5.33) must coincide term by term.

Equating the coefficients of $d\widetilde{W}_t$: $\sigma\,S_t\,\partial_S V = \sigma\,S_t\,\phi_t$, whence $\phi_t = \partial_S V$.

Equating the coefficients of dt:

$$\partial_t V + r\,S_t\,\partial_S V + \tfrac{1}{2}\,\sigma^2\,S_t^2\,\partial_{SS}V = r\,V.$$

Since this identity must hold for all (t,S_t) with S_t ranging over $(0,\infty)$ with positive probability, it translates into a partial differential equation for the function $V(t,S)$:

Theorem 5.10 (Black-Scholes equation)**.** *The value function $V(t,S)$ of the European* call *in the*

Black-Scholes model satisfies the partial differential equation

$$\boxed{\partial_t V + r\,S\,\partial_S V + \tfrac{1}{2}\,\sigma^2\,S^2\,\partial_{SS} V = r\,V,} \tag{5.34}$$

on the domain $(0,T)\times(0,\infty)$*, with the terminal boundary condition*

$$V(T,S) = (S-K)^+. \tag{5.35}$$

Identity (5.34) is the classical *Black-Scholes equation*, in its original form published in 1973. By what has been proved, the function defined by (5.31) is a solution of this boundary value problem; a uniqueness argument shows that it is in fact the unique sufficiently regular solution (see [5], Section 5.8).

The connection via Feynman-Kac. The equivalence between the two derivation routes, martingales and PDE, can be expressed abstractly through the *Feynman-Kac formula*, which establishes the identity between the solution of the PDE (5.34)–(5.35) and the probabilistic formula (5.23). Its rigorous proof lies outside the scope of this work and can be found in [8], Chapter 6.

Chapter 6

Conclusions

6.1 Summary of the work

This work has focused on the problem of *pricing the European derivative*, addressed in its discrete version (Chapter 2, binomial model) and in its continuous version (Chapters 3 to 5, Black-Scholes model). In both settings the same three-step structure is observed: change of measure to a risk-neutral measure, construction of the $\mathbb{Q}$-martingale associated with the discounted *payoff*, and application of a representation theorem. The translation between the two settings is not trivial (combinatorial and constructive in the discrete case, via Girsanov and abstract arguments in the continuous one), but the logical structure is identical.

The structural relationship between the discrete binomial model and the Black-Scholes model admits, beyond this conceptual analogy, a precise quantitative formulation: the Black-Scholes formula is *the limit* of the multi-period binomial formula as the number of steps tends to infinity. If in the binomial tree on $[0,T]$ with N periods of length $\Delta t = T/N$ the up and down parameters are defined as

$$u = e^{\mu\Delta t+\sigma\sqrt{\Delta t}}, \qquad d = e^{\mu\Delta t-\sigma\sqrt{\Delta t}}, \qquad q = \frac{e^{r\Delta t}-d}{u-d},$$

and one lets $N \to \infty$, the binomial random walk converges in distribution to the GBM under $\mathbb{Q}$ (by the central limit theorem), and the price of the *call* obtained by finite recursion on the tree converges to the Black-Scholes formula (5.30). Analogously, the quantity ϕ_k^A of the discrete replicating portfolio, given by the explicit formula (2.22), converges to $N(d_1)$. The rigorous proof of these convergences can be found in [3], Chapter 2. The Black-Scholes formula is therefore not an object independent of the binomial model, but its limit as the time steps become infinitely small.

6.2 Reflection on the approach

The fundamental structural decision of this work has been to approach Black-Scholes via martingales rather than via PDEs. Both formulations are equivalent, and their consistency is expressed through Feynman-Kac (Section 5.4); however, the martingale route allows the passage from the discrete to the continuous model to be presented as a natural extension, preserving the conceptual ideas of Chapter 2. This transition, less evident in a PDE-based presentation, justifies the

approach adopted.

It is worth stressing, finally, that the apparatus developed to pursue this route (the change of measure, Girsanov's theorem and martingale representation) is of interest well beyond the pricing problem: these are central tools of modern probability and stochastic analysis. The financial problem has been, in a sense, the thread that motivated the introduction of that machinery.

6.3 Further directions

This work has been restricted to European options on a single underlying in the Black-Scholes model with constant parameters. Each of these restrictions can be relaxed in various ways:

1. *American options*: incorporating the notion of optimal stopping time into the scheme developed here.
2. *More realistic models*: stochastic volatility (Heston), jumps (Merton), stochastic interest rates (Vasicek, Hull-White) and combinations thereof.
3. *Numerical methods*: Monte Carlo simulation and finite differences for *payoffs* without a closed-form formula (Asian, *barrier*, *basket options*, etc.).
4. *Calibration and implied volatility*: practical estimation of σ from market prices of liquid options.

Each of them presupposes the foundations developed in the present work and constitutes an independent chapter of contemporary mathematical finance.

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